

COVER REVIEW PLACEHOLDER

THE LEIMBACH METHOD

Powerbuilding, Nutrition, Recovery, and Performance

Version 1.0 Review Manuscript

Generated 2026-07-12

Review draft - not final publication copy

Final cover artwork and final cover copy pending author approval

Table of Contents

Table of Contents	2
Front Matter	5
Title Page	5
Copyright Page	6
Dedication	7
Foreword	8
Foundation	9
The Leimbach Philosophy	9
How to Use This Manual	10
Implementation Guide	11
Start Here	11
First-Week Setup	13
Mid-Block Entry	15
Weekly Operating Rhythm	17
Common Scenarios	19
User Checklists	21
Volume I - Nutrition	24
Athlete Profile	24
Goals and Nutrition Philosophy	26
Macro Targets	28
Day-Shift Meal Timing	31
Night-Shift Meal Timing	34
Training, Cardio, and Rest-Day Nutrition	37
Fourteen-Day Meal Plan	40
Meal-Prep Workflow	48
Costco and H-E-B Grocery Guide	54
Hydration and Electrolytes	59
Supplement Framework	62
Progress and Macro Adjustments	64
Volume II - Training	67
Program Overview	67

Readiness and Autoregulation	69
Warm-Up and Movement Preparation	71
Squat Development	74
Bench Press Development	76
Deadlift Development	78
Accessory Training	80
Core Training	82
Conditioning	84
Deloads and Fatigue Management	86
Six-Week Strength Block	88
Progression Rules	91
Volume III - Recovery	94
Recovery Overview	94
Sleep System	96
Shift-Work Recovery	98
Heat-Stress Recovery	100
Soreness and Pain Management	102
Weekly Recovery Planning	104
Recovery Checklists	106
Volume IV - Tracking	108
Tracking Overview	108
Bodyweight and Waist Tracking	110
Training Log	112
Recovery Log	114
Weekly Review	116
Dashboard Template	118
Exercise Library	121
Library Overview	121
Squat Variations	123
Bench Variations	125
Deadlift Variations	127
Accessories	129
Core and Conditioning	131
Substitution Rules	133
Back Matter	135

The Promise	135
About the Author	136
References	137
Version History	139
PDF Export Checklist	140
Final Review Checklist	141

Front Matter

Title Page

Source: Publishing/03-front-matter.md | Status: review

Title Page Review Placeholder

The Leimbach Method Powerbuilding, Nutrition, Recovery, and Performance Review Draft Manuscript

Prepared as a linked publication assembly for review. Final title page copy, edition number, publication date, and release notes should be added only after author approval.

Placeholder status: Final publication copy has not been approved. Do not replace this section with invented copy.

Safety Disclaimer

This manual is not medical advice and does not diagnose, treat, or prevent medical conditions. Training, nutrition, hydration, supplement, heat-stress, and recovery decisions should be individualized and reviewed with appropriate professionals when needed.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention. Pain, injury concerns, persistent joint symptoms, or symptoms that do not improve with normal load management should be reviewed with an appropriate coach and clinician.

Copyright Page

Source: Generated review placeholder | Status: review

Review placeholder. Approved copyright copy is not present in the repository.

Review placeholder - final copy needed.

Approved copyright, rights, permissions, ISBN, disclaimer, and edition language are not present yet.

Do not insert legal or publication copy until the author supplies or approves it.

This placeholder is intentionally preserved for review and does not mark the manuscript final.

Dedication

Source: Generated review placeholder | Status: review

Review placeholder. Approved dedication copy is not present in the repository.

Review placeholder - final copy needed.

Approved dedication copy is not present yet.

Leave this section as a placeholder until the author supplies final wording.

This placeholder is intentionally preserved for review and does not mark the manuscript final.

Foreword

Source: Generated review placeholder | Status: review

Review placeholder. Approved foreword copy is not present in the repository.

Review placeholder - final copy needed.

Approved foreword copy is not present yet.

Leave this section as a placeholder until final foreword text is supplied and approved.

This placeholder is intentionally preserved for review and does not mark the manuscript final.

Foundation

The Leimbach Philosophy

Source: README.md | Status: review draft

Who This Method Is For

This method is written for a large strength athlete working rotating oilfield shifts who wants a practical system for building size, strength, conditioning, and recovery capacity. It assumes long workdays, hot environments, inconsistent sleep, heavy barbell training, and the need for simple weekly decisions.

How to Use This Manual

Source: Publishing/03-front-matter.md | Status: review

How To Use This Manual

- Start with the Implementation Guide if you need the fastest path into the method.
- Use the Nutrition, Training, Recovery, and Tracking volumes together rather than as isolated systems.
- Use the Exercise Library when a lift, accessory, equipment option, readiness state, or time constraint requires a substitution.
- Use the weekly review process to decide whether to maintain, push, reduce stress, repeat a week, or seek review.
- Use the Publishing checklists only for review and PDF preparation until the final publication process begins.

Implementation Guide

Start Here

Source: *Implementation/01-start-here.md* | Status: review

Purpose and Scope

This chapter explains how to begin The Leimbach Method without reading every chapter first. Use it as the onboarding map: set the minimum pieces, start training, track the week, then use the weekly review to adjust.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Who This Method Is For

This method is built for a large strength athlete working rotating oilfield shifts who wants to move from roughly 260 lb toward a lean, strong 275 lb while building the squat, bench, and deadlift. It assumes unpredictable work stress, hot shifts, uneven sleep, and the need for simple decisions.

What the Method Includes

System	What It Does	Read First
Nutrition	Sets macros, meal timing, meal prep, hydration, and adjustment rules	Macro targets and meal prep workflow
Training	Runs the 6-week strength block with RPE-based loading	Program overview and current block
Recovery	Manages sleep, shift work, heat stress, soreness, and weekly recovery planning	Recovery overview and shift-work recovery
Tracking	Turns bodyweight, waist, training, and recovery logs into weekly decisions	Tracking overview and dashboard template
Exercise Library	Gives substitutions when equipment, readiness, or time changes	Library overview and substitution rules

What to Read First

Time Available	Read These
20 minutes	This chapter, first-week setup, weekly operating rhythm
45 minutes	Add macro targets, current training block, tracking dashboard
90 minutes	Add recovery overview, exercise library overview, substitution rules
Ongoing	Read deeper chapters only when the weekly review shows a need

What to Set Up Before Beginning

Setup Item	Minimum Standard
Baselines	7-day bodyweight start, waist, current shift schedule
Training week	Monday squat, Tuesday conditioning, Wednesday bench, Thursday conditioning, Friday deadlift
Nutrition	Protein, carb, fat targets and first meal-prep plan
Hydration	Cooler, fluids, electrolytes, heat-shift plan
Recovery	Sleep anchors, caffeine cutoff, recovery day plan
Tracking	Weekly dashboard copied and ready
Substitutions	One backup option for each main lift and conditioning day

Choose Your Starting Point

Situation	Starting Point
New to the current plan or coming off inconsistent training	Start at Week 1
Already training hard and roughly around Week 4 loading	Start around Week 4 after readiness check
Coming back after time off	Restart at Week 1 or run a build week
One bad week from work, sleep, heat, or missed sessions	Repeat the week
Pain changes technique or red flags appear	Stop the session and seek coach/clinician review or urgent care when warranted

Copy-Ready Start-Here Checklist

Item	Done
Read Start Here, First-Week Setup, and Weekly Operating Rhythm	
Record starting bodyweight and waist	
Pick training start point: Week 1 / Week 4 / repeat week / restart	
Set nutrition targets and meal-prep plan	
Set hydration/cooler plan for work shifts	
Set sleep anchors and recovery days	
Copy the weekly dashboard	
Choose backup exercises for squat, bench, deadlift, and conditioning	
Note any pain, persistent symptoms, or red flags	

Coach Review Notes

- Review pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms with a qualified coach and clinician.
- Use urgent medical attention for heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms.
- Do not use this guide as a substitute for medical diagnosis or treatment.

References

- ACSM and NSCA guidance for conservative exercise entry, progression, readiness, and weekly planning.
- ISSN sports nutrition guidance for bodyweight trend decisions, fueling, and supplement caution.
- CDC/NIOSH/OSHA heat-stress guidance for hot-shift planning and red-flag escalation.
- ACSM behavior-change and exercise-adherence guidance for keeping onboarding simple and repeatable.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

First-Week Setup

Source: [Implementation/02-first-week-setup.md](#) | Status: review

Purpose and Scope

The first week is not for perfection. It is for setting baselines, proving the schedule works, and creating the dashboard data needed for the first weekly review.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Set Baseline Bodyweight and Waist

Baseline	How to Set It
Bodyweight	Weigh in each morning when practical; calculate the first 7-day average
Waist	Measure once under the same morning conditions each week
Notes	Log heat exposure, sodium changes, poor sleep, or missed meals that may distort weight

Set Training Days and Shift Context

Day	Default Plan	Shift Adjustment
Monday	Squat emphasis	Move or trim if coming off a brutal night shift
Tuesday	Conditioning	Keep low impact if legs or heat exposure are high
Wednesday	Bench emphasis	Use substitutions if bench station or readiness is limited
Thursday	Conditioning	Reduce intensity before a hard Friday deadlift
Friday	Deadlift emphasis	Cap RPE if sleep, heat, or soreness is poor
Saturday	Recovery/prep	Meal prep, light movement, sleep catch-up
Sunday	Review/planning	Complete dashboard and set next week

Set Nutrition Targets and Meal Prep

Setup Item	First-Week Action
Protein	Hit the daily target consistently
Carbs	Start at the planned target; adjust only after trend review
Fats	Keep steady enough to make appetite and calories predictable
Meal prep	Cook portable work-shift meals before the week starts
Missed meals	Log them before changing macros

Set Hydration and Cooler Plan

Item	First-Week Action
Fluids	Pack enough for the shift and commute
Electrolytes	Include a work-shift option, especially for heat exposure
Cooler	Pack meals, backup carbs, protein, fluids, and salt/electrolyte option
Heat shifts	Log heat exposure and bodyweight changes before cutting food

Set Recovery and Sleep Anchors

Anchor	First-Week Standard
Main sleep window	Protect the longest available sleep block
Caffeine cutoff	Set a realistic stop time for day and night shifts
Wind-down	Use a repeatable pre-sleep routine
Recovery days	Keep Saturday/Sunday easy unless the week was unusually light

Set Tracking Dashboard

Copy the weekly dashboard before the week starts. Fill in bodyweight, waist, training log, recovery notes, nutrition compliance, heat exposure, and the three weekly decisions: nutrition, training, recovery.

Choose Equipment and Substitution Options

Slot	Backup Option
Squat	Safety-bar squat, high-bar squat, leg press, or split squat
Bench	Paused bench, dumbbell press, floor press, machine press, or push-up
Deadlift	Trap-bar deadlift, Romanian deadlift, hip thrust, or block pull
Conditioning	Bike, incline walk, sled, rower/air bike if tolerated

First-Week Decision Rules

Signal	First Move
Weight trend unclear	Maintain food; collect more weigh-ins
Waist jumps quickly	Check sodium, heat, sleep, and compliance before changing food
RPE too high	Repeat load or trim volume
Sleep poor or shift brutal	Move session, cap RPE, or reduce conditioning
Pain changes technique	Stop/modify and seek coach/clinician review

Copy-Ready First-Week Setup Checklist

Item	Done
Record 7-day bodyweight plan and first waist measurement	
Map work shifts against training days	
Set nutrition targets	
Meal prep work-shift meals and backup snacks	
Pack hydration/electrolyte/cooler setup	
Set sleep anchors and caffeine cutoff	
Copy weekly dashboard	
Choose backup exercises and conditioning options	
Note red flags or pain issues for review	

Coach Review Notes

- Review pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms with a qualified coach and clinician.
- Red-flag symptoms require urgent medical attention rather than weekly troubleshooting.

References

- ACSM and NSCA guidance for conservative exercise entry, progression, readiness, and weekly planning.
- ISSN sports nutrition guidance for bodyweight trend decisions, fueling, and supplement caution.
- CDC/NIOSH/OSHA heat-stress guidance for hot-shift planning and red-flag escalation.
- ACSM behavior-change and exercise-adherence guidance for keeping onboarding simple and repeatable.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Mid-Block Entry

Source: Implementation/03-mid-block-entry.md | Status: review

Purpose and Scope

This chapter explains how to enter the 6-week block when the athlete is already training around Week 4 intensity. The goal is to choose a productive entry point without using old maxes or forcing a test.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Entering Around Week 4

Start around Week 4 only if recent training is consistent, technique is repeatable, readiness is green or stable yellow, and current RPE-based loads are known. Week 4 should feel like strong productive work, not a max test.

Readiness Check Before Choosing a Week

Signal	Start Week 4	Start Week 5	Repeat/Build Week
Recent lifting	Consistent and close to planned block	Consistent but fatigue is high	Inconsistent or uncertain
Sleep/shift stress	Manageable	Mixed	Poor or chaotic
Bar speed	Normal	Slightly slow	Slow across main lifts
Soreness	Normal	Elevated but manageable	High or movement-limiting
Pain	No technique change	Mild and monitored	Changes technique or persistent

Select Loads With RPE

Step	Rule
Warm-up	Build gradually and watch speed/position
Top work	Choose the load that lands at the planned RPE that day
Back-offs	Reduce load enough to keep reps clean
Old maxes	Use only as context, not as today's prescription
Missed reps	Do not chase; reduce load or repeat the week

Poor Sleep, Hot Shifts, or Soreness

Constraint	Adjustment
Poor sleep	Cap RPE, trim accessories, or move the heavy session
Hot shift	Prioritize hydration/recovery; reduce conditioning intensity
High soreness	Extend warm-up, reduce volume, choose lower-fatigue variation
Multiple constraints	Repeat/build week instead of entering Week 4

Using Weeks 5-6 After Mid-Block Entry

Week	Purpose
Week 5	Consolidate: hold load, clean technique, trim fatigue if needed
Week 6	Lower stress: reduce volume/intensity enough to recover and review

When to Restart at Week 1

Restart at Week 1 after time off, multiple missed main sessions, unclear current loading, repeated slow bar speed, poor readiness, or pain that changes technique. Restarting is a cleaner decision than pretending Week 4 data exists.

Copy-Ready Mid-Block Entry Decision Table

Question	Answer	Decision
Have the last 2-3 weeks been consistent?		Week 4 if yes; build week if no
Are current RPE loads known?		Week 4 if yes; Week 1/build if no
Is readiness green or stable yellow?		Week 4/5 if yes; repeat/build if no
Any pain changing technique?		Coach/clinician review before progressing
Sleep/heat/soreness manageable?		Week 4 if yes; Week 5 or repeat if no
Is Week 6 planned as lower stress?		Keep it lower stress after entry

Coach Review Notes

- Review pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms with a qualified coach and clinician.
- Do not use mid-block entry to justify max testing.
- Use urgent medical attention for heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms.

References

- ACSM and NSCA guidance for conservative exercise entry, progression, readiness, and weekly planning.
- ISSN sports nutrition guidance for bodyweight trend decisions, fueling, and supplement caution.
- CDC/NIOSH/OSHA heat-stress guidance for hot-shift planning and red-flag escalation.
- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for symptom escalation.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Weekly Operating Rhythm

Source: [Implementation/04-weekly-operating-rhythm.md](#) | Status: review

Purpose and Scope

The weekly rhythm turns the method into a repeatable operating system. Each week connects Nutrition, Training, Recovery, Tracking, and the Exercise Library with one simple review loop.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Monday Through Sunday Rhythm

Day	Main Job	Minimum Standard	Log
Monday	Squat emphasis	Warm up, squat work, one support movement	Readiness, sets/ reps/ load, RPE, pain/ soreness
Tuesday	Conditioning	Low-impact intervals or adjusted conditioning	Mode, duration, readiness effect
Wednesday	Bench emphasis	Bench work, one upper support movement	RPE, bar speed, technique note
Thursday	Conditioning	Conditioning that does not hurt Friday	Mode, intensity, recovery effect
Friday	Deadlift emphasis	Hinge work with RPE cap	Load, bar speed, low-back fatigue
Saturday	Recovery/ prep	Meal prep, light movement, sleep support	Recovery notes, food prep
Sunday	Review/ planning	Dashboard, weekly decisions, next-week plan	Nutrition/ training/ recovery decisions

Daily Minimums When Work Is Chaotic

Area	Minimum
Nutrition	Hit protein and pack work-shift food
Hydration	Bring fluids/ electrolyte option for heat exposure
Training	Complete the highest-priority lift or choose a same-intent substitution
Recovery	Protect the biggest sleep window available
Tracking	Log bodyweight when practical, readiness, training, and red flags

Weekly Review Rhythm

- Calculate 7-day average bodyweight.
- Record waist.
- Review main lift performance, RPE, bar speed, and substitutions.
- Review sleep, heat exposure, soreness, pain, and stress.
- Review nutrition compliance before changing macros.
- Choose one nutrition decision, one training decision, and one recovery decision.

One Decision Per System

System	Decision Options
Nutrition	Maintain, add carbs, reduce carbs
Training	Progress, repeat, trim, deload
Recovery	Maintain, adjust sleep, move sessions, reduce conditioning
Exercise Library	Keep programmed lifts or choose substitutions that preserve intent
Tracking	Keep dashboard simple enough to complete weekly

Copy-Ready Weekly Rhythm Table

Day	Plan	Shift Adjustment	Must Log
Monday	Squat emphasis	Move/trim after brutal shift	RPE, bar speed, pain
Tuesday	Conditioning	Low impact after heat or soreness	Mode, duration, readiness
Wednesday	Bench emphasis	Substitute if station unavailable	Load, RPE, technique
Thursday	Conditioning	Protect Friday deadlift	Intensity, soreness
Friday	Deadlift emphasis	Cap RPE if sleep/heat poor	Low-back fatigue, bar speed
Saturday	Recovery/prep	Meal prep and sleep catch-up	Prep completed
Sunday	Review/planning	Dashboard and weekly decisions	Nutrition/training/recovery decisions

Coach Review Notes

- Review pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms with a qualified coach and clinician.
- Red-flag symptoms require urgent medical attention instead of schedule manipulation.

References

- ACSM and NSCA guidance for conservative exercise entry, progression, readiness, and weekly planning.
- ISSN sports nutrition guidance for bodyweight trend decisions, fueling, and supplement caution.
- CDC/NIOSH/OSHA heat-stress guidance for hot-shift planning and red-flag escalation.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Common Scenarios

Source: *Implementation/05-common-scenarios.md* | Status: review

Purpose and Scope

This chapter gives quick decisions for the problems most likely to interrupt the method: missed sessions, heat, sleep, bodyweight swings, slow bar speed, pain flags, unavailable equipment, and conditioning that affects lifting.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Scenario Decision Table

Scenario	First Move	Avoid	What to Log	Next-Week Decision
Missed Monday squat	Move squat to Tuesday if recovery allows, or run a shortened squat session	Cramming squat and bench heavy on the same day	Reason missed, sleep, shift, readiness	Repeat or adjust week if squat quality suffered
Missed Wednesday bench	Move bench to Thursday or use a shorter bench session	Adding extra volume to compensate	Reason missed, available equipment	Progress if quality holds; repeat if not
Missed Friday deadlift	Move to Saturday only if recovery is good; otherwise repeat/build next week	Heavy deadlift after severe fatigue	Shift context, soreness, readiness	Repeat week if deadlift was skipped
Worked a hot 16-hour shift	Prioritize fluids, food, sleep, and lower-stress training	Hard intervals or heavy grinders	Heat exposure, bodyweight, hydration, symptoms	Move sessions or reduce conditioning
Slept poorly	Extend warm-up, cap RPE, trim accessories	Forcing planned loads	Sleep duration/quality, caffeine, readiness	Adjust sleep plan or repeat loads
Bodyweight dropped suddenly	Check heat, hydration, sodium, missed meals, and scale timing	Cutting food immediately	Weight, heat, fluids, meals	Maintain until trend confirms
Waist jumped quickly	Recheck measurement and review sodium/carbs/sleep	Panic-cutting calories after one measure	Waist site, bodyweight trend, meals	Use weekly trend before reducing carbs
Bar speed was slow	Reduce load or repeat work at lower RPE	Adding sets to prove toughness	Lift, load, RPE, sleep/heat context	Repeat week, trim, or deload if repeated
Pain changed technique	Stop or modify the movement	Training through altered mechanics	Location, movement affected, severity	Coach/clinician review before progression
Gym equipment unavailable	Use Exercise Library substitution with same pattern/intent	Random exercise swap	Equipment issue, substitution, RPE	Keep plan if intent was preserved
Conditioning hurt lifting readiness	Reduce intensity, switch to low-impact, or move timing	More intervals on low readiness	Mode, duration, next-day readiness	Reduce conditioning or move sessions

Quick Rules

Problem Type	Rule
Missed training	Preserve the main lift priority; do not cram everything
Recovery disruption	Adjust training before changing the whole plan
Scale surprise	Look for heat, hydration, sodium, sleep, and missed meals
Pain or red flags	Stop troubleshooting and seek appropriate review or urgent care
Equipment problem	Substitute same pattern, same intent, similar fatigue cost

Coach Review Notes

- Review pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms with a qualified coach and clinician.
- Use urgent medical attention for heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms.
- Do not use scenario rules to bypass red-flag care.

References

- ACSM and NSCA guidance for conservative exercise entry, progression, readiness, and weekly planning.
- ISSN sports nutrition guidance for bodyweight trend decisions, fueling, and supplement caution.
- CDC/NIOSH/OSHA heat-stress guidance for hot-shift planning and red-flag escalation.
- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for symptom escalation.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

User Checklists

Source: *Implementation/06-user-checklists.md* | Status: review

Purpose and Scope

These checklists turn the method into repeatable actions. Copy them into a dashboard, note app, printed binder, or weekly review sheet.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Start-Here Checklist

Item	Done
Read Start Here, First-Week Setup, and Weekly Operating Rhythm	
Choose start point: Week 1 / Week 4 / restart / repeat	
Copy weekly dashboard	
Set first review day	
List current work shifts	
Note pain, persistent symptoms, or red flags	

First-Week Setup Checklist

Item	Done
Record daily weigh-ins for 7-day average	
Measure waist once under consistent conditions	
Set macro targets and meal-prep plan	
Pack cooler meals, fluids, and electrolyte option	
Set training days around shifts	
Choose backup exercises and conditioning options	
Set sleep anchors and caffeine cutoff	

Daily Checklist

Item	Done
Morning bodyweight if practical	
Protein target planned	
Work meals packed	
Fluids/electrolytes planned	
Sleep duration/quality logged	
Readiness color selected	
Pain/red flags checked	

Training-Day Checklist

Item	Done
Check sleep, heat exposure, soreness, and readiness	
Warm up gradually	
Use RPE for loading instead of old maxes	
Log sets, reps, load, RPE, and bar speed	
Use substitution only if it preserves training intent	
Trim accessories if readiness is low	
Stop/modify if pain changes technique	

Cardio-Day Checklist

Item	Done
Check leg soreness and next lifting day	
Choose low-impact option if readiness is low	
Reduce intensity after hot shifts or poor sleep	
Log mode, duration, intensity, and readiness effect	
Avoid sprints when joints or recovery are not ready	

Rest-Day Checklist

Item	Done
Meal prep or restock work food	
Protect sleep window	
Use light movement if helpful	
Review soreness and pain notes	
Prepare next training day	

Weekly Review Checklist

Item	Done
Calculate 7-day bodyweight average	
Record waist	
Review training performance, RPE, and bar speed	
Review conditioning effect on lifting	
Review sleep, heat, soreness, pain, and stress	
Review nutrition compliance before macro changes	
Choose one nutrition decision	
Choose one training decision	
Choose one recovery decision	

Mid-Block Entry Checklist

Item	Done
Confirm recent training consistency	
Confirm current RPE loads are known	
Check sleep, heat, soreness, and readiness	
Decide Week 4 / Week 5 / repeat-build / restart Week 1	
Plan Week 6 as lower stress	
Avoid max testing	

Red-Flag Checklist

Red Flag	Action
Heat illness symptoms	Urgent medical attention
Chest pain	Urgent medical attention
Fainting	Urgent medical attention
Severe shortness of breath	Urgent medical attention
Neurological symptoms	Urgent medical attention
Rapidly worsening symptoms	Urgent medical attention
Pain changes technique	Stop/modify and seek coach/clinician review
Persistent joint issues	Coach/clinician review

Coach Review Notes

- Review pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms with a qualified coach and clinician.

- Do not use checklists to continue training through red-flag symptoms.

References

- ACSM and NSCA guidance for conservative exercise entry, progression, readiness, and weekly planning.
- ISSN sports nutrition guidance for bodyweight trend decisions, fueling, and supplement caution.
- CDC/NIOSH/OSHA heat-stress guidance for hot-shift planning and red-flag escalation.
- ACSM behavior-change and exercise-adherence guidance for keeping onboarding simple and repeatable.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Volume I - Nutrition

Athlete Profile

Source: Nutrition/01-athlete-profile.md | Status: review

Purpose

Document the athlete's starting context so the nutrition plan is built around the actual body, schedule, training load, preferences, and constraints instead of a generic bulking template.

Chapter Contents

- Profile Snapshot (#profile-snapshot)
- Work and Schedule Demands (#work-and-schedule-demands)
- Training Background (#training-background)
- Nutrition Preferences (#nutrition-preferences)
- Health and Recovery Context (#health-and-recovery-context)
- Primary Outcomes (#primary-outcomes)

Profile Snapshot

This manual is written for a male strength athlete, age 33, standing 6'4" and currently weighing approximately 260 lb. The long-term objective is to build toward a lean, strong 275 lb while keeping waist gain controlled, training performance high, and recovery stable across rotating day and night shifts.

Item	Current Working Value	Planning Use
Age	33 years	Adult athlete nutrition planning
Height	6'4"	Portion size and bodyweight context
Bodyweight	~260 lb	Weekly trend baseline
Goal bodyweight	275 lb	Long-term lean mass target
Primary goal	Lean strength gain	Calorie surplus with performance monitoring

Unconfirmed assumptions: body-fat percentage, current waist measurement, food allergies, medications, lab markers, digestive limitations, supplement use, and exact training times are not yet confirmed. Until those are recorded, this chapter should be treated as a practical draft, not a final prescription.

Work and Schedule Demands

The athlete works rotating 3-to-7 shifts that may run either day or night. That creates two major nutrition problems: long work windows and inconsistent sleep timing. The plan therefore favors portable meals, repeatable meal timing, and predictable carbohydrate placement around training instead of relying on appetite alone.

Shift Type	Work Window	Nutrition Priority
Day shift	3:00 a.m. to 7:00 p.m.	Front-load prepared food and protect post-shift training fuel
Night shift	3:00 p.m. to 7:00 a.m.	Eat enough before and during shift while protecting morning sleep
Off/rest day	Variable	Keep protein fixed and adjust carbohydrate lower

A successful plan must be easy to pack: cooked rice or potatoes, lean proteins, fruit, soups, Greek yogurt, oats, beans, tortillas, sauces, and shakes all fit the schedule better than meals that require fresh cooking during the shift.

Training Background

The current weekly training structure is heavy lifting 3 days per week and cardio intervals 2 days per week. Nutrition should support high-output lifting first, maintain enough carbohydrate for intervals, and keep rest days productive without forcing training-day calories when activity is lower.

Day Type	Weekly Frequency	Primary Fuel Need
Heavy lifting	3 days	Highest carbohydrate and total calories
Cardio intervals	2 days	Moderate-high carbohydrate with stable protein
Rest/recovery	2 days	Protein anchored, carbohydrate reduced

The plan assumes lifting sessions are the main growth stimulus. Cardio is treated as conditioning support, not as a fat-loss phase. If cardio volume increases, calorie targets should be reassessed rather than quietly absorbed by appetite.

Nutrition Preferences

The food system should be built from meals the athlete already likes: steak, chicken, fish, avocados, fruit, soups, overnight oats, beans, rice, Mexican, Italian, Asian, BBQ, protein shakes, and Greek-yogurt protein desserts. This gives enough variety to hit high calories without turning the plan into a short list of bland foods.

Food Category	Practical Examples	Use in the Plan
Proteins	Chicken, steak, fish, Greek yogurt, protein shakes	Hit 260 g protein daily
Carbs	Rice, oats, beans, fruit, pasta, tortillas, potatoes	Drive training output and weight gain
Fats	Avocado, olive oil, fattier steak, sauces	Add calories without excessive food volume
Meal prep bases	Soups, rice bowls, burrito bowls, pasta bowls	Portable meals for long shifts

Portions should be gram-based when possible. Cups may be added for convenience, but gram weights are the default because they make weekly adjustments easier.

Health and Recovery Context

No medical conditions are confirmed in the issue context. The nutrition plan therefore stays within general performance coaching: consistent protein, sufficient calories, hydration, fiber, fruits and vegetables, and planned adjustments based on measurable response.

The main recovery risks are missed meals during long shifts, under-eating before training, eating too heavily immediately before sleep, and drifting into a surplus that is larger than needed. The weekly review process in Chapter 3 is the control system for those risks.

Primary Outcomes

The first phase is successful when the athlete can repeat the plan across both shift types, maintain strong gym performance, and gain weight at a controlled pace without a disproportionate waist increase.

Outcome	Target / Standard	Review Signal
Bodyweight	Gradual trend toward 275 lb	7-day average weight
Protein	260 g daily	Daily tracking average
Training	Heavy sessions fueled	Strength, reps, pump, and readiness
Waist control	No rapid waist jump	Weekly waist measurement
Schedule adherence	Meals packed before shift	Missed-meal count

Coach Review Notes

- Confirm starting waist measurement, average morning bodyweight, and usual lifting time on day shift and night shift.
- Confirm any food allergies, digestion issues, or medical nutrition restrictions.
- Confirm whether the athlete prefers 4, 5, or 6 feedings on the longest workdays.

References

- ISSN and ACSM sports nutrition guidance for strength athletes and body-composition goals.
- NSCA strength-training references for aligning nutrition support with progressive training.
- CDC/NIOSH/OSHA heat-stress guidance for oilfield shift and hydration context.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Goals and Nutrition Philosophy

Source: Nutrition/02-goals-and-nutrition-philosophy.md | Status: review

Purpose

Define the principles that guide the nutrition system and translate the 260-to-275-lb objective into realistic performance outcomes.

Chapter Contents

- Mission (#mission)
- Desired Rate of Gain (#desired-rate-of-gain)
- Performance Priorities (#performance-priorities)
- Food Quality and Flexibility (#food-quality-and-flexibility)
- Consistency Standards (#consistency-standards)
- Success Criteria (#success-criteria)

Mission

The mission of Volume I is to build a repeatable nutrition base for a large strength athlete who wants to move from approximately 260 lb toward a lean, strong 275 lb. The plan should make eating enough feel organized instead of random: fixed protein, calorie targets by day type, practical meal prep, and shift-specific timing.

The nutrition system has three jobs:

- Support heavy lifting performance 3 days per week.
- Recover from cardio intervals 2 days per week without treating them like a cutting phase.
- Keep the athlete gaining slowly enough that most of the added weight supports performance, not just scale weight.

Desired Rate of Gain

The first draft target is a controlled gain of roughly 0.25 to 0.75 lb per week based on the 7-day average bodyweight. Faster gain may be useful for a short block if training is climbing and waist is stable, but it should not become the default.

Weekly Trend	Interpretation	Default Action
Down or flat for 2 weeks	Not enough total intake or adherence gaps	Add 150-250 kcal/day, mostly carbs
Up 0.25-0.75 lb/week	Productive gaining range	Hold targets steady
Up >1.0 lb/week with waist jump	Surplus may be too aggressive	Remove 150-250 kcal/day, usually carbs/fats
Weight up, performance down	Poor timing, sleep, hydration, or food quality	Fix schedule before adding more calories

Unconfirmed assumption: the exact rate of gain should be refined after 2-4 weeks of morning weigh-ins and waist measurements.

Performance Priorities

Nutrition decisions should be judged by gym output, recovery, and repeatability. A macro target that looks perfect on paper but causes missed meals during a 16-hour shift is not the right target yet.

Priority order:

- Hit daily protein.
- Eat enough total calories for the day type.
- Place most carbohydrates before and after lifting or intervals.
- Keep meals digestible during work and before sleep.
- Adjust based on the weekly trend, not one noisy weigh-in.

Food Quality and Flexibility

The plan should be mostly whole-food meals with enough flexibility to survive real workdays. Steak, chicken, fish, rice, beans, fruit, oats, soups, Greek yogurt, avocados, pasta, tortillas, and sauces can cover most needs. Protein shakes and Greek-yogurt desserts are tools, especially when appetite drops or the schedule compresses.

A practical plate for this athlete usually includes:

- 180-250 g cooked lean protein or a shake/dairy equivalent.
- 250-450 g cooked carbohydrate base such as rice, potatoes, pasta, beans, or oats.
- 100-200 g vegetables or fruit when digestion allows.
- 10-30 g fat from avocado, olive oil, cheese, sauce, or fattier meat.

This is not a clean-food-only plan. It is a performance plan with enough structure to keep the surplus controlled.

Consistency Standards

The athlete does not need perfect days. The standard is enough repeatable days that the weekly average becomes predictable.

Standard	Target
Protein adherence	6-7 days/week within 20 g of target
Calorie adherence	5-6 days/week within 250 kcal of target
Prepared work meals	Packed before shift starts
Bodyweight tracking	4-7 morning weigh-ins/week
Waist tracking	1 measurement/week, same conditions
Meal timing	No more than 5-6 waking hours without protein

Success Criteria

A successful first nutrition block produces a stable rhythm before it produces a perfect physique outcome. The athlete should know what to eat on lifting, cardio, and rest days; how to pack food for day and night shifts; and how to adjust intake without guessing.

This phase is working when:

- The 7-day average bodyweight trends upward at a controlled rate.
- Heavy lifts feel fueled, not flat.
- Interval days do not create next-day energy crashes.
- Waist gain stays proportionate to bodyweight gain.
- Appetite is manageable across both day and night shifts.
- The athlete can describe the plan without needing a spreadsheet open every hour.

Coach Review Notes

- Confirm preferred weekly check-in day.
- Confirm whether target bodyweight is tied to a performance goal, visual goal, or both.
- Confirm whether any foods should be limited for digestion, budget, or work access.

References

- ISSN and ACSM sports nutrition guidance for strength athletes and body-composition goals.
- NSCA strength-training references for aligning nutrition support with progressive training.
- CDC/NIOSH/OSHA heat-stress guidance for oilfield shift and hydration context.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Macro Targets

Source: Nutrition/03-macro-targets.md | Status: review

Purpose

Establish starting calorie and macronutrient targets with clear assumptions, practical meal distribution, and weekly adjustment rules.

Chapter Contents

- Starting Assumptions (#starting-assumptions)
- Training-Day Targets (#training-day-targets)
- Cardio-Day Targets (#cardio-day-targets)
- Rest-Day Targets (#rest-day-targets)
- Meal Distribution (#meal-distribution)
- Monitoring and Recalculation (#monitoring-and-recalculation)

Starting Assumptions

These are starting targets for a 33-year-old, 6'4", approximately 260-lb male athlete lifting heavily 3 days per week, doing cardio intervals 2 days per week, and working rotating 3-to-7 shifts. The targets are intentionally high because the athlete is large, active, and trying to gain toward 275 lb.

Day Type	Calories	Protein	Carbs	Fat
Training day	4,200 kcal	260 g	510 g	95 g
Cardio day	3,850 kcal	260 g	420 g	95 g
Rest day	3,500 kcal	260 g	320 g	100 g

Unconfirmed assumptions: maintenance calories, current body composition, exact lifting schedule, step count, sweat rate, sodium needs, and digestion tolerance are not yet confirmed. Use these as starting targets and adjust from observed response.

Training-Day Targets

Training days get the highest calorie and carbohydrate target because heavy lifting is the main growth driver.

Target	Amount
Calories	4,200 kcal
Protein	260 g
Carbohydrate	510 g
Fat	95 g

Training-day priorities:

- Keep protein steady across 4-6 feedings.
- Put 120-200 g carbs in the pre- and post-workout window when possible.
- Use lower-fat meals close to training if digestion feels slow.
- Add calorie-dense extras such as avocado, olive oil, cheese, or sauces away from the workout window if appetite is low.

Cardio-Day Targets

Cardio interval days stay high enough to support conditioning and recovery, but carbohydrates are lower than heavy lifting days.

Target	Amount
Calories	3,850 kcal
Protein	260 g
Carbohydrate	420 g
Fat	95 g

Cardio-day priorities:

- Eat a carb-and-protein meal 1-3 hours before intervals when schedule allows.

- Keep post-cardio protein steady even if appetite is lower.
- Do not turn interval days into accidental low-calorie days.
- If intervals are late in the shift, use a shake plus fruit or oats rather than forcing a heavy meal.

Rest-Day Targets

Rest days keep protein high while reducing carbohydrates and slightly increasing fat. The purpose is recovery, not restriction.

Target	Amount
Calories	3,500 kcal
Protein	260 g
Carbohydrate	320 g
Fat	100 g

Rest-day priorities:

- Keep meal timing consistent enough to avoid under-eating.
- Use soups, bowls, and Greek-yogurt desserts to maintain protein without feeling stuffed.
- Keep carbs mostly around the most active part of the day.
- Use rest days to prep food for the next work block.

Meal Distribution

The easiest starting structure is 5 feedings per day on workdays and 4-5 feedings on non-workdays. Protein stays evenly distributed; carbs flex up or down by day type.

Feeding	Protein Target	Carb Target	Fat Target	Example
Meal 1	45-60 g	60-110 g	10-25 g	Overnight oats with whey and fruit
Meal 2	45-60 g	70-120 g	10-25 g	Chicken, rice, beans, salsa, avocado
Meal 3	45-60 g	70-120 g	10-25 g	Steak or fish bowl with potatoes or rice
Meal 4	45-60 g	60-120 g	10-25 g	Pasta, lean meat sauce, vegetables
Meal 5	40-60 g	30-80 g	10-25 g	Greek yogurt protein dessert or shake

For gram-based prep, start with repeatable components:

- 200-250 g cooked chicken, steak, fish, or lean ground meat per main meal.
- 250-450 g cooked rice, potatoes, pasta, or beans depending on day type.
- 150-300 g fruit or vegetables per meal when practical.
- 15-30 g fats from avocado, olive oil, cheese, nuts, or sauce as needed.

Monitoring and Recalculation

Use a weekly check-in, not daily emotion, to adjust the targets. Review bodyweight trend, waist change, appetite, recovery, and gym performance together.

Weekly adjustment rule:

Check-In Result	Adjustment
Bodyweight flat/down for 2 weeks, waist stable, appetite manageable, recovery/gym performance good or flat	Add 150-250 kcal/day, mostly from 35-60 g carbs
Bodyweight up 0.25-0.75 lb/week, waist stable, appetite and performance good	Hold targets steady
Bodyweight up >1.0 lb/week and waist rising quickly	Remove 150-250 kcal/day from carbs and/or fats
Appetite poor but weight not rising	Reduce meal size per feeding and add a shake, smoothie, soup, or liquid carbs
Recovery poor or gym performance falling	First check sleep, missed meals, hydration, sodium, and workout timing; adjust calories only after those are addressed

Track the following each week:

- 7-day average morning bodyweight.
- Waist measurement under the same conditions.
- Appetite rating from 1-5.

- Recovery rating from 1-5.
- Heavy-lift performance notes.
- Missed meals or low-calorie days.

Coach Review Notes

- Confirm whether macro targets should be tracked exactly or with ranges.
- Confirm preferred tracking app or spreadsheet.
- Confirm sodium, caffeine, and hydration habits before writing final guidance.

References

- ISSN position stands on protein, nutrient timing, and body-composition support for athletes.
- Academy of Nutrition and Dietetics / ACSM / Dietitians of Canada sports nutrition position guidance.
- ACSM and NSCA strength-training guidance for matching fueling to training demand.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Day-Shift Meal Timing

Source: Nutrition/04-day-shift-meal-timing.md | Status: review

Purpose

Create a practical meal and hydration schedule for a 3:00 a.m. to 7:00 p.m. workday.

Chapter Contents

- Shift Timeline (#shift-timeline)
- Wake-Up Meal (#wake-up-meal)
- Work Meals (#work-meals)
- Training Window (#training-window)
- Post-Training Meal (#post-training-meal)
- Bedtime Nutrition (#bedtime-nutrition)

Shift Timeline

The day-shift plan assumes the athlete is awake before 3:00 a.m., works until 7:00 p.m., and may train after the shift. The schedule is long enough that food must be packed before work starts.

Time	Feeding	Purpose	Example
2:00-2:30 a.m.	Wake-up meal	Start protein and carbs early	Overnight oats with whey, Greek yogurt, berries, and banana
5:30-6:30 a.m.	Work meal 1	Prevent early energy crash	Chicken, rice, beans, salsa, avocado
9:30-10:30 a.m.	Work meal 2	Keep calories moving	Steak or fish bowl with rice/potatoes and fruit
1:30-2:30 p.m.	Work meal 3	Main mid-shift fuel	Soup plus rice, lean meat, beans, and tortillas
5:00-6:00 p.m.	Pre-training meal or snack	Prepare for post-shift training	Shake plus fruit, rice cakes, oats, or a lower-fat bowl
7:30-9:00 p.m.	Post-training meal	Refill carbs and protein	Pasta with lean meat sauce or BBQ chicken and rice
9:30-10:30 p.m.	Bedtime protein	Finish protein without heavy digestion	Greek-yogurt protein dessert or casein-style shake

Unconfirmed assumption: exact wake time, commute time, break access, and training time are not confirmed. Move feedings 30-60 minutes as needed while keeping the same order.

Wake-Up Meal

The wake-up meal should be prepared the night before. The athlete should not be making complicated food before a 3:00 a.m. start.

Good wake-up meal targets:

- 45-60 g protein.
- 70-120 g carbs on training days.
- 10-25 g fat depending on appetite.
- Fluids and electrolytes if the shift starts dehydrated.

Practical options:

- 100 g dry oats mixed with whey, Greek yogurt, berries, and banana.
- 300 g Greek yogurt with 40 g whey, 80-100 g cereal or granola, and fruit.
- Rice bowl with 200 g cooked chicken, 300 g cooked rice, salsa, and 50-75 g avocado.

Work Meals

The workday needs 2-3 packed meals plus a backup. Each main meal should be easy to eat cold or reheated and should include a known protein amount.

Meal Build	Training Day	Cardio Day	Rest Day
Cooked protein	200-250 g	200-250 g	200-250 g
Cooked rice/potatoes/pasta	350-450 g	275-375 g	200-300 g
Beans or fruit	100-200 g	100-200 g	100-150 g
Avocado/oil/cheese/sauce	10-25 g fat	10-25 g fat	15-30 g fat

Backup foods for missed breaks:

- Whey shake plus banana.
- Greek yogurt cup plus granola.
- Rice cakes and jerky.
- Ready-to-drink protein plus fruit.
- Soup in a thermos with cooked rice added.

Training Window

If lifting happens after the 7:00 p.m. shift, the pre-training feeding should be lighter and lower in fat. The goal is to arrive at the gym fueled but not sluggish.

Best pre-training options 60-120 minutes before lifting:

- Whey shake plus banana and oats.
- Chicken and rice with salsa, little added fat.
- Greek yogurt, honey, berries, and cereal.
- Soup with rice and lean protein if digestion is good.

If training must happen before the shift, use a smaller carb-and-protein meal before training and move the larger breakfast to the first work break.

Post-Training Meal

The post-training meal is the largest opportunity to finish the day. It should include a full protein serving, a large carbohydrate serving, and enough fat to close the calorie gap without making sleep uncomfortable.

Examples:

- 250 g cooked chicken, 450 g cooked rice, beans, salsa, and avocado.
- 225 g lean steak, 500 g potatoes, vegetables, and olive-oil-based sauce.
- 250 g lean meat sauce, 350-450 g cooked pasta, parmesan, and fruit.
- BBQ chicken bowl with rice, beans, corn, and sauce.

Bedtime Nutrition

Bedtime nutrition should finish protein and calories while protecting sleep. If dinner was large and late, keep this small. If the day ran short, use a dense but easy option.

Good bedtime options:

- 300-400 g Greek yogurt mixed with whey and berries.
- Protein shake with milk and a banana.
- Cottage cheese or Greek yogurt with honey and granola.
- Small steak/chicken leftovers if appetite is still strong.

Avoid turning bedtime into a second dinner if it disrupts sleep. Poor sleep will make the next day's appetite, training, and bodyweight trend harder to interpret.

Coach Review Notes

- Confirm actual break times and whether reheating food is available.
- Confirm whether post-shift training is the default or only one option.
- Confirm caffeine cutoff time for day shift.

References

- ISSN nutrient-timing guidance for protein/carbohydrate distribution around training.
- NIOSH work-schedule and shift-work education for fatigue-aware planning.
- CDC/NIOSH/OSHA heat-stress guidance where shift heat exposure affects hydration and recovery.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Night-Shift Meal Timing

Source: Nutrition/05-night-shift-meal-timing.md | Status: review

Purpose

Create a practical meal and hydration schedule for a 3:00 p.m. to 7:00 a.m. work night.

Chapter Contents

- Shift Timeline (#shift-timeline)
- Pre-Shift Meal (#pre-shift-meal)
- Overnight Meals (#overnight-meals)
- Training Window (#training-window)
- Post-Shift Meal (#post-shift-meal)
- Sleep Preparation (#sleep-preparation)

Shift Timeline

The night-shift plan assumes a 3:00 p.m. to 7:00 a.m. work window. The goal is to eat enough before and during the shift without creating a heavy meal right before daytime sleep.

Time	Feeding	Purpose	Example
12:00-1:00 p.m.	Wake/pre-shift meal	First major protein and carb anchor	Chicken, rice, beans, salsa, avocado, and fruit
2:00-2:30 p.m.	Pre-shift top-off	Prevent early-shift hunger	Greek yogurt protein dessert or shake plus oats
6:00-7:00 p.m.	Work meal 1	Main evening meal	Steak, potatoes or rice, vegetables, sauce
10:00-11:00 p.m.	Work meal 2	Keep calories steady overnight	Mexican, Italian, Asian, or BBQ rice bowl
2:00-3:00 a.m.	Overnight snack	Protein plus easy carbs	Shake, fruit, cereal, yogurt, or soup
6:00-6:30 a.m.	Post-shift small meal	Finish protein without hurting sleep	Greek yogurt, shake, eggs/lean meat, or soup
7:30-8:30 a.m.	Sleep wind-down	Hydration and digestion control	Light fluids; avoid a large high-fat meal

Unconfirmed assumption: sleep timing after the shift, commute length, break access, and night-shift training preference are not confirmed. Adjust times while keeping the same feeding sequence.

Pre-Shift Meal

The pre-shift meal should be one of the largest meals of the night-shift day. It gives the athlete a calorie base before work demands start and reduces the need to force oversized meals at 3:00 a.m.

Targets:

- 50-70 g protein.
- 90-150 g carbs on training days.
- 15-30 g fat.
- 500-1,000 ml fluid before the shift if tolerated.

Practical examples:

- 250 g cooked chicken, 400 g cooked rice, 150 g beans, salsa, and 75 g avocado.
- 225 g steak, 450 g potatoes, fruit, and vegetables.
- 350-450 g cooked pasta with lean meat sauce and parmesan.
- Large bowl of overnight oats plus Greek yogurt and whey if appetite is better for breakfast-style food.

Overnight Meals

Overnight meals should be repeatable and digestible. The athlete may need more liquid or soft foods after midnight because appetite can drop when sleep pressure rises.

Overnight Need	Best Tools
Full meal	Rice bowl, pasta bowl, soup with rice, burrito bowl

Overnight Need	Best Tools
Low appetite	Shake, Greek yogurt, smoothie, soup
High hunger	Steak/chicken bowl with avocado or olive oil
No break	Ready-to-drink protein, fruit, rice cakes, jerky
Cold meal only	Greek yogurt bowl, chicken rice salad, tuna rice bowl

The 2:00-3:00 a.m. feeding should usually be smaller than the 6:00-11:00 p.m. meals. It exists to protect protein, carbs, and energy, not to become a sleep-disrupting feast.

Training Window

There are two practical training windows on night shift: before work or after work.

Pre-shift training works best when the athlete can eat a real meal afterward before clocking in. Use a lighter pre-workout meal 60-120 minutes before lifting, then make the 12:00-2:30 p.m. meals larger.

Post-shift training is possible but should be treated cautiously. If performance is consistently poor after a 16-hour night shift, move lifting to before the shift or to a non-shift day. If post-shift training is unavoidable, use a shake plus fast carbs near the end of shift and eat the recovery meal immediately after training.

Post-Shift Meal

The post-shift meal should match the next priority. If the athlete is going straight to sleep, keep it lighter. If the athlete trained after the shift, it must be a real recovery meal.

Sleep-first option:

- 300-400 g Greek yogurt with whey and berries.
- Protein shake with milk and banana.
- Soup with lean protein and rice.

Post-training option:

- 250 g cooked chicken or fish with 350-450 g cooked rice.
- 225 g steak with potatoes and fruit.
- Pasta with lean meat sauce and a small amount of added fat.

Sleep Preparation

The last hour after the shift should reduce digestive load and stimulation. The athlete should avoid using a massive meal to make up for missed calories if it ruins sleep. Missed calories are better handled by improving the next night's packing plan.

Sleep-prep standards:

- Finish the final substantial meal before the commute or immediately after arriving home.
- Keep fluids reasonable if waking to urinate disrupts sleep.
- Stop caffeine early enough that daytime sleep is protected.
- Use a dark, cool sleep environment whenever possible.

Coach Review Notes

- Confirm whether the athlete normally trains before night shift, after night shift, or on off days.
- Confirm caffeine timing and tolerance during overnight work.
- Confirm whether meals can be refrigerated or reheated overnight.

References

- ISSN nutrient-timing guidance for protein/carbohydrate distribution around training.
- NIOSH work-schedule and shift-work education for fatigue-aware planning.
- CDC/NIOSH/OSHA heat-stress guidance where shift heat exposure affects hydration and recovery.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.

- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Training, Cardio, and Rest-Day Nutrition

Source: Nutrition/06-day-type-nutrition.md | Status: review

Purpose

Explain how food quantity and meal timing change by activity type while keeping the plan practical for rotating 3-to-7 shifts.

Chapter Contents

- Training Days (#training-days)
- Cardio Days (#cardio-days)
- Rest Days (#rest-days)
- Pre-Workout Nutrition (#pre-workout-nutrition)
- Post-Workout Nutrition (#post-workout-nutrition)
- Substitution Rules (#substitution-rules)

Training Days

Training days support heavy lifting and carry the highest target: 4,200 kcal, 260 g protein, 510 g carbs, and 95 g fat. These days should feel fueled. If the athlete is dragging through warmups, losing pump, or cutting sets because energy is low, the first suspects are missed meals, poor pre-workout carbs, sleep, hydration, and sodium.

Training-day structure:

Meal Focus	Goal
First meal	Start protein and carbs early
Pre-workout	Easy protein plus 60-120 g carbs
Post-workout	Full protein plus 100-160 g carbs
Later meals	Finish calories without forcing uncomfortable volume

Good training-day meals include burrito bowls, chicken and rice, steak and potatoes, pasta with lean meat sauce, BBQ chicken rice bowls, overnight oats, shakes, and Greek-yogurt desserts.

Cardio Days

Cardio interval days use 3,850 kcal, 260 g protein, 420 g carbs, and 95 g fat. These days need enough carbs to support intervals and recovery, but not as much as heavy lifting days.

Cardio-day structure:

Meal Focus	Goal
Pre-cardio	Protein plus 40-90 g carbs
Post-cardio	Protein plus 60-100 g carbs
Work meals	Moderate carb portions with steady protein
Final meal	Finish protein and calories without sleep disruption

If intervals are short but intense, keep carbohydrates close to the session. If the session is easier than planned, do not slash food the same day; review trends weekly.

Rest Days

Rest days use 3,500 kcal, 260 g protein, 320 g carbs, and 100 g fat. Rest days are lower in carbs but still high enough to recover and maintain the gaining phase.

Rest-day structure:

Meal Focus	Goal
Protein	Stay at 260 g for recovery and appetite control
Carbs	Lower than training days, mostly around active hours
Fat	Slightly higher to keep calories workable
Meal prep	Cook proteins and carbs for upcoming shifts

Rest-day meals can be more flexible: soups with lean meat and beans, steak with avocado and potatoes, fish with rice and vegetables, Greek yogurt bowls, and lower-carb Mexican or Asian bowls.

Pre-Workout Nutrition

Pre-workout nutrition should make training feel easier, not heavier. The best option depends on timing.

Time Before Training	Target	Examples
3-4 hours	Full meal	Chicken, rice, beans, fruit; steak and potatoes
1-2 hours	Moderate meal	Greek yogurt, oats, whey, banana; chicken and rice with little fat
30-60 minutes	Small top-off	Whey shake, banana, rice cakes, sports drink if needed

Keep fat and fiber lower close to training if digestion is slow. On long shifts, the pre-workout meal may be packed and eaten at work before leaving for the gym.

Post-Workout Nutrition

Post-workout nutrition should close the loop: protein for recovery, carbs to restore training fuel, and enough calories to keep the gaining phase on track.

Post-workout targets:

- 50-70 g protein.
- 100-160 g carbs on heavy lifting days.
- 60-100 g carbs on cardio days.
- Fat based on remaining daily target and sleep timing.

Examples:

- 250 g cooked chicken, 450 g cooked rice, beans, salsa, avocado.
- 225 g steak, 500 g potatoes, vegetables, and sauce.
- 350-450 g cooked pasta with lean meat sauce.
- Whey shake plus cereal or oats when appetite is low, followed by a full meal later.

Substitution Rules

Substitutions should preserve the role of the food, not just the flavor. Swap protein for protein, carb for carb, and fat for fat.

Replace	With	Approximate Swap
200 g cooked chicken	Fish, lean steak, lean ground meat, Greek yogurt plus whey	Match protein grams
300 g cooked rice	Potatoes, pasta, oats, tortillas, beans	Match carb grams
75 g avocado	Olive oil, cheese, fattier meat, nuts	Match fat grams
Full meal when rushed	Shake plus fruit/oats, then a later meal	Protect protein and carbs
Low appetite meal	Soup, smoothie, Greek yogurt bowl	Reduce chewing burden

Practical meal-prep rule: build 2 protein bases, 2 carbohydrate bases, 1 soup or bowl option, and 1 dessert/shake option for each work block. This keeps variety high without requiring a new recipe every day.

Weekly Adjustment Rule

The day-type plan should be adjusted only after reviewing the weekly pattern.

- If bodyweight is flat or down for 2 weeks, waist is stable, appetite is manageable, recovery is acceptable, and gym performance is not improving, add 150-250 kcal/day, mostly from carbohydrates.
- If bodyweight is rising 0.25-0.75 lb/week, waist is stable, appetite is manageable, recovery is good, and gym performance is improving or steady, hold targets.
- If bodyweight jumps more than 1 lb/week and waist rises quickly, remove 150-250 kcal/day from carbs and/or fats.
- If appetite is poor, keep calories similar but shift some food to shakes, soups, smoothies, or Greek-yogurt desserts.
- If gym performance falls, first review sleep, hydration, sodium, missed meals, and training timing before changing macros.

Coach Review Notes

- Confirm which weekdays usually map to lifting, cardio, and rest.
- Confirm whether the athlete prefers high-carb meals before or after training.
- Confirm any foods that cause bloating during lifting or overnight shifts.

References

- ISSN position stands on protein, nutrient timing, and body-composition support for athletes.
- Academy of Nutrition and Dietetics / ACSM / Dietitians of Canada sports nutrition position guidance.
- ACSM and NSCA strength-training guidance for matching fueling to training demand.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Fourteen-Day Meal Plan

Source: Nutrition/07-fourteen-day-meal-plan.md | Status: review

Purpose

Provide the first usable Week One meal plan for the rotating 3-to-7 shift athlete. The title remains Fourteen-Day Meal Plan, but this draft fills Week One only. Week Two will be built later from the same structure after Week One is reviewed.

Chapter Contents

- How to Use Week One (#how-to-use-week-one)
- Meal Builder Rule (#meal-builder-rule)
- Macro Calculation Assumptions (#macro-calculation-assumptions)
- Shift Timing Template (#shift-timing-template)
- Week One Overview (#week-one-overview)
- Week One Schedule (#week-one-schedule)
- Week One Meal Prep (#week-one-meal-prep)
- Week Two Placeholder (#week-two-placeholder)
- Daily Macro Totals (#daily-macro-totals)
- Meal Swaps (#meal-swaps)
- Validation Note (#validation-note)
- Coach Review Notes (#coach-review-notes)

How to Use Week One

This plan uses the Volume I starting macro targets as the starting point, then keeps every meal row calculated from the assumptions table below. Daily totals are summed from the six listed meals. The original calorie targets are higher than the exact calories produced by 260 g protein, 510/420/320 g carbs, and 95/95/100 g fat, so Week One keeps totals inside the issue's allowed practical ranges instead of forcing exact-looking numbers.

Day Type	Original Target	Accepted Week One Range
Training Day	4,200 kcal, 260 g protein, 510 g carbs, 95 g fat	4,050-4,350 kcal, 240-280 g protein, 480-540 g carbs, 80-110 g fat
Cardio Day	3,850 kcal, 260 g protein, 420 g carbs, 95 g fat	3,700-4,000 kcal, 240-280 g protein, 390-450 g carbs, 80-110 g fat
Rest Day	3,500 kcal, 260 g protein, 320 g carbs, 100 g fat	3,350-3,650 kcal, 240-280 g protein, 290-350 g carbs, 85-115 g fat

Use cooked weights for prepared proteins and rice unless a note clearly says raw or dry. Macro numbers are practical planning estimates. Final numbers depend on the exact brands used, especially for granola, peanut butter, BBQ sauce, teriyaki sauce, Alfredo, marinara, pudding mix, tortillas, cheese, and protein powder.

Meal Builder Rule

The meal plan is a working example, not a prison sentence. The daily target comes first; the meals are modular. Build the day by choosing meals that preserve the same role: wake-up meal, work-shift meal, pre-training meal, post-training meal, bedtime meal, or no-cook backup.

Use this rule before changing the plan:

- Match the day type first: training, cardio, or rest.
- Keep protein steady.
- Move carbs toward training and away from rest-day meals.
- Keep fats lower before training and higher when the meal is farther from training.
- Recalculate the meal from the assumptions table when portions change.

Breakfast Options - Choose One

Nobody needs to build this method around two dozen eggs. Eggs can be useful, but breakfast should rotate based on shift timing, appetite, prep time, and digestion.

Option	Best Use	Build	Macro Role
Overnight oats	Fast wake-up meal, cooler-friendly	Oats, whey, Greek yogurt, berries or banana, peanut butter if needed	High-carb training or moderate-carb rest option
Greek yogurt bowl	No-cook, low appetite, before sleep	Greek yogurt, whey, berries, granola, pudding mix, peanut butter or jam	High-protein snack or dessert-style meal
Breakfast burrito	Meal prep, long shift, bigger appetite	Tortilla, egg whites, 1-2 whole eggs, potatoes, cheese, salsa, avocado	Higher-calorie work meal
Protein smoothie	Travel, pre-training, hot weather	Whey, oats, banana, berries, water or low-calorie milk	Lower-fat carb/protein option
Steak and eggs	Weekend or home meal	Lean steak, 1-2 whole eggs, potatoes or rice, salsa	Higher-satiety meal; adjust fats carefully
Avocado toast with eggs	Quick home meal	Toast or tortilla, eggs/egg whites, avocado, fruit or potatoes	Moderate-carb, higher-fat option
Cottage cheese bowl	No-cook backup	Cottage cheese or high-protein dairy, fruit, granola, peanut butter if needed	High-protein snack or breakfast swap
Protein pancakes	Training-day option	Protein pancake mix or oats/egg whites/whey, berries, syrup or jam	Higher-carb meal; label-check mix

Lunch and Dinner Options - Choose One

Lunch and dinner work the same way. Pick the cuisine that sounds edible, then keep the protein and carb role close to the meal being replaced.

Option	Best Use	Macro Role
Chicken Alfredo	Post-training or high-calorie dinner	Higher-carb, moderate-fat
Steak and rice bowl	Heavy lifting day	High-protein, high-carb
Salmon bowl	Cardio or rest day	Higher-fat protein meal
Chicken fajitas	Work-shift meal	High-protein, flexible carbs
Hibachi chicken	Prep-friendly bowl	High-carb, lower-fat if sauce is controlled
Korean beef bowl	Higher-satiety work meal	Higher-fat protein; adjust rice as needed
BBQ chicken bowl	Training or cardio day	High-carb, low-to-moderate fat
Turkey chili	Cooler or thermos meal	High-protein, moderate-carb

Snacks and Dessert Options

Snacks are not random grazing. Choose one or two to patch the day without blowing up the macro target.

Category	Options	Use
Protein snacks	Protein shake, Greek yogurt, cottage cheese, beef jerky, protein pudding	Patch protein when a full meal is not practical
Carb snacks	Fruit, rice cakes, oats packet, tortilla, jam or honey	Add carbs around training or long shifts
Higher-fat snacks	Peanut butter sandwich, trail mix, avocado add-on	Add calories away from training
Protein desserts	Cheesecake bowl, protein pudding, frozen yogurt bark, Oreo-style protein fluff, chocolate peanut butter mousse, high-protein ice cream, brownie mug cake	Keep dessert in the system instead of pretending it does not exist

The meal-builder standard is simple: if the swap helps adherence and the day still lands near the target range, it belongs in the method.

Macro Calculation Assumptions

These working values keep the plan internally consistent. Calories are derived from the listed macros using 4 kcal/g for protein, 4 kcal/g for carbohydrate, and 9 kcal/g for fat. Confirm final numbers against the actual labels used during meal prep.

Food	Portion Basis	Calories	Protein	Carbs	Fat
Cooked chicken breast	100 g cooked	160	31 g	0 g	4 g
Cooked 93% lean beef	100 g cooked	189	27 g	0 g	9 g
Cooked lean steak	100 g cooked	206	29 g	0 g	10 g
Cooked salmon	100 g cooked	192	22 g	0 g	12 g
Cooked white fish	100 g cooked	122	26 g	0 g	2 g
Cooked lean turkey	100 g cooked	156	29 g	0 g	5 g
Cooked white/jasmine rice	100 g cooked	124	3 g	28 g	0 g
Cooked pasta	100 g cooked	157	6 g	31 g	1 g
Cooked potatoes	100 g cooked	88	2 g	20 g	0 g
Dry oats	100 g dry	387	13 g	68 g	7 g

Food	Portion Basis	Calories	Protein	Carbs	Fat
Nonfat Greek yogurt	100 g	56	10 g	4 g	0 g
Whey protein	35 g scoop	142	28 g	3 g	2 g
Egg whites	100 g cooked	48	11 g	1 g	0 g
Whole egg	1 large / 50 g	69	6 g	0 g	5 g
Black/white beans	100 g cooked/drained	133	8 g	23 g	1 g
Banana	100 g peeled	96	1 g	23 g	0 g
Berries	100 g	52	1 g	12 g	0 g
Avocado	100 g	179	2 g	9 g	15 g
Peanut butter	16 g / 1 tbsp	100	4 g	3 g	8 g
Large flour tortilla	70 g	218	6 g	35 g	6 g
Corn tortillas	120 g	308	8 g	60 g	4 g
Granola	30 g	149	4 g	22 g	5 g
Salsa	80 g	28	1 g	6 g	0 g
Peppers/onions	100 g	32	1 g	7 g	0 g
Broccoli	100 g	40	3 g	7 g	0 g
Mixed vegetables	100 g	60	3 g	12 g	0 g
Corn	100 g	105	3 g	21 g	1 g
Tortilla strips	40 g	175	3 g	25 g	7 g
Reduced-fat shredded cheese	30 g	85	8 g	2 g	5 g
Parmesan	15 g	60	5 g	1 g	4 g
Mozzarella	30 g	90	7 g	2 g	6 g
BBQ sauce	50 g	140	0 g	35 g	0 g
Teriyaki sauce	50 g	104	1 g	25 g	0 g
Marinara	125 g	99	3 g	15 g	3 g
Light Alfredo	120 g	152	3 g	8 g	12 g
Enchilada sauce	125 g	98	2 g	18 g	2 g
Stir-fry sauce	30 g	44	1 g	10 g	0 g
Honey	15 g	48	0 g	12 g	0 g
Jam	60 g	160	0 g	40 g	0 g
Pudding mix	20 g	72	0 g	18 g	0 g
Pudding mix	25 g	88	0 g	22 g	0 g
Cocoa powder	10 g	37	2 g	5 g	1 g
Olive oil	10 g	90	0 g	0 g	10 g
Soup broth/tomato base	300 g	90	3 g	15 g	2 g
Soup broth/tomato base	250 g	78	3 g	12 g	2 g
Cucumber/carrots	120 g	28	1 g	6 g	0 g
Zucchini	150 g	32	2 g	6 g	0 g

Shift Timing Template

Use the same six-meal order on day shift or night shift. Move meals by 30-60 minutes when breaks, commute, or training time require it.

Meal	Day Shift Timing: 3:00 a.m. to 7:00 p.m.	Night Shift Timing: 3:00 p.m. to 7:00 a.m.
Meal 1	2:00-2:30 a.m., pre-shift / wake-up	12:00-1:00 p.m., wake-up / pre-shift
Meal 2	5:30-6:30 a.m., early shift	6:00-7:00 p.m., early shift
Meal 3	9:30-10:30 a.m., mid-shift	10:00-11:00 p.m., midnight meal
Meal 4	1:30-2:30 p.m., late shift	2:00-3:00 a.m., late shift
Meal 5	7:00-7:30 p.m., post-shift or pre-training	7:00-7:30 a.m., post-shift or pre-training
Meal 6	8:30-9:30 p.m., post-training / before bed	8:00-9:00 a.m., before sleep

For training days, Meal 5 should usually be the lighter pre-training meal and Meal 6 should usually be the larger post-training meal. If training happens before shift, swap those roles while keeping the same daily food totals.

Week One Overview

Day	Training Type	Day Type	Target	Planned Total	Main Food Themes
Monday	Heavy lifting	Training	4,200 kcal	4,125 kcal	Overnight oats, Mexican bowls, Asian chicken, lean beef pasta
Tuesday	Cardio intervals	Cardio	3,850 kcal	3,709 kcal	Breakfast burrito, salmon rice bowl, soup, BBQ chicken
Wednesday	Heavy lifting	Training	4,200 kcal	4,057 kcal	Protein oats, steak fajitas, chicken fried rice, high-protein Alfredo
Thursday	Cardio intervals	Cardio	3,850 kcal	3,703 kcal	Eggs and potatoes, beef and broccoli, chili, Greek-yogurt dessert
Friday	Heavy lifting	Training	4,200 kcal	4,213 kcal	Breakfast burrito, BBQ bowl, teriyaki chicken, chicken parmesan bowl
Saturday	Rest	Rest	3,500 kcal	3,354 kcal	Eggs, white chicken chili, salmon avocado bowl, protein pudding
Sunday	Rest or light recovery	Rest	3,500 kcal	3,498 kcal	Overnight oats, beef vegetable soup, enchilada-style prep, yogurt bowl

Week One Schedule

Monday - Heavy Lifting / Training Day

Day-shift timing: Meal 1 pre-shift, Meal 2 early shift, Meal 3 mid-shift, Meal 4 late shift, Meal 5 post-shift or pre-training, Meal 6 post-training / before bed.

Night-shift timing: Meal 1 wake-up / pre-shift, Meal 2 early shift, Meal 3 midnight meal, Meal 4 late shift, Meal 5 post-shift or pre-training, Meal 6 before sleep.

Meal	Food and Portions	Calories	Protein	Carbs	Fat	Notes
Meal 1	Overnight oats: 60 g dry oats, 35 g whey, 50 g nonfat Greek yogurt, 100 g banana, 100 g berries, 16 g peanut butter	650	47 g	84 g	14 g	Oats are dry weight; yogurt and fruit are ready-to-eat weights. About 2/3 cup dry oats.
Meal 2	Chicken burrito bowl: 100 g cooked chicken breast, 235 g cooked white rice, 80 g black beans, 80 g salsa, 50 g avocado	675	46 g	95 g	12 g	Chicken, rice, and beans are cooked weights. About 1.3 cups cooked rice.
Meal 3	Steak fajita bowl: 130 g cooked lean steak, 250 g cooked rice, 100 g peppers/onions, 50 g black beans, 60 g avocado	784	51 g	94 g	22 g	Steak and rice are cooked weights. Use fajita seasoning and salsa as preferred.
Meal 4	Teriyaki chicken rice bowl: 90 g cooked chicken breast, 200 g cooked jasmine rice, 150 g broccoli, 50 g teriyaki sauce	556	39 g	92 g	4 g	Chicken and rice are cooked weights. About 1 cup cooked rice.
Meal 5	Pre-training shake: 35 g whey, 50 g dry oats, 100 g banana, 15 g honey	480	36 g	72 g	6 g	Oats are dry weight. Blend with water or ice.
Meal 6	Lean beef pasta plus Greek-yogurt cheesecake bowl: 100 g cooked 93% lean beef, 180 g cooked pasta, 125 g marinara, 15 g parmesan, 100 g nonfat Greek yogurt, 20 g cheesecake pudding mix, 80 g berries, 20 g olive oil	980	57 g	103 g	38 g	Beef and pasta are cooked weights. Pasta is about 1.25 cups cooked.
Daily Total		4,125	276 g	539 g	96 g	Sum of the six meal rows.

Tuesday - Cardio Intervals / Cardio Day

Day-shift timing: Meal 1 pre-shift, Meal 2 early shift, Meal 3 mid-shift, Meal 4 late shift, Meal 5 post-shift or pre-cardio, Meal 6 post-cardio / before bed.

Night-shift timing: Meal 1 wake-up / pre-shift, Meal 2 early shift, Meal 3 midnight meal, Meal 4 late shift, Meal 5 post-shift or pre-cardio, Meal 6 before sleep.

Meal	Food and Portions	Calories	Protein	Carbs	Fat	Notes
Meal 1	Breakfast burrito: 100 g cooked egg whites, 2 whole eggs, 180 g cooked potatoes, 70 g tortilla, 30 g reduced-fat cheese, 80 g salsa, 30 g avocado	729	42 g	83 g	26 g	Eggs and potatoes are cooked weights. Tortilla weight is ready-to-eat.
Meal 2	Salmon rice bowl: 150 g cooked salmon, 180 g cooked rice, 120 g cucumber/carrots, 55 g avocado	644	40 g	61 g	26 g	Salmon and rice are cooked weights. About 1 cup cooked rice.
Meal 3	Chicken tortilla soup: 90 g cooked chicken breast, 300 g broth/tomato base, 100 g black beans, 80 g corn, 40 g tortilla strips, 30 g avocado	680	45 g	82 g	19 g	Chicken, beans, and corn are cooked/drained weights. Soup base is prepared weight.
Meal 4	BBQ chicken bowl: 90 g cooked chicken breast, 185 g cooked rice, 80 g corn, 40 g BBQ sauce, 20 g shredded cheese	626	41 g	98 g	8 g	Chicken and rice are cooked weights. Label-check BBQ sauce.
Meal 5	Cardio smoothie: 35 g whey, 40 g dry oats, 100 g banana, 100 g berries	445	35 g	65 g	5 g	Oats are dry weight. Use water or low-calorie milk if preferred.

Meal	Food and Portions	Calories	Protein	Carbs	Fat	Notes
Meal 6	Greek-yogurt PB&J bowl: 200 g nonfat Greek yogurt, 35 g whey, 20 g granola, 16 g peanut butter, 30 g jam, 100 g berries	585	56 g	61 g	13 g	Yogurt and fruit are ready-to-eat weights. Granola and jam vary by brand.
Daily Total		3,709	260 g	450 g	97 g	Sum of the six meal rows.

Wednesday - Heavy Lifting / Training Day

Day-shift timing: Meal 1 pre-shift, Meal 2 early shift, Meal 3 mid-shift, Meal 4 late shift, Meal 5 post-shift or pre-training, Meal 6 post-training / before bed.

Night-shift timing: Meal 1 wake-up / pre-shift, Meal 2 early shift, Meal 3 midnight meal, Meal 4 late shift, Meal 5 post-shift or pre-training, Meal 6 before sleep.

Meal	Food and Portions	Calories	Protein	Carbs	Fat	Notes
Meal 1	Chocolate banana protein oats: 55 g dry oats, 35 g whey, 50 g nonfat Greek yogurt, 100 g banana, 10 g cocoa, 16 g peanut butter, 5 g honey	632	47 g	77 g	15 g	Oats are dry weight. Add water for texture before chilling.
Meal 2	Steak fajita bowl: 130 g cooked lean steak, 250 g cooked rice, 100 g peppers/onions, 50 g black beans, 60 g avocado	784	51 g	94 g	22 g	Steak and rice are cooked weights.
Meal 3	Chicken burrito bowl: 100 g cooked chicken breast, 235 g cooked white rice, 80 g black beans, 80 g salsa, 50 g avocado	675	46 g	95 g	12 g	Chicken, rice, and beans are cooked weights.
Meal 4	Chicken fried rice: 90 g cooked chicken breast, 200 g cooked rice, 100 g mixed vegetables, 1 whole egg, 10 g olive oil, 30 g stir-fry sauce	655	44 g	78 g	19 g	Chicken and rice are cooked weights. Egg is cooked into the rice.
Meal 5	Pre-training shake: 35 g whey, 50 g dry oats, 100 g banana, 15 g honey	480	36 g	72 g	6 g	Oats are dry weight. Use before lifting when a full meal is too heavy.
Meal 6	High-protein Alfredo bowl: 100 g cooked chicken breast, 180 g cooked pasta, 120 g light Alfredo sauce, 150 g broccoli, 15 g parmesan, 13 g olive oil	832	54 g	75 g	35 g	Chicken and pasta are cooked weights. Pasta is about 1.25 cups cooked; label-check Alfredo.
Daily Total		4,057	279 g	491 g	109 g	Sum of the six meal rows.

Thursday - Cardio Intervals / Cardio Day

Day-shift timing: Meal 1 pre-shift, Meal 2 early shift, Meal 3 mid-shift, Meal 4 late shift, Meal 5 post-shift or pre-cardio, Meal 6 post-cardio / before bed.

Night-shift timing: Meal 1 wake-up / pre-shift, Meal 2 early shift, Meal 3 midnight meal, Meal 4 late shift, Meal 5 post-shift or pre-cardio, Meal 6 before sleep.

Meal	Food and Portions	Calories	Protein	Carbs	Fat	Notes
Meal 1	Eggs with potatoes: 100 g cooked egg whites, 2 whole eggs, 180 g cooked potatoes, 70 g tortilla, 30 g reduced-fat cheese, 80 g salsa, 30 g avocado	729	42 g	83 g	26 g	Eggs and potatoes are cooked weights.
Meal 2	Beef and broccoli rice bowl: 130 g cooked 93% lean beef, 180 g cooked rice, 180 g broccoli, 30 g stir-fry sauce, 15 g olive oil	720	47 g	73 g	27 g	Beef and rice are cooked weights. About 1 cup cooked rice.
Meal 3	Turkey chili: 120 g cooked lean turkey, 100 g kidney beans, 250 g tomato/broth base, 80 g corn, 20 g reduced-fat cheese	545	54 g	53 g	13 g	Turkey, beans, and corn are cooked/drained weights.
Meal 4	BBQ lean beef bowl: 100 g cooked 93% lean beef, 185 g cooked rice, 80 g corn, 40 g BBQ sauce, 20 g shredded cheese	671	40 g	98 g	13 g	Beef and rice are cooked weights. Label-check BBQ sauce.
Meal 5	Cardio smoothie: 35 g whey, 40 g dry oats, 100 g banana, 100 g berries	445	35 g	65 g	5 g	Oats are dry weight. Blend thin if used close to intervals.
Meal 6	Chocolate protein pudding: 200 g nonfat Greek yogurt, 35 g whey, 25 g cocoa pudding mix, 20 g granola, 16 g peanut butter, 100 g berries	593	56 g	63 g	13 g	Yogurt and fruit are ready-to-eat weights. Granola and pudding mix vary by brand.
Daily Total		3,703	274 g	435 g	97 g	Sum of the six meal rows.

Friday - Heavy Lifting / Training Day

Day-shift timing: Meal 1 pre-shift, Meal 2 early shift, Meal 3 mid-shift, Meal 4 late shift, Meal 5 post-shift or pre-training, Meal 6 post-training / before bed.

Night-shift timing: Meal 1 wake-up / pre-shift, Meal 2 early shift, Meal 3 midnight meal, Meal 4 late shift, Meal 5 post-shift or pre-training, Meal 6 before sleep.

Meal	Food and Portions	Calories	Protein	Carbs	Fat	Notes
Meal 1	Breakfast burrito plus fruit: 120 g cooked egg whites, 2 whole eggs, 200 g cooked potatoes, 70 g tortilla, 30 g reduced-fat cheese, 100 g banana, 30 g avocado	824	45 g	104 g	26 g	Eggs and potatoes are cooked weights. Tortilla is ready-to-eat.
Meal 2	BBQ chicken rice bowl: 110 g cooked chicken breast, 230 g cooked rice, 80 g corn, 40 g BBQ sauce, 20 g reduced-fat cheese, 40 g avocado	785	50 g	114 g	15 g	Chicken and rice are cooked weights. Label-check BBQ sauce.
Meal 3	Salmon rice bowl: 150 g cooked salmon, 240 g cooked rice, 120 g cucumber/carrots, 50 g avocado	709	42 g	78 g	26 g	Salmon and rice are cooked weights.
Meal 4	Teriyaki chicken bowl: 90 g cooked chicken breast, 200 g cooked jasmine rice, 150 g broccoli, 50 g teriyaki sauce	556	39 g	92 g	4 g	Chicken and rice are cooked weights. Label-check teriyaki sauce.
Meal 5	Pre-training shake: 35 g whey, 50 g dry oats, 100 g banana, 15 g honey	480	36 g	72 g	6 g	Oats are dry weight.
Meal 6	Chicken parmesan bowl: 100 g cooked chicken breast, 180 g cooked pasta, 125 g marinara, 30 g mozzarella, 15 g parmesan, 150 g zucchini, 15 g olive oil	859	59 g	80 g	34 g	Chicken and pasta are cooked weights. Use baked or air-fried chicken; label-check marinara.
Daily Total		4,213	270 g	539 g	108 g	Sum of the six meal rows.

Saturday - Rest / Rest Day

Day-shift timing: Meal 1 pre-shift, Meal 2 early shift, Meal 3 mid-shift, Meal 4 late shift, Meal 5 post-shift, Meal 6 before bed.

Night-shift timing: Meal 1 wake-up / pre-shift, Meal 2 early shift, Meal 3 midnight meal, Meal 4 late shift, Meal 5 post-shift, Meal 6 before sleep.

Meal	Food and Portions	Calories	Protein	Carbs	Fat	Notes
Meal 1	Eggs and potatoes: 100 g cooked egg whites, 2 whole eggs, 150 g cooked potatoes, 70 g tortilla, 20 g reduced-fat cheese, 40 g avocado	664	38 g	71 g	25 g	Eggs and potatoes are cooked weights.
Meal 2	White chicken chili: 100 g cooked chicken breast, 100 g white beans, 250 g broth/chile base, 20 g reduced-fat cheese, 40 g avocado	499	48 g	40 g	16 g	Chicken and beans are cooked/drained weights. Soup base is prepared weight.
Meal 3	Beef vegetable soup with rice: 120 g cooked 93% lean beef, 250 g broth/tomato base, 120 g cooked rice, 150 g mixed vegetables	544	44 g	64 g	13 g	Beef and rice are cooked weights. About 2/3 cup cooked rice.
Meal 4	Salmon avocado bowl: 130 g cooked salmon, 120 g cooked rice, 120 g mixed vegetables, 70 g avocado, 7 g olive oil	664	37 g	54 g	33 g	Salmon and rice are cooked weights.
Meal 5	Shake and fruit: 35 g whey, 30 g dry oats, 100 g banana, 100 g berries	406	34 g	58 g	4 g	Oats are dry weight. Use when appetite is low on rest day.
Meal 6	Chocolate protein pudding: 200 g nonfat Greek yogurt, 35 g whey, 20 g cocoa pudding mix, 20 g granola, 16 g peanut butter, 100 g berries	577	56 g	59 g	13 g	Yogurt and fruit are ready-to-eat weights. Granola and pudding mix vary by brand.
Daily Total		3,354	257 g	346 g	105 g	Sum of the six meal rows.

Sunday - Rest or Light Recovery / Rest Day

Day-shift timing: Meal 1 pre-shift, Meal 2 early shift, Meal 3 mid-shift, Meal 4 late shift, Meal 5 post-shift, Meal 6 before bed.

Night-shift timing: Meal 1 wake-up / pre-shift, Meal 2 early shift, Meal 3 midnight meal, Meal 4 late shift, Meal 5 post-shift, Meal 6 before sleep.

Meal	Food and Portions	Calories	Protein	Carbs	Fat	Notes
Meal 1	Rest-day overnight oats: 40 g dry oats, 35 g whey, 150 g nonfat Greek yogurt, 100 g berries, 16 g peanut butter	533	53 g	51 g	13 g	Oats are dry weight. Use less oats than training days.
Meal 2	Beef vegetable soup: 120 g cooked 93% lean beef, 250 g broth/tomato base, 100 g cooked potatoes, 150 g mixed vegetables, 20 g olive oil	663	42 g	50 g	33 g	Beef and potatoes are cooked weights. Soup base is prepared weight.
Meal 3	Enchilada-style chicken prep: 80 g cooked chicken breast, 80 g black beans, 70 g corn tortillas, 125 g enchilada sauce, 20 g reduced-fat cheese	569	43 g	73 g	12 g	Chicken and beans are cooked/drained weights. Tortillas are ready-to-eat weight.
Meal 4	Fish avocado rice bowl: 140 g cooked white fish, 120 g cooked rice, 120 g mixed vegetables, 100 g avocado, 20 g olive oil	751	46 g	57 g	38 g	Fish and rice are cooked weights.
Meal 5	Shake and fruit: 35 g whey, 30 g dry oats, 100 g banana, 100 g berries	406	34 g	58 g	4 g	Oats are dry weight.
Meal 6	Greek-yogurt cheesecake bowl: 200 g nonfat Greek yogurt, 35 g whey, 20 g cheesecake pudding mix, 20 g granola, 16 g peanut butter, 100 g berries	577	56 g	59 g	13 g	Yogurt and fruit are ready-to-eat weights. Granola and pudding mix vary by brand.
Daily Total		3,498	273 g	348 g	112 g	Sum of the six meal rows.

Week One Meal Prep

Cook once, portion in containers, and keep sauces separate when possible. Freeze extra servings if they will not be eaten within normal food-safety storage windows.

Batch Item	Week One Amount	Use
Cooked chicken breast	1,200-1,600 g cooked	Burrito bowls, BBQ bowls, teriyaki bowls, soups, Alfredo, parmesan bowl
Cooked lean steak / beef	900-1,200 g cooked	Steak fajita bowls, beef and broccoli, chili, beef vegetable soup
Cooked lean turkey	150-250 g cooked	Turkey chili
Cooked salmon or white fish	600-900 g cooked	Salmon rice bowls and fish avocado bowls
Cooked white or jasmine rice	3,200-4,000 g cooked	Bowls, soups, and post-training meals
Cooked potatoes	700-1,000 g cooked	Breakfast meals and soup
Beans	900-1,300 g cooked/drained	Burrito bowls, chili, soups, enchilada-style prep
Overnight oat packs	3-4 dry packs	Monday, Wednesday, Sunday, backup meals
Greek-yogurt dessert base	5-7 servings	Protein desserts and bedtime meals
Vegetables and fruit	4,500+ g total	Bowls, soups, smoothies, and snacks
Sauces	Portion in 30-80 g cups	Salsa, BBQ, teriyaki, marinara, enchilada sauce, light Alfredo

Week Two Placeholder

Week Two is intentionally not filled in this draft. Build it later from the same template after reviewing Week One adherence, appetite, digestion, training performance, and bodyweight trend.

Daily Macro Totals

Day	Day Type	Calories	Protein	Carbs	Fat	Target Range Check
Monday	Training	4,125	276 g	539 g	96 g	Within range; total is summed from six meal rows
Tuesday	Cardio	3,709	260 g	450 g	97 g	Within range; total is summed from six meal rows
Wednesday	Training	4,057	279 g	491 g	109 g	Within range; total is summed from six meal rows
Thursday	Cardio	3,703	274 g	435 g	97 g	Within range; total is summed from six meal rows
Friday	Training	4,213	270 g	539 g	108 g	Within range; total is summed from six meal rows
Saturday	Rest	3,354	257 g	346 g	105 g	Within range; total is summed from six meal rows
Sunday	Rest	3,498	273 g	348 g	112 g	Within range; total is summed from six meal rows

Meal Swaps

Swap meals by day type first. A training-day meal can move to another training day with no change. A cardio-day meal can move to another cardio day with no change. Rest-day meals should stay on rest days unless the carb portion is adjusted.

Swap Need	Rule
Swap a protein	Match the cooked protein grams and keep the same macro role: chicken for fish, lean beef for turkey, Greek yogurt plus whey for a lighter meal
Swap a carb	Match cooked carb portions: rice, potatoes, pasta, oats, tortillas, beans, or fruit can trade places when carbs stay close
Lower appetite	Use a shake, smoothie, soup, or Greek-yogurt bowl for one meal instead of skipping food
Training moved earlier	Move Meal 5 before training and Meal 6 after training, even if the clock changes
Training moved to night shift	Keep the six meals; make the meal before lifting lower-fat and the meal after lifting the largest carb meal
Need a rest-day reduction	Remove 75-125 g cooked rice or potatoes from two meals rather than cutting protein
Need more calories	Add 150-250 kcal with carbs from rice, oats, fruit, potatoes, or a shake, then recalculate the meal row from the assumptions table

Validation Note

Monday was spot-checked from the assumptions table. Meal 1 uses 60 g dry oats, 35 g whey, 50 g nonfat Greek yogurt, 100 g banana, 100 g berries, and 16 g peanut butter. From the table, that equals 46.8 g protein, 83.8 g carbs, and 14.2 g fat, which rounds to 650 kcal / 47 g protein / 84 g carbs / 14 g fat. The six Monday rows sum to 4,124.9 kcal, 276.1 g protein, 539.3 g carbs, and 95.9 g fat, shown as 4,125 kcal / 276 g protein / 539 g carbs / 96 g fat.

Coach Review Notes

- Confirm exact training times for day shift and night shift so Meal 5 and Meal 6 can be pinned more tightly.

- Confirm whether dairy-heavy protein desserts digest well before sleep.
- Confirm preferred sauce brands because BBQ, teriyaki, Alfredo, marinara, granola, pudding mix, tortillas, and protein powder macros vary by label.
- Confirm whether the athlete wants Week Two to repeat these meals with different cuisines or introduce more variety.

References

- ISSN sports nutrition position stands for protein and carbohydrate priorities.
- Academy of Nutrition and Dietetics / ACSM / Dietitians of Canada sports nutrition guidance.
- CDC/NIOSH/OSHA heat-stress guidance for work-shift hydration and cooler planning context.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Meal-Prep Workflow

Source: Nutrition/08-meal-prep-workflow.md | Status: review

Purpose

Turn the Week One meal plan into a repeatable prep system for a 260 lb rotating-shift athlete building toward a lean, strong 275 lb. The goal is simple: cook the high-effort foods once, portion the work meals, keep sauces and fragile items separate, and make the cooler reliable enough for long oilfield shifts.

Chapter Contents

- Weekly Prep Overview (#weekly-prep-overview)
- Modular Meal-Builder Prep (#modular-meal-builder-prep)
- Best Prep Day Options (#best-prep-day-options)
- 90-Minute Fast Prep Workflow (#90-minute-fast-prep-workflow)
- 2-Hour Full Prep Workflow (#2-hour-full-prep-workflow)
- Batch Cooking Instructions (#batch-cooking-instructions)
- Storage Guidance (#storage-guidance)
- Packing Examples (#packing-examples)
- Emergency No-Cook Fallback Meals (#emergency-no-cook-fallback-meals)
- Copy-Ready Prep Checklist (#copy-ready-prep-checklist)

Weekly Prep Overview

Prep the base proteins and carbs first, then build meals from the same parts. Week One needs chicken, lean beef or turkey, steak or fajita meat, fish, rice, potatoes, beans or chili, overnight oats, and Greek-yogurt protein desserts.

Prep Target	Weekly Amount	Main Use
Cooked chicken breast	1,200-1,600 g cooked	Burrito bowls, BBQ bowls, teriyaki bowls, soup, Alfredo, parmesan bowl
Cooked lean beef, turkey, and steak	1,200-1,600 g cooked total	Fajita bowls, beef and broccoli, chili, soup, pasta
Cooked salmon or white fish	600-900 g cooked	Salmon bowls, fish avocado bowls
Cooked rice	3,200-4,000 g cooked	Bowls, soups, training meals
Cooked potatoes	700-1,000 g cooked	Breakfast meals, soup
Beans, chili, or soup base	900-1,300 g cooked/drained beans plus broth base	Chili, tortilla soup, white chicken chili, enchilada prep
Overnight oats	3-4 jars or dry packs	Wake-up meals, shift backups
Greek-yogurt protein desserts	5-7 servings	Bedtime meals, no-cook protein

Use cooked weights for prepared meats, rice, potatoes, and beans unless the recipe says dry or raw. Portion by meal role: training days get the biggest rice and pasta portions; rest days keep protein steady and pull carbs down through rice, potatoes, oats, and tortillas.

Modular Meal-Builder Prep

Prep should create options, not force the same breakfast every day. Build a few base components, then assemble meals by role.

Meal Role	Prep Components	Options
Wake-up breakfast	Oat jars, burrito filling, yogurt bowls, smoothie packs	Overnight oats, Greek yogurt bowl, breakfast burrito, protein smoothie
Bigger home breakfast	Cooked potatoes, lean steak, eggs/egg whites, avocado, tortillas	Steak and eggs, avocado toast with eggs, protein pancakes
Work-shift bowl	Cooked protein, rice/potatoes, beans, vegetables, sauce cups	Chicken fajitas, steak rice bowl, salmon bowl, BBQ chicken, hibachi chicken, Korean beef
Thermos or cooler meal	Chili, soup base, cooked rice, tortillas, shredded cheese	Turkey chili, tortilla soup, white chicken chili, beef vegetable soup
Snack or dessert	Greek yogurt, whey, pudding mix, berries, granola, peanut butter	Cheesecake bowl, protein pudding, frozen yogurt bark, mousse-style bowl
Emergency backup	Whey, oats packets, jerky, tuna/salmon pouches, rice cakes, fruit	No-cook protein and carb patch when the planned meal breaks

Prep two breakfast paths each week: one no-cook option and one hot option. For example, build overnight oats plus breakfast burrito filling, or Greek yogurt bowls plus steak/potato packs. This keeps breakfast from turning into an egg-only routine.

Breakfast Option Prep Notes

Option	Prep Move	Storage
Overnight oats	Build 3-4 jars or dry packs; add berries/banana close to eating if texture matters	Refrigerate assembled jars 3-4 days
Greek yogurt bowl	Mix yogurt and whey; keep granola, cereal, cookies, or crunchy toppings separate	Refrigerate 3-4 days
Breakfast burrito	Batch potatoes, egg whites/eggs, cheese, salsa; wrap only what will be eaten soon	Refrigerate 2-3 days or freeze wrapped burritos
Protein smoothie	Build dry bags with whey/oats; pack fruit separately or use frozen fruit	Dry packs are shelf-stable; blend when needed
Steak and eggs	Prep steak and potatoes; cook eggs fresh when possible	Steak/potatoes refrigerate 3-4 days
Avocado toast with eggs	Keep bread/tortillas, avocado, eggs, and fruit ready; cook eggs fresh	Assemble day-of
Cottage cheese bowl	Portion cottage cheese or high-protein dairy with fruit; keep granola separate	Refrigerate according to label date
Protein pancakes	Portion dry mix or oats/whey; cook fresh or batch and freeze	Freeze cooked pancakes with parchment between servings

Best Prep Day Options

Pick the prep day that protects sleep. A perfect plan that costs recovery is a bad trade.

Work Pattern	Best Prep Window	What To Prep
Day shift starting around 3:00 a.m.	Late morning or early afternoon on the day before the shift block	Full cook, cooler setup, first two shift days packed
Night shift starting around 3:00 p.m.	Morning after the last sleep before the shift block	Rice, meats, oats, desserts, cooler reset
Switching from days to nights	Split prep: proteins and rice before the switch, fresh produce after sleep	Freeze extra cooked protein; do not leave all cooking for the flip day
Switching from nights to days	Use the first recovery day for groceries only; cook after the next full sleep	Pantry restock, freezer pull, lighter 90-minute prep
Seven-day run	Full prep before Day 1, freezer reload on Day 4	Keep only 3-4 days of cooked meals in the fridge

90-Minute Fast Prep Workflow

Use this when time is tight. It covers the highest-return foods and leaves fish or specialty meals for quick cooking later.

Time	Action	Output
0:00-0:10	Start rice cooker, heat oven or air fryer, set out containers	Rice cooking; station ready
0:10-0:25	Season chicken breast and ground beef or turkey	Two proteins ready to cook
0:25-0:45	Bake or air-fry chicken; brown ground meat	6-8 protein servings
0:45-0:55	Microwave or boil potatoes; rinse beans; portion sauces	Carb base and sauce cups
0:55-1:10	Build overnight oat jars and Greek-yogurt desserts	3-4 oat jars; 4-5 desserts
1:10-1:25	Portion rice, chicken, beef/turkey, beans, and vegetables	Main work meals packed
1:25-1:30	Label dates, vent hot lids, load fridge, freeze overflow	Food stored safely

Fast-prep rule: make chicken, one ground meat, rice, oats, desserts, and sauce cups. Buy ready-to-eat produce and frozen vegetables to fill gaps.

2-Hour Full Prep Workflow

Use this before a long shift block or two-week hitch. It adds steak, fish, potatoes, soup/chili, and more freezer coverage.

Time	Action	Output
0:00-0:10	Start rice cooker and oven; wash produce; set out sheet pans	Base cook started
0:10-0:25	Season chicken, steak/fajita meat, and fish; dice potatoes	Proteins and potatoes ready
0:25-0:50	Bake chicken and potatoes; brown beef or turkey; start chili or soup base	Bulk proteins cooking
0:50-1:05	Sear steak/fajita meat; cook fish last or prep it raw for next-day cooking	Steak cooked; fish controlled
1:05-1:20	Finish rice; cool proteins; portion beans, vegetables, sauces	Assembly station ready
1:20-1:40	Build bowls, soups, burrito parts, and breakfast packs	10-14 meal components
1:40-1:55	Make overnight oats and Greek-yogurt protein desserts	7-10 no-cook meals/snacks

Time	Action	Output
1:55-2:00	Label, separate fridge/freezer items, pack first cooler	Week ready

Fish quality drops faster than chicken or beef. Cook enough fish for 1-2 days, then freeze raw portions or use frozen fish fillets for later in the week.

Batch Cooking Instructions

Chicken Breast

Step	Standard
Buy	4-5 lb raw chicken breast for the week; cook extra if freezing
Season	Salt, pepper, garlic, fajita seasoning, BBQ rub, Italian seasoning, or teriyaki marinade
Cook	Bake, grill, air-fry, or pressure cook until the thickest pieces reach a safe poultry temperature
Portion	80-110 g cooked portions for bowls; 90-100 g cooked portions for soup meals
Use	Keep plain chicken for flexible sauces; sauce the meal, not the whole batch

Lean Ground Beef or Turkey

Step	Standard
Buy	2-3 lb raw 93% lean beef, lean turkey, or a mix
Cook	Brown in a skillet, drain if needed, season after browning
Split	Half taco/chili seasoning, half Italian or stir-fry seasoning
Portion	100-130 g cooked portions
Use	Pasta, beef and broccoli, BBQ bowls, chili, soup

Steak or Fajita Meat

Step	Standard
Buy	1.5-2 lb raw lean steak, sirloin, flank, or fajita meat
Prep	Slice across the grain after cooking unless using pre-sliced fajita meat
Cook	Sear hot and fast; avoid overcooking because reheating will finish it more
Portion	120-140 g cooked portions
Use	Fajita bowls with rice, peppers, onions, beans, salsa, and avocado

Salmon or White Fish

Step	Standard
Buy	1.5-2 lb salmon or white fish, fresh or frozen
Cook	Bake, air-fry, or pan-cook until flaky; keep seasoning simple
Portion	130-150 g cooked portions
Store	Keep 1-2 days in the fridge; freeze extra cooked fish or keep raw frozen portions
Use	Rice bowls, avocado bowls, cucumber/carrot bowls

Rice

Step	Standard
Buy	Jasmine, white, or long-grain rice in bulk
Cook	Use a rice cooker when possible; cook 1,100-1,400 g dry rice for a large week
Portion	120-250 g cooked rice per meal based on day type
Cool	Spread in shallow containers before sealing
Use	Bowls, soups, fried rice, post-training meals

Potatoes

Step	Standard
Buy	3-5 lb potatoes
Cook	Roast diced potatoes, microwave whole potatoes, or boil cubes
Portion	150-200 g cooked for breakfast meals; 100 g cooked for soup
Store	Keep plain; add salsa, cheese, eggs, or chili at assembly

Beans, Chili, or Soup Base

Step	Standard
Buy	Canned black beans, white beans, kidney beans, broth, tomatoes, enchilada sauce
Build	Combine drained beans with broth/tomato base, corn, peppers, lean meat, and seasoning
Portion	250-300 g soup base; 80-100 g drained beans for bowls
Store	Freeze soup and chili flat in 1-2 serving containers
Use	Chicken tortilla soup, turkey chili, white chicken chili, beef vegetable soup

Overnight Oats

Step	Standard
Base	40-60 g dry oats, 35 g whey, Greek yogurt, fruit, water or milk as needed
Add	Banana, berries, cocoa, peanut butter, honey, or cinnamon
Portion	Use smaller 40 g oat jars for rest days and 55-60 g jars for training days
Store	Keep 3-4 days refrigerated; build dry packs for later in the week

Greek-Yogurt Protein Desserts

Step	Standard
Base	200 g nonfat Greek yogurt plus 35 g whey
Flavor	Cheesecake pudding mix, cocoa pudding mix, berries, granola, peanut butter, or jam
Portion	Build 5-7 sealed servings; keep granola separate until eating
Use	Bedtime meal, shift dessert, emergency no-cook protein

Storage Guidance

Keep food safety conservative. Cool hot food quickly in shallow containers, refrigerate once steam drops, and do not leave packed meals sitting warm. Reheat cooked leftovers until steaming hot when a microwave is available. When in doubt on smell, texture, or time, throw it out.

Item	Fridge Plan	Freezer Plan	Keep Separate
Chicken, beef, turkey, steak	3-4 days	Freeze extra portions for later in the shift block	Sauces, avocado, fresh vegetables
Fish	1-2 days best quality	Freeze cooked portions or raw fillets	Avocado, cucumber, sauces
Rice and potatoes	3-4 days	Freeze rice in flat portions; potatoes are best fresh	Wet sauces until reheating
Chili, soup, beans	3-4 days	Freeze in 1-2 serving containers	Tortilla strips, cheese toppings
Overnight oats	3-4 days	Freeze dry oat packs only, not assembled jars	Crunchy toppings
Greek-yogurt desserts	3-4 days	Do not freeze assembled yogurt desserts	Granola, cereal, nuts

Sauce Handling

Use 30-80 g sauce cups. Salsa, BBQ sauce, teriyaki, marinara, enchilada sauce, light Alfredo, and stir-fry sauce vary a lot by label, so portion them instead of free-pouring. Pack sauce cold, add after reheating, and keep creamy sauces separate until the meal is hot.

Cooler Packing for Long Shifts

Use a hard cooler or insulated lunch bag with enough ice packs to keep meals cold through the whole shift. Pack cold meals straight from the fridge. Put the meals eaten last deepest in the cooler and open it as little as possible.

Cooler Layer	What Goes There
Bottom	Large ice pack, frozen water bottle, meals for the second half of shift
Middle	Meal containers, Greek-yogurt dessert, fruit, sauce cups
Top	First meal, spoon/fork, napkins, electrolyte packets, quick shake
Side pocket	Whey scoop, oats packet, tuna or salmon pouch, tortillas, shelf-stable backup

Packing Examples

Day-Shift Packing Example

Timing	Pack	Notes
2:00-2:30 a.m.	Overnight oats or breakfast burrito before leaving	Eat at home if possible
5:30-6:30 a.m.	Chicken burrito bowl	Sauce cup separate
9:30-10:30 a.m.	Steak fajita bowl or salmon rice bowl	Avocado packed separately
1:30-2:30 p.m.	Soup, chili, or BBQ bowl	Tortilla strips separate
7:00-7:30 p.m.	Whey/oats shake or smoothie	Use after shift or before training
8:30-9:30 p.m.	Larger post-training dinner or Greek-yogurt dessert	Eat at home when possible

Night-Shift Packing Example

Timing	Pack	Notes
12:00-1:00 p.m.	Breakfast burrito, eggs and potatoes, or oats	Wake-up meal
6:00-7:00 p.m.	Chicken or beef rice bowl	First cooler meal
10:00-11:00 p.m.	Soup, chili, or fajita bowl	Main midnight meal
2:00-3:00 a.m.	Teriyaki chicken, BBQ bowl, or yogurt dessert	Keep easy to eat
7:00-7:30 a.m.	Shake and fruit	Post-shift or pre-training
8:00-9:00 a.m.	Greek-yogurt dessert or lighter home meal	Before sleep

Emergency No-Cook Fallback Meals

Use these when a shift runs long, a microwave is unavailable, or the planned meal gets missed. They are not fancy; they keep the day from collapsing.

Fallback	Build
Whey oats shake	35 g whey, 40-60 g dry oats, banana, water
Greek-yogurt bowl	200 g Greek yogurt, 35 g whey, berries, granola or peanut butter
Tuna wrap	Tuna or salmon pouch, tortilla, salsa packet, avocado if available
Rotisserie chicken bowl	Pulled chicken, microwave rice cup, salsa, beans
Beef jerky plus carbs	Jerky, fruit, oats packet, rice cakes, or tortillas
Cottage cheese style swap	High-protein dairy cup, fruit, granola, peanut butter
Ready soup plus protein	Canned chili or soup plus tuna pouch or leftover chicken
Protein bar backup	Bar plus fruit; use only when real food is not practical

Copy-Ready Prep Checklist

- Check the next 7 days: lifting days, cardio days, rest days, day shifts, night shifts.
- Choose prep day: before the shift block, after full sleep, or split around a rotation change.
- Cook 1,200-1,600 g cooked chicken breast.
- Cook 1,200-1,600 g cooked lean beef, turkey, and steak/fajita meat combined.
- Cook 600-900 g cooked salmon or white fish, or freeze fish portions for later.
- Cook 3,200-4,000 g cooked rice.
- Cook 700-1,000 g cooked potatoes.
- Build 900-1,300 g beans, chili, or soup base.
- Make 3-4 overnight oat jars or dry oat packs.
- Make 5-7 Greek-yogurt protein desserts.
- Portion sauces into 30-80 g cups.
- Keep avocado, crunchy toppings, fresh vegetables, and creamy sauces separate.
- Pack the first day-shift or night-shift cooler before bed.
- Freeze any cooked meals that will not be eaten within 3-4 days.
- Add one no-cook emergency meal to the cooler or truck bag.

Coach Review Notes

- Confirm exact shift-block length and access to a microwave on site.
- Confirm whether fish should be cooked fresh more often for quality.
- Confirm preferred sauce brands before final macro lock because sauce labels vary widely.
- Confirm cooler size and number of ice packs needed for the longest shift.

References

- ISSN sports nutrition position stands for protein and carbohydrate priorities.
- Academy of Nutrition and Dietetics / ACSM / Dietitians of Canada sports nutrition guidance.
- CDC/NIOSH/OSHA heat-stress guidance for work-shift hydration and cooler planning context.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Costco and H-E-B Grocery Guide

Source: Nutrition/09-grocery-guide.md | Status: review

Purpose

Give a practical shopping system for the Week One meal plan and a two-week hitch. Use Costco for bulk staples and H-E-B for flexible weekly produce, sauces, tortillas, canned goods, and smaller items.

Chapter Contents

- Shopping Strategy (#shopping-strategy)
- One-Week Grocery List (#one-week-grocery-list)
- Meal-Builder Add-Ons (#meal-builder-add-ons)
- Two-Week Hitch Grocery List (#two-week-hitch-grocery-list)
- Costco List (#costco-list)
- H-E-B List (#h-e-b-list)
- Pantry Staples (#pantry-staples)
- Freezer Staples (#freezer-staples)
- Cooler Staples (#cooler-staples)
- Budget Swaps (#budget-swaps)
- High-Protein Emergency Foods (#high-protein-emergency-foods)

Shopping Strategy

Buy the big repeat foods in bulk when the price and storage make sense. Buy the short-life and flavor items weekly so the food still tastes like something a human agreed to eat.

Store	Best Use	Examples
Costco	Bulk proteins, carbs, frozen staples, high-volume dairy	Chicken breast, lean beef/turkey, eggs, egg whites, Greek yogurt, rice, oats, frozen berries, frozen vegetables, salmon/fish
H-E-B	Flexible weekly items, produce, sauces, smaller specialty items	Avocados, fresh vegetables, tortillas, beans, broth, tomato bases, low-fat cheeses, marinara, BBQ, teriyaki, enchilada sauce

Quantity notes: meat amounts are listed as raw purchase weights unless marked cooked. The Week One plan uses cooked weights, so buy extra raw weight to account for cooking loss.

One-Week Grocery List

This list covers Week One with a small buffer for label differences, cooking loss, and one missed portion.

Category	Item	One-Week Quantity	Notes
Protein	Chicken breast	4-5 lb raw	Yields roughly 1,200-1,600 g cooked
Protein	93% lean ground beef	2 lb raw	Bowls, soup, pasta
Protein	Lean ground turkey	1 lb raw	Turkey chili or beef swap
Protein	Lean steak or fajita meat	1.5-2 lb raw	Steak fajita bowls
Protein	Salmon or white fish	1.5-2 lb raw/frozen	Cook 1-2 days at a time if quality matters
Protein	Whey protein	14-16 scoops / about 500-600 g	Shakes, oats, yogurt desserts
Protein	Nonfat Greek yogurt	2 large tubs / 64-96 oz total	Oats and desserts
Protein	Egg whites	1 carton / 32 oz	Breakfast meals
Protein	Whole eggs	1 dozen	Breakfast meals and fried rice
Carbs	White or jasmine rice	3-4 lb dry bag	Produces more than 3,200-4,000 g cooked
Carbs	Dry oats	1 large canister / 42 oz	Oats, shakes, backup meals
Carbs	Potatoes	3-5 lb bag	Breakfast meals and soup
Carbs	Pasta	1 lb box	Alfredo, marinara, parmesan bowl
Carbs	Large flour tortillas	1 pack / 8-10 count	Breakfast burritos
Carbs	Corn tortillas	1 pack	Enchilada-style prep

Category	Item	One-Week Quantity	Notes
Beans	Black beans	4 cans	Bowls, tortilla soup, enchilada prep
Beans	White beans	2 cans	White chicken chili
Beans	Kidney beans	1-2 cans	Turkey chili
Produce	Bananas	10-12 medium	Oats and shakes
Produce	Berries	2-3 lb fresh or frozen	Oats, smoothies, desserts
Produce	Avocados	6-8 medium	Bowls and breakfast
Produce	Broccoli	2-3 lb fresh or frozen	Teriyaki, Alfredo, beef bowls
Produce	Mixed vegetables	2-3 lb frozen	Soup and fish bowls
Produce	Peppers and onions	2-3 lb fresh or frozen	Fajitas, chili, bowls
Produce	Cucumber/carrots	2-3 lb total	Salmon bowls
Produce	Zucchini	2-3 medium	Chicken parmesan bowl
Canned/frozen	Corn	3 cans or 1 frozen bag	BBQ bowls, chili, soup
Sauces	Salsa	1 large jar	Bowls and breakfast
Sauces	BBQ sauce	1 bottle	Label-check carbs
Sauces	Teriyaki sauce	1 bottle	Label-check carbs
Sauces	Stir-fry sauce	1 bottle	Beef and fried rice
Sauces	Marinara	1 jar	Pasta bowl
Sauces	Light Alfredo	1 jar	Alfredo bowl
Sauces	Enchilada sauce	1 can/jar	Enchilada prep
Soup base	Broth and tomato/chile base	4 cartons/cans total	Chili and soups
Dairy/toppings	Reduced-fat shredded cheese	1 bag / 8 oz	Burritos, chili, bowls
Dairy/toppings	Parmesan	1 small tub	Pasta bowls
Dairy/toppings	Mozzarella	1 small bag	Chicken parmesan bowl
Extras	Peanut butter	1 jar	Oats and desserts
Extras	Granola	1 bag	Yogurt desserts
Extras	Pudding mix	2 boxes	Cheesecake and chocolate desserts
Extras	Cocoa powder	1 container	Chocolate oats/desserts
Extras	Honey and jam	1 bottle/jar each	Shakes and yogurt bowls
Extras	Olive oil	1 bottle	Cooking and planned fats
Cooler	Ice packs and bottled water	Enough for full shift	Use frozen water bottles as backups

Meal-Builder Add-Ons

Use these when rotating breakfast, snacks, and protein desserts. They are optional; buy only the items that match the week's chosen meals.

Category	Item	Use
Breakfast options	Cottage cheese or high-protein dairy cups	Cottage cheese bowls, no-cook breakfast swaps
Breakfast options	Protein pancake mix or pancake ingredients	Training-day breakfast option
Breakfast options	Bread, English muffins, or extra tortillas	Avocado toast with eggs, breakfast sandwiches, burritos
Breakfast options	Frozen fruit	Smoothies, oats, yogurt bowls
Bowl options	Chicken thigh trimmed, sirloin, canned salmon, tuna pouches	Protein swaps when chicken breast gets old
Lunch/dinner options	Fajita seasoning, Korean-style sauce, hibachi-style sauce, Alfredo, BBQ	Flavor rotation without changing the base macro structure
Snack options	Rice cakes, fruit cups, pretzels, oats packets	Carb patches for training days and long shifts
Snack options	Beef jerky, protein bars, tuna/salmon pouches	Emergency protein when a cooler meal fails
Higher-calorie add-ons	Trail mix, peanut butter, avocado, olive oil	Add calories away from training when needed
Dessert options	Sugar-free or regular pudding mix, cocoa, Greek yogurt, berries, granola	Cheesecake bowls, protein pudding, mousse-style bowls
Dessert options	Oreo-style cookies, frozen yogurt, low-fat whipped topping	Optional protein dessert rotation; portion and label-check

Two-Week Hitch Grocery List

Double the durable staples, but do not force all fresh produce into one trip if it will spoil. Buy fresh avocados, bananas, and delicate vegetables weekly when possible.

Category	Item	Two-Week Quantity	Storage Notes
Protein	Chicken breast	8-10 lb raw	Cook half now; freeze half raw or cooked
Protein	Lean ground beef/turkey	5-6 lb raw total	Cook and freeze in 100-130 g portions

Category	Item	Two-Week Quantity	Storage Notes
Protein	Steak/fajita meat	3-4 lb raw	Freeze cooked portions or raw strips
Protein	Salmon/white fish	3-4 lb raw/frozen	Prefer frozen fillets for Week Two
Protein	Whey protein	1 tub or 1,000-1,200 g	Shelf-stable
Protein	Greek yogurt	4 large tubs / 128-192 oz total	Buy second half mid-hitch if possible
Protein	Egg whites	2 cartons / 64 oz	Check dates
Protein	Whole eggs	2 dozen	Good durable protein
Carbs	Rice	6-8 lb dry	Bulk bag is fine
Carbs	Oats	2 canisters or 1 bulk bag	Shelf-stable
Carbs	Potatoes	8-10 lb	Store cool and dry
Carbs	Pasta	2 lb	Shelf-stable
Carbs	Tortillas	2 packs flour, 1 pack corn	Freeze extra tortillas
Beans/cans	Black, white, kidney beans	12-16 cans total	Shelf-stable
Beans/cans	Broth, tomato/chile base	8-10 cartons/cans	Soup and chili
Produce	Bananas	10-12 now plus 10-12 later	Buy green for later days
Produce	Berries	4-6 lb frozen	Best two-week option
Produce	Avocados	6-8 now plus 6-8 later	Buy mixed ripeness
Produce	Frozen vegetables	6-8 lb	Broccoli, mixed veg, peppers/onions
Produce	Fresh vegetables	5-7 lb	Buy second half if possible
Sauces	Salsa, BBQ, teriyaki, stir-fry, marinara, Alfredo, enchilada	1-2 containers each	Portion; label-check calories
Dairy/toppings	Reduced-fat cheese, parmesan, mozzarella	2 bags plus small tubs	Freeze extra shredded cheese if needed
Extras	Peanut butter, granola, pudding mix, cocoa, honey, jam	1-2 each	Shelf-stable
Cooler	Ice packs, disposable utensils, shaker bottles	2-week supply	Keep one backup shaker in the vehicle

Costco List

Item	Buy	Why
Chicken breast	8-10 lb pack if freezing; 4-5 lb if one week	Best bulk protein anchor
Lean ground beef or turkey	5-6 lb pack	Cook once, freeze portions
Eggs	2 dozen	Breakfast meals and fried rice
Egg whites	2 cartons	Easy lean protein
Nonfat Greek yogurt	2-4 large tubs	Oats and protein desserts
Rice	10-25 lb bag	Low-cost carb base
Oats	Bulk box/canister	Oats, shakes, backups
Frozen berries	4-6 lb bag	Smoothies, oats, desserts
Frozen vegetables	5-8 lb total	Broccoli, mixed veg, peppers/onions
Salmon or white fish	3-4 lb frozen	Better value when available
Whey protein	1 tub if brand works	Label-check serving size and macros
Peanut butter	1 large jar	High-calorie add-on

H-E-B List

Item	Buy	Why
Avocados	6-8 per week	Fresh fats for bowls
Fresh produce	5-7 lb per week	Cucumber, carrots, zucchini, peppers, onions
Bananas	10-12 per week	Oats and shakes
Tortillas	1 flour pack, 1 corn pack	Burritos and enchilada prep
Beans and canned goods	8-10 cans per week	Black, white, kidney beans, corn
Broth and tomato bases	4-5 cartons/cans per week	Chili, soup, tortilla soup
Sauces	1 each as needed	Salsa, BBQ, teriyaki, stir-fry, marinara, Alfredo, enchilada
Low-fat cheeses	1-2 bags per week	Reduced-fat shredded cheese, mozzarella
Parmesan	1 small tub	Pasta meals
Pudding mix and cocoa	1-2 boxes plus cocoa	Greek-yogurt desserts
Granola and jam	1 each	Dessert toppings
Smaller specialty items	As needed	Seasoning packets, salsa verde, hot sauce

Pantry Staples

Staple	Minimum To Keep	Use
Rice	5 lb	Bowl base
Oats	42 oz	Oats, shakes, emergency meals
Pasta	2 lb	Italian meals
Beans	8 cans	Bowls, chili, soups
Broth/tomatoes	4 cans/cartons	Soup base
Tortillas	1 backup pack	Wraps and no-cook meals
Whey protein	2-week supply	Shakes, oats, desserts
Peanut butter	1 jar	Calories and fats
Olive oil	1 bottle	Cooking and planned fats
Sauces	2-3 bottles	Flavor rotation
Seasonings	Salt, pepper, garlic, taco, fajita, Italian, BBQ rub	Fast variety

Freezer Staples

Staple	Minimum To Keep	Use
Chicken breast	4-5 lb	Backup cook
Lean beef/turkey	2-3 lb	Chili, bowls, pasta
Fish fillets	2 lb	Fish bowls
Frozen rice portions	4-6 servings	Emergency bowls
Frozen vegetables	4-6 lb	Any bowl or soup
Frozen berries	2-3 lb	Oats, smoothies, desserts
Frozen cooked chili/soup	4 servings	Long-shift meals
Tortillas	1 pack	Backup wraps

Cooler Staples

Staple	Quantity	Notes
Ice packs	2 large plus 1 backup	More for hot job sites
Frozen water bottles	2-4	Ice pack plus hydration
Shaker bottle	1-2	Keep one backup
Sauce cups	7-14	Prevent soggy meals
Disposable utensils	1 pack	Keep in side pocket
Napkins/wipes	1 pack	Cleaner eating in the truck
Electrolyte packets	7-14	Use as needed for heat and sweat
Shelf-stable protein	3-5 servings	Tuna/salmon pouches, jerky, protein bars
Shelf-stable carbs	3-5 servings	Oats packets, rice cakes, tortillas, fruit cups

Budget Swaps

Higher-Cost Item	Budget Swap	Notes
Steak/fajita meat	Lean ground beef or chicken thigh trimmed well	Keep cooked grams close
Salmon	White fish, canned salmon, tuna pouches	Watch sodium and sauce calories
Fresh berries	Frozen berries	Usually better value
Fresh broccoli/vegetables	Frozen vegetables	Faster prep, less waste
Greek-yogurt dessert toppings	Banana, cocoa, pudding mix	Granola and specialty toppings add up
Name-brand sauces	Store-brand sauce	Check calories and carbs
Single rice cups	Bulk rice cooked ahead	Keep rice cups for emergencies only
Rotisserie chicken	Bulk chicken breast	Rotisserie is useful when time breaks

High-Protein Emergency Foods

Food	Practical Unit	How To Use
Whey protein	35 g scoop	Shake, oats, yogurt
Tuna or salmon pouch	1 pouch	Wrap, rice cup, salad kit
Greek yogurt	1 tub or single-serve cups	Dessert bowl or snack
Beef jerky	1 bag	Pair with fruit or oats
Rotisserie chicken	1 chicken	Pull meat for bowls and wraps
Canned chili	1 can	Add rice or tortillas
Cottage cheese or high-protein dairy cup	1 tub/cup	Pair with fruit and granola
Protein bars	2-4 bars	Backup only, not the main plan
Microwave rice cups	2-4 cups	Emergency carb base
Pre-cooked chicken strips	1 pack	No-cook bowl or wrap

Copy-Ready Shopping Checklist

- Costco: chicken breast, lean beef/turkey, eggs, egg whites, Greek yogurt, rice, oats.
- Costco: frozen berries, frozen vegetables, fish if value is better, whey if the label works.
- H-E-B: avocados, bananas, fresh vegetables, tortillas, beans, corn, broth, tomato bases.
- H-E-B: salsa, BBQ, teriyaki, stir-fry, marinara, light Alfredo, enchilada sauce.
- H-E-B: reduced-fat cheese, mozzarella, parmesan, pudding mix, cocoa, granola, jam.
- Pantry: rice, oats, pasta, beans, sauces, seasoning, whey, peanut butter, olive oil.
- Freezer: chicken, beef/turkey, fish, rice portions, frozen vegetables, berries, chili/soup.
- Cooler: ice packs, frozen water bottles, shaker, utensils, sauce cups, emergency protein.

Coach Review Notes

- Confirm preferred H-E-B and Costco brands before final macro lock.
- Confirm whether the athlete has freezer space for two-week hitch quantities.
- Confirm which sauces are worth keeping in the plan after label review.
- Confirm grocery budget range so swaps can be prioritized.

References

- ISSN sports nutrition position stands for protein and carbohydrate priorities.
- Academy of Nutrition and Dietetics / ACSM / Dietitians of Canada sports nutrition guidance.
- CDC/NIOSH/OSHA heat-stress guidance for work-shift hydration and cooler planning context.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Hydration and Electrolytes

Source: Nutrition/10-hydration-electrolytes.md | Status: review

Purpose and Scope

Hydration is a daily performance system for long oilfield shifts, West Texas heat, heavy lifting, cardio intervals, and rotating day/night work. This chapter gives practical starting targets and check-ins. It does not diagnose, treat, or override medical advice.

Heat illness symptoms, confusion, fainting, chest pain, severe weakness, stopped sweating with overheating, or worsening nausea require urgent medical attention. Do not try to solve those symptoms with a nutrition checklist.

Daily Baseline Hydration Target

Start with a steady baseline before adding heat or training fluids.

Situation	Practical Target	Notes
Normal rest day indoors	3-4 liters fluid/day	Spread across the waking window
Training day	4-5 liters fluid/day	Add fluids before, during, and after lifting
Hot outdoor shift	5-7+ liters fluid/day	Adjust to sweat, heat index, PPE, and urine/bodyweight checks
Very heavy sweat day	Individualized	Add electrolytes and food; consider coach/clinician guidance

Do not force water far beyond thirst, food intake, and electrolyte intake. Water-only intake can become a problem when sweat losses are high and meals are delayed.

Hot-Shift Hydration Strategy

Timing	Action
Before shift	Drink 500-750 ml with a meal and pack the full cooler
First half of shift	Sip steadily; do not wait until headache or cramps show up
Mid-shift meal	Pair fluids with sodium-containing food or an electrolyte drink
Second half of shift	Keep a bottle visible; use electrolyte packets if sweat is heavy
After shift	Rehydrate with water, food, and electrolytes before sleep or training

For long hot shifts, pack more than the minimum: a water jug, 2-4 cooler bottles, and at least 2 electrolyte packets. Frozen bottles can double as ice packs.

Training-Day Hydration Strategy

Training Window	Strategy
Before lifting	Drink 500-750 ml with the meal before training
During lifting	Sip 500-1,000 ml depending on session length and sweat
After lifting	Drink with the post-training meal and include sodium from food
Training after shift	Start rehydrating before leaving work; do not walk into lifting already depleted

If performance is flat and the day was hot, review hydration, sodium, sleep, and total food before assuming the training program is the problem.

Cardio-Day Hydration Strategy

Cardio intervals raise sweat rate faster than easy work. Keep the stomach comfortable, but do not start dry.

Timing	Strategy
1-2 hours before	Drink 500 ml with a meal or snack
During	Sip water as tolerated; use electrolytes if hot or sweat is heavy
After	Replace fluids with the next meal, fruit, rice/potatoes/oats, and sodium-containing food

Electrolyte Guidance for Long, Sweaty Shifts

Electrolytes are useful when sweat is high, food is delayed, or water intake is large. Sodium is usually the main practical concern during hard sweating. Potassium and magnesium are best covered through regular meals unless a clinician advises otherwise.

Need	Practical Move
Heavy sweat plus long shift	Use 1 electrolyte packet in a bottle during the shift
Salt stains, repeated cramps, headache with lots of sweating	Add electrolytes and eat a real meal or salty snack

Need	Practical Move
Water tastes bad or intake drops	Use a low-sugar flavor packet or electrolyte packet
Blood pressure, kidney issues, heart issues, or relevant medications	Ask a clinician before increasing sodium or electrolyte products

Water Jugs, Cooler Bottles, and Packets

Tool	How To Use
1-gallon jug	Main shift supply; mark halfway point for mid-shift check
Cooler bottles	2-4 bottles kept cold and easy to grab
Frozen bottle	Backup fluid plus ice pack
Electrolyte packets	Keep 2-4 in the cooler bag; use when sweat and heat justify them
Shaker bottle	Backup for whey, oats, and fluids after shift

When To Back Off Water-Only Intake

Add electrolytes and food instead of only more plain water when these show up during a sweaty shift:

- You have been drinking water steadily but still feel wiped out, crampy, or headachy.
- Urine is completely clear all day while meals and salt intake are low.
- You are sweating heavily for hours and only drinking plain water.
- You feel bloated or sloshy from water but still low-energy.
- You missed a meal and are trying to replace food with fluid.

These are practical prompts, not diagnoses. Severe or worsening symptoms need medical attention.

Morning Check-Ins

Check-In	What To Look For	What To Do
Morning bodyweight	Sudden drop after a hot shift	Rehydrate with meals; review previous day's fluids and sodium
Urine color	Pale straw is usually fine; very dark suggests low fluids	Add fluids early and monitor heat exposure
Thirst and mouth dryness	Persistent thirst on waking	Start with water plus breakfast
Hands/feet swelling or unusual symptoms	Not a normal coaching target	Consider medical review, especially with blood pressure or kidney history

Caution Notes

Get medical guidance before deliberately increasing sodium, electrolyte supplements, or fluid targets if there is a history of high blood pressure, kidney disease, heart disease, liver disease, fluid restriction, diuretic use, blood pressure medication, or any condition where fluid or sodium intake is monitored.

Heat illness symptoms require urgent medical attention. Stop work/training, get cooled, and follow site safety procedures.

Copy-Ready Daily Hydration Checklist

- Start the day with water and a real meal.
- Pack a water jug, 2-4 cooler bottles, and 2 electrolyte packets.
- Use electrolytes on long hot shifts, heavy sweat days, or when food is delayed.
- Drink before training, sip during training, and rehydrate with the post-training meal.
- Check morning bodyweight trend after hot shifts.
- Check urine color without chasing perfectly clear urine all day.
- Add food/electrolytes when water-only intake is not working.
- Treat heat illness symptoms as urgent, not as a hydration tweak.

Coach Review Notes

- Confirm job-site heat exposure, PPE, and access to breaks.
- Confirm blood pressure history and medications before final electrolyte guidance.
- Confirm whether the athlete prefers jugs, bottles, or hydration packs.

References

- ACSM exercise fluid-replacement guidance for practical hydration during training.
- CDC/NIOSH/OSHA workplace heat-stress guidance for heat exposure, rest, and red-flag escalation.
- NIH MedlinePlus general medical information for dehydration and urgent symptom context.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Supplement Framework

Source: Nutrition/11-supplements.md | Status: review

Purpose

Supplements are optional tools. They should make the food plan easier, support training execution, or fill a confirmed gap. They should not replace meals, sleep, hydration, medical care, or consistent training.

Food-First Principle

The base plan comes first: protein at each meal, enough carbs for training, fats kept consistent, fruits and vegetables, water, electrolytes when needed, and packed meals for long shifts. If a supplement does not help that system work better, skip it.

Supplement Decision Rules

Question	Decision
Does food already solve the problem?	Use food first
Is the label simple and dose clear?	Prefer simple products
Is there third-party testing where relevant?	Prefer tested products
Does it affect sleep, heart rate, blood pressure, digestion, or medications?	Get medical review or avoid
Is the product making extreme claims?	Do not rely on it

Core Supplements To Discuss Cautiously

Supplement	Practical Role	Conservative Notes
Whey protein	Convenient protein for oats, shakes, and Greek-yogurt desserts	Use to hit protein targets when meals are hard; label-check serving size
Creatine monohydrate	Common strength-training supplement	Discuss with clinician if kidney issues, relevant medications, or medical concerns exist
Electrolytes	Useful for long, sweaty shifts and heat exposure	Match use to sweat, meals, and medical context
Caffeine	Alertness and training support	Timing matters; avoid stacking stimulant products
Vitamin D	Only if deficient or clinician-advised	Confirm with lab work or clinician guidance
Fish oil	Optional if fatty fish intake is low or clinician-advised	Review with clinician if using blood thinners, surgery is planned, or bleeding concerns exist

Do not add everything at once. Add one product at a time, note the reason, and track whether it actually helps.

What Not To Rely On

Product Type	Why It Does Not Belong At The Center
Fat burners	Often stimulant-heavy and not needed for a lean-gain phase
Detox products	Do not replace normal nutrition or medical care
Unverified hormone or peptide claims	High risk of unsupported claims, contamination, or legal/medical issues
Stimulant-heavy pre-workouts	Can interfere with sleep, blood pressure, heart symptoms, and shift recovery
Proprietary blends	Hide exact ingredient amounts

Shift-Work Caffeine Timing

Use caffeine to protect performance, not to bulldoze through bad sleep indefinitely.

Scenario	Timing Rule
Day shift	Use early in the shift; avoid late-day caffeine if sleep follows
Night shift	Use early in the night shift; stop far enough before the planned sleep window
Training after shift	Use the lowest effective amount, or skip if it will wreck sleep
Hot shift	Do not use caffeine as a replacement for fluids, food, and electrolytes

If caffeine causes palpitations, chest discomfort, panic-like symptoms, dizziness, or unusual shortness of breath, stop using it and seek medical review.

Supplement Label Checklist

- Product name and exact serving size are clear.

- Active ingredients are listed with amounts.
- No proprietary blend hides stimulant or herb amounts.
- Caffeine amount is listed if the product is caffeinated.
- No hormone, peptide, detox, or disease-treatment claims.
- Allergen information fits the athlete.
- Third-party testing is shown where relevant.
- Expiration date and lot number are present.

Third-Party Testing Note

For performance supplements, especially powders and pre-workouts, prefer products tested by a reputable third-party program when available. Testing does not prove a product is necessary or risk-free, but it can reduce contamination and label-accuracy concerns.

Medical Review Note

Get clinician review before using supplements if there are medications, high blood pressure, heart symptoms, kidney issues, liver issues, TRT or hormone use, stimulant sensitivity, planned surgery, or any condition being monitored by a clinician.

Copy-Ready Supplement Checklist

- Food plan is consistent before adding supplements.
- Whey is used only to fill protein gaps or support convenience meals.
- Creatine is discussed cautiously and reviewed if medical concerns exist.
- Electrolytes are matched to sweat, heat, and shift length.
- Caffeine is timed away from the sleep window.
- Vitamin D is used only if deficient or clinician-advised.
- Fish oil is used only if fatty fish intake is low or clinician-advised.
- Fat burners, detox products, hormone/peptide claims, and stimulant-heavy blends are avoided.
- Labels are checked before buying.
- One supplement is added at a time and reviewed for benefit or side effects.

Coach Review Notes

- Confirm current caffeine use on day shift and night shift.
- Confirm any medications, blood pressure concerns, TRT/hormone use, kidney or liver history.
- Confirm preferred whey brand and whether third-party testing matters for the athlete.

References

- ISSN position stands on creatine, caffeine, protein, and sports nutrition evidence hierarchy.
- IOC consensus statements on dietary supplements and high-performance athletes.
- NIH Office of Dietary Supplements fact sheets for supplement safety and medication-interaction cautions.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Progress and Macro Adjustments

Source: Nutrition/12-adjustment-protocol.md | Status: review

Purpose and Scope

This protocol turns weekly data into small nutrition adjustments while building toward a lean, strong 275 lb. The goal is not to react to one weigh-in. The goal is to watch the trend, protect training performance, and make the smallest useful change.

Weekly Check-In Rhythm

Day	Action
Daily	Weigh after waking and bathroom, before food or fluid when practical
1x/week	Measure waist under the same conditions
1x/week	Review gym performance, appetite, digestion, sleep, shift difficulty, and heat exposure
Same day weekly	Decide: maintain, add carbs, reduce carbs, or troubleshoot recovery first

Use the 7-day average bodyweight, not a single high or low day after heat, travel, a salty meal, or a hard shift.

What To Measure

Metric	What To Record	Why It Matters
7-day average bodyweight	Average of daily weigh-ins	Main rate-of-gain signal
Waist measurement	Same location, same time weekly	Helps separate lean gain from fast waist gain
Gym performance	Top sets, reps, pump, readiness	Shows whether food is supporting training
Appetite	Low, normal, high	Guides whether more food is realistic
Digestion	Bloating, reflux, stool changes, meal tolerance	Prevents forcing food that is not working
Sleep/recovery	Sleep length, quality, soreness, mood	Explains performance changes
Shift difficulty	Day/night shift, overtime, missed meals	Prevents overreacting to work stress
Heat exposure	Hot outdoor work, heavy sweat, dehydration signs	Explains bodyweight drops and fatigue

Starting Adjustment Rules

Weekly Signal	Adjustment
Average gain is less than 0.25 lb and performance/recovery are not improving	Add 25-40 g carbs per day
Average gain is 0.5-1.0 lb and waist is stable or only slightly up	Maintain
Average gain is more than 1.0 lb with noticeable waist jump	Reduce 25-40 g carbs per day
Performance is dropping, appetite is poor, or sleep is bad	Do not automatically cut calories; review recovery, hydration, and shift stress first

Hold changes for at least one full week unless digestion, heat illness symptoms, or medical concerns require faster review.

Day-Type Adjustment Examples

Need	First Move
Need more calories for lean gain	Add carbs to training days first
Training performance is flat	Add rice, oats, potatoes, fruit, or pasta around training
Waist is climbing too fast	Reduce carbs from rest days first
Protein target is inconsistent	Fix protein consistency before changing carbs/fats
Carbs are already appropriate but calories need a small bump	Add a small fat serving with meals
Digestion is struggling	Spread carbs across meals, use lower-fiber carbs around training, review dairy/sauce tolerance

Protein stays stable. Carbs are the main adjustment lever. Fats move after carbs are already appropriate.

Simple Decision Table

Bodyweight Trend	Waist	Performance/Recovery	Decision
Down or flat	Stable	Flat or worse	Add 25-40 g carbs/day
Up less than 0.25 lb/week	Stable	Improving	Maintain one more week or add small carbs if appetite is good
Up 0.5-1.0 lb/week	Stable/slightly up	Good	Maintain
Up more than 1.0 lb/week	Noticeably up	Good	Reduce 25-40 g carbs/day, usually from rest days
Any trend	Any	Sleep bad, heat high, shifts chaotic	Troubleshoot recovery/hydration/meal execution before macro change

Recovery First Troubleshooting

Before cutting calories because a week felt bad, check:

- Were meals missed during day or night shifts?
- Was the cooler packed with enough real food?
- Was hydration low after hot outdoor work?
- Were electrolytes and salty meals adequate for sweat?
- Did sleep get shortened by shift rotation?
- Was caffeine too late in the work window?
- Did digestion suffer from rushed meals, too much sauce, or too much dairy at once?

Red Flags Requiring Coach or Clinician Review

Red Flag	Review Needed
Rapid unexplained weight change	Coach/clinician review
Persistent vomiting, diarrhea, or inability to keep fluids down	Clinician review
Heat illness symptoms	Urgent medical attention
Chest pain, fainting, severe shortness of breath, confusion	Urgent medical attention
Blood pressure, kidney, liver, heart, or medication concerns	Clinician review before aggressive changes
Ongoing binge/restrict cycle or anxiety around tracking	Coach/clinician support

Monthly Review Process

Every four weeks, zoom out.

Review Item	Question
Bodyweight	Is the trend moving toward 275 lb at a controlled pace?
Waist	Is waist gain proportional and acceptable?
Training	Are heavy lifts and cardio work improving or at least holding under shift stress?
Recovery	Is sleep, soreness, and mood sustainable?
Execution	Are groceries, prep, cooler packing, and meal timing actually happening?
Adjustments	Did small carb changes work, or does the plan need a bigger review?

Copy-Ready Weekly Check-In Template

Check-In Item	Entry
Week dates	
7-day average bodyweight	
Change from last week	
Waist measurement	
Training performance	
Cardio performance	
Appetite	
Digestion	
Sleep/recovery	
Shift difficulty	
Heat exposure/sweat	
Meals missed	
Hydration/electrolytes notes	
Decision	Maintain / add 25-40 g carbs / reduce 25-40 g carbs / troubleshoot recovery
Where change goes	Training days / rest days / all days / no macro change
Coach notes	

Copy-Ready Adjustment Checklist

- Use 7-day average bodyweight, not one weigh-in.
- Compare waist measurement under the same conditions.
- Review gym performance before changing calories.
- Review appetite, digestion, sleep, shift difficulty, and heat exposure.
- If gain is under 0.25 lb/week and performance/recovery are not improving, add 25-40 g carbs/day.
- If gain is 0.5-1.0 lb/week and waist is stable or slightly up, maintain.
- If gain is over 1.0 lb/week with noticeable waist jump, reduce 25-40 g carbs/day.
- Add carbs to training days first.
- Reduce carbs from rest days first when waist gain is too fast.
- Keep protein stable.
- Adjust fats only after carbs are appropriate.
- Review red flags before pushing harder.

Coach Review Notes

- Confirm target monthly rate of gain after first two review weeks.
- Confirm exact waist measurement location.
- Confirm whether shift heat exposure needs its own adjustment note.

References

- ISSN position stands on protein, nutrient timing, and body-composition support for athletes.
- Academy of Nutrition and Dietetics / ACSM / Dietitians of Canada sports nutrition position guidance.
- ACSM and NSCA strength-training guidance for matching fueling to training demand.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Volume II - Training

Program Overview

Source: Training/01-program-overview.md | Status: review

Purpose and Scope

Volume II turns the athlete's strength goals into a repeatable weekly training system. The plan is built for a 33-year-old, 6'4", roughly 260 lb rotating-shift oilfield athlete building toward a lean, strong 275 lb while chasing a 600+ lb squat, 405 lb bench, and 700 lb deadlift.

This volume is not a rehab plan or medical protocol. Pain, injury concerns, neurological symptoms, or persistent joint issues should be reviewed with a qualified coach and clinician.

Athlete Profile Summary

Item	Current Context
Athlete	Male, 33 years old, 6'4", approximately 260 lb
Long-term bodyweight goal	Lean, strong 275 lb
Previous meet numbers	Squat 235 kg, bench 157.5 kg, deadlift 255 kg
Long-term strength targets	Squat 600+ lb, bench 405 lb, deadlift 700 lb
Work constraint	Rotating 3-to-7 day/night shifts, long oilfield workdays
Weekly training frame	3 heavy lifting days, 2 cardio interval days, 2 recovery days

Main Goals

Goal	Training Meaning
Build the big three	Practice squat, bench, and deadlift with enough frequency to improve skill
Stay durable	Progress only as fast as joints, sleep, work stress, and readiness allow
Keep volume repeatable	Use work that can be completed across day shift, night shift, and heat exposure
Support lean gain	Connect training quality with nutrition, hydration, recovery, and bodyweight tracking

Weekly Structure

Day	Primary Focus	Notes
Monday	Heavy lifting	Squat emphasis, bench practice, lower-body accessories
Tuesday	Cardio intervals	Short, repeatable conditioning; avoid burying legs before Wednesday
Wednesday	Heavy lifting	Bench emphasis, squat/deadlift variation, upper accessories
Thursday	Cardio intervals	Keep output honest but recoverable
Friday	Heavy lifting	Deadlift emphasis, bench volume, posterior-chain accessories
Saturday	Rest/recovery	Walking, mobility, food prep, sleep catch-up
Sunday	Rest/recovery or light movement	Easy movement only if it improves readiness

Training Philosophy

Principle	How It Shows Up
Heavy strength practice	Heavy but controlled top work; no constant max testing
Repeatable volume	Enough sets to grow without requiring perfect life conditions
Recovery-first progression	Add load or volume when readiness and performance support it
Shift-work flexibility	Move heavy sessions around work blocks without guilt
Build while durable	Technique, warm-ups, and autoregulation keep the long game intact

Day-Shift and Night-Shift Considerations

Schedule	Training Move
Day shift with early start	Train after shift only if readiness is green/yellow; otherwise move heavy work to the next better window
Night shift	Train after waking when possible, before the shift drains focus and hydration
Training after shift	Keep expectations conservative; use readiness rules before top sets
Hot outdoor day	Review hydration, electrolytes, and warm-up bar speed before loading heavy
Rotation change	Use a lighter technique day or recovery day instead of forcing a peak session

Connection to Other Volumes

Volume	Connection
Nutrition	Fuels heavy days, cardio days, recovery days, and lean weight gain
Recovery	Sleep, heat management, soreness, and shift stress drive readiness
Tracking	Bodyweight, waist, performance, pain notes, and session RPE guide adjustments

Copy-Ready Weekly Training Overview

Day	Session	Main Lift Focus	Conditioning/Recovery	Readiness Note
Monday	Heavy Lift 1	Squat	Optional easy walk	Green: full work; yellow: trim volume; red: technique/recovery
Tuesday	Cardio	None	Intervals	Keep recoverable
Wednesday	Heavy Lift 2	Bench	Optional mobility	Adjust for shift fatigue
Thursday	Cardio	None	Intervals	Stop before it hurts Friday pulls
Friday	Heavy Lift 3	Deadlift	Optional easy walk	Manage lower-back fatigue
Saturday	Recovery	None	Mobility/walk	Food prep and sleep
Sunday	Recovery	None	Light movement optional	Prepare for Monday

Coach Review Notes

- Confirm preferred training days when the work schedule flips from day shift to night shift.
- Confirm current best gym lifts before percentages or exact loading are finalized.
- Review pain, injury concerns, neurological symptoms, or persistent joint issues with a coach and clinician before progressing load.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Readiness and Autoregulation

Source: Training/02-readiness-autoregulation.md | Status: review

Purpose

Autoregulation keeps heavy training productive when sleep, long shifts, heat exposure, and soreness change week to week. The goal is simple: push when the body is ready, hold steady when readiness is mixed, and reduce load or volume when the signal is clearly off.

Pain, injury concerns, neurological symptoms, or persistent joint issues require coach/clinician review. This chapter does not diagnose or treat injuries.

Daily Readiness Score

Score each input 0-2 before training.

Input	0 Points	1 Point	2 Points
Sleep	Under 5 hours or poor quality	5-7 hours or broken	7+ hours or solid for schedule
Work stress	Brutal shift or overtime	Normal hard shift	Manageable day
Heat exposure	Heavy heat/sweat	Moderate heat	Low heat
Soreness/joint pain	Limiting or sharp	Noticeable but manageable	Normal or minimal
Motivation/focus	Distracted or flat	Average	Ready to train
Warm-up bar speed	Slow/heavy early	Normal	Snappy

Total	Readiness Color	Meaning
9-12	Green	Train as planned
5-8	Yellow	Train, but manage load or volume
0-4	Red	Reduce to technique work, recovery, or move the session

Green/Yellow/Red Rules

Color	Main Lift Rule	Volume Rule	Accessories
Green	Work to planned top sets at target RPE	Complete planned back-off work	Complete normal accessories
Yellow	Cap top sets 1 RPE lower or reduce 2.5-5%	Drop 1-2 back-off sets	Keep only high-value accessories
Red	Technique sets only or move session	Cut volume heavily	Mobility, easy pump work, or stop

RPE and RIR in Plain Language

Term	Meaning	Example
RPE 6	About 4 reps in reserve	Fast, easy work
RPE 7	About 3 reps in reserve	Productive but controlled
RPE 8	About 2 reps in reserve	Heavy training work
RPE 9	About 1 rep in reserve	Very heavy; use sparingly
RPE 10	No reps left	Not a routine training target
RIR	Reps in reserve	How many clean reps were left

When to Push

Push when sleep is acceptable, warm-ups move well, joints feel normal, focus is good, and the previous heavy session did not leave unusual fatigue. Pushing usually means completing the planned top set and back-off work, not maxing out.

When to Hold Steady

Hold steady when readiness is average, work stress is high but manageable, or the warm-up feels normal after a slow start. Use the planned load if bar speed and technique stay clean. Do not add bonus work.

When to Reduce Load or Volume

Reduce when warm-up bar speed is poor, soreness changes technique, the shift was long/hot, sleep was short, or focus is not there. First cut optional accessories, then back-off sets, then top-set load.

Long Shifts and Poor Sleep

Situation	Rule
Under 5 hours sleep	No grinders; technique or reduced-volume session
Hot shift with heavy sweat	Hydrate, eat, warm up longer, and cap load if bar speed is slow
Training after 12+ hour workday	Start yellow unless warm-ups prove green
Two bad readiness days in a row	Move the heaviest lift or reduce total weekly volume

Red Flags for Review

Red Flag	Action
Sharp pain, worsening pain, or pain changing technique	Stop the aggravating work; coach/clinician review
Numbness, tingling, weakness, radiating symptoms	Clinician review
Persistent joint swelling or repeated same-joint irritation	Coach/clinician review
Dizziness, chest pain, fainting, or severe shortness of breath	Urgent medical review

Copy-Ready Readiness Checklist

- Sleep score recorded.
- Work stress score recorded.
- Heat exposure score recorded.
- Soreness/joint pain score recorded.
- Motivation/focus score recorded.
- Warm-up bar speed score recorded.
- Green/yellow/red decision made before top sets.
- Load or volume adjusted if needed.
- Pain, neurological symptoms, or persistent joint issues flagged for review.

Coach Review Notes

- Confirm whether readiness scoring should be logged in the training sheet.
- Confirm top-set RPE targets once the first training block is written.
- Review recurring pain or technique changes with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- Peer-reviewed RPE/autoregulation literature for using session readiness and perceived effort without max testing.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Warm-Up and Movement Preparation

Source: Training/03-warm-up-mobility.md | Status: review

Purpose

The warm-up prepares the athlete to lift heavy with better positions, better bar speed, and fewer surprises after long work shifts. It should raise temperature, open the positions needed for the main lift, rehearse bracing, and ramp load without wasting energy.

Pain, injury concerns, neurological symptoms, or persistent joint issues should be reviewed by a coach and clinician.

10-Minute Minimum Warm-Up

Time	Work
0:00-3:00	Easy bike, walk, row, or sled drag
3:00-5:00	Hips and ankles: squat pry, ankle rocks, bodyweight lunges
5:00-7:00	T-spine, shoulders, lats: rotations, band pull-aparts, lat reach
7:00-9:00	Bracing: dead bug, plank, or loaded carry
9:00-10:00	Empty-bar practice for the main lift

Full Warm-Up for Heavy Days

Step	Focus
General heat	5 minutes easy movement
Mobility	Hips, ankles, T-spine, shoulders, lats based on lift
Activation	Upper back, glutes, trunk brace
Empty bar	2-4 sets to groove the pattern
Ramp sets	Small jumps until the work weight is ready
Readiness check	Bar speed and joint feedback decide green/yellow/red

Squat Warm-Up Flow

Step	Work
Hips/ankles	Ankle rocks, squat pry, hip airplanes or lateral lunges
Bracing	Dead bug or plank, then beltless goblet squat
Bar practice	Empty-bar squats, pause in bottom, controlled descent
Ramp sets	Add load in even jumps; keep depth and foot pressure consistent

Bench Warm-Up Flow

Step	Work
Shoulders/T-spine	Band pull-aparts, scap push-ups, T-spine rotations
Lats/upper back	Lat pulldown or band row, light face pulls
Setup practice	Arch, shoulder blades down/back, feet set
Ramp sets	Empty bar, then controlled paused reps as load builds

Deadlift Warm-Up Flow

Step	Work
Hips/hamstrings	Hip hinge drill, hamstring sweeps, glute bridge
Lats/bracing	Straight-arm pulldown, plank, suitcase carry
Start position	Wedge practice with empty bar or light plates
Ramp sets	Singles or triples; stop jumps if position breaks

Mobility Priorities

Priority	Why It Matters	Simple Drill
Hips	Squat depth and deadlift start	Squat pry, 90/90 transitions
Ankles	Squat balance and depth	Knee-to-wall rocks
T-spine	Squat shelf and bench arch	Open books, foam roller extension
Shoulders	Bench setup and squat bar position	Band pull-aparts, wall slides
Lats	Bench tightness and deadlift bar path	Lat reach, straight-arm pulldown
Bracing	Force transfer on all big lifts	Dead bug, plank, loaded carry

Shift-Work Stiffness Reset

Use this after a long drive, long shift, or before training after sleep debt.

Time	Reset
2 minutes	Walk or bike easy
2 minutes	Hip flexor stretch and ankle rocks
2 minutes	T-spine rotations and lat reach
2 minutes	Bodyweight squat or hinge pattern
2 minutes	Empty-bar main lift

What Not To Do Before Heavy Lifting

Avoid	Why
Long exhausting cardio	Drains the heavy work
Max-effort stretching into pain	Can irritate positions instead of preparing them
Huge jumps in ramp sets	Hides readiness signals
Skipping empty-bar practice	Misses the cheapest technique work
Ignoring pain that changes movement	Needs coach/clinician review

Copy-Ready Warm-Up Checklist

- 3-5 minutes easy movement.
- Hips and ankles prepared.
- T-spine, shoulders, and lats prepared.
- Bracing drill completed.
- Empty-bar lift pattern practiced.
- Ramp sets built with clean technique.
- Readiness color confirmed from bar speed and joint feedback.
- Pain or neurological symptoms flagged for review.

Coach Review Notes

- Confirm which mobility drills produce the best squat depth and bench setup.
- Confirm whether work-shift stiffness changes after day shift versus night shift.
- Review pain, injury concerns, neurological symptoms, or persistent joint issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- ACSM warm-up and mobility guidance for preparing movement without overstating injury-prevention claims.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.

- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Squat Development

Source: Training/04-squat-system.md | Status: review

Purpose

The squat system builds a bigger, more consistent competition squat while managing fatigue from heavy lifting, cardio intervals, and rotating oilfield shifts. The target is 600+ lb long term, built from technical consistency and repeatable weekly exposure.

Pain, injury concerns, neurological symptoms, or persistent hip, knee, or back issues should be reviewed with a qualified coach and clinician.

Baseline and Target

Item	Number
Past meet squat	235 kg / about 518 lb
Long-term target	600+ lb

Technical Priorities

Priority	Standard
Bracing	Big breath, ribs down, trunk locked before descent
Foot pressure	Tripod foot; pressure through midfoot without heel/toe rocking
Bar position	Consistent shelf; wrists and elbows support tight upper back
Depth consistency	Hit the same legal depth in training reps
Controlled descent	Fast enough to be athletic, controlled enough to keep position
Drive out of the hole	Chest and hips rise together; push floor away

Weekly Squat Exposure

Day	Squat Role
Monday heavy lift	Main squat day
Wednesday heavy lift	Secondary squat or pause/tempo variation
Friday heavy lift	Light squat patterning only if recovery allows

Main Squat Day Example

Block	Work
Warm-up	Full squat warm-up and ramp sets
Top work	1-3 top sets of 1-5 reps at planned RPE
Back-off volume	3-5 sets of 3-6 reps with clean speed
Secondary lift	Bench practice or upper-body volume
Accessories	Quads, hamstrings, trunk, upper back

Secondary Squat or Variation Example

Variation	Use
Pause squat	Depth, tightness, drive from bottom
Tempo squat	Position control and descent discipline
Front squat	Upper-back strength and quad focus
Safety-bar squat	Squat volume with less shoulder demand
Beltless squat	Bracing practice at lower loads

Accessory Priorities

Priority	Options
Quads	Leg press, split squat, hack squat, step-up
Hamstrings/glutes	RDL, hamstring curl, back extension
Trunk	Plank, dead bug, weighted carry, cable chop
Upper back	Rows, pulldowns, rear-delt work
Adductors	Copenhagen variation, adductor machine, lateral lunge

Common Weak Points and Fixes

Weak Point	Likely Training Fix
Folding forward	Bracing, upper-back work, front squat, pause squat
Losing depth	Ankle/hip warm-up, pause squats, consistent stance
Knees caving	Foot pressure, tempo work, single-leg work
Slow out of hole	Pause squat, pin squat, quad volume
Shifting side to side	Reduce load, film reps, unilateral accessories

Readiness-Based Adjustments

Readiness	Squat Adjustment
Green	Complete planned top sets and back-off work
Yellow	Cap top set, remove 1-2 back-off sets, keep technique crisp
Red	Technique squat, light variation, or move squat day

Copy-Ready Squat Day Template

Section	Prescription	Notes
Readiness color	Green / Yellow / Red	
Warm-up	10-minute minimum or full squat warm-up	
Main squat	Top set(s): _ x _ @ RPE _	
Back-off squat	_ sets x _ reps	
Variation	Pause / tempo / front / safety-bar	
Accessories	Quad, hamstring, trunk, upper back	
Pain/technique notes		Review if pain changes movement

Coach Review Notes

- Confirm current squat training max before loading targets are finalized.
- Confirm stance, bar position, and depth standard with video review.
- Review pain, injury concerns, neurological symptoms, or persistent hip/knee/back issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- NSCA technique and exercise-analysis references for main lift coaching cues and variation selection.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Bench Press Development

Source: Training/05-bench-system.md | Status: review

Purpose

The bench system builds a stronger, more repeatable press through setup, upper-back tightness, pressing volume, triceps strength, and shoulder-friendly progression. The long-term target is a 405 lb bench.

Pain, injury concerns, neurological symptoms, or persistent shoulder, elbow, wrist, or pec issues should be reviewed with a qualified coach and clinician.

Baseline and Target

Item	Number
Past meet bench	157.5 kg / about 347 lb
Long-term target	405 lb

Technical Priorities

Priority	Standard
Setup and arch	Stable arch that keeps shoulders pinned and feet useful
Upper-back tightness	Shoulder blades set before unrack and stay set
Leg drive	Feet drive the body back into the bench without hips rising
Touch point	Consistent lower-chest/sternum touch
Bar path	Slight back-toward-rack press path, not straight up from touch
Pause control	Still bar on the chest without relaxing
Lockout	Finish with elbows locked and shoulders still tight

Weekly Bench Exposure

Day	Bench Role
Monday heavy lift	Bench technique after squat or moderate volume
Wednesday heavy lift	Main bench day
Friday heavy lift	Secondary bench, close-grip, or volume work

Main Bench Day Example

Block	Work
Warm-up	Bench warm-up, upper back, ramp sets
Top work	1-3 top sets of 1-5 reps at planned RPE
Back-off volume	3-6 sets of 3-8 reps
Secondary press	Close-grip, Spoto, incline, or dumbbell press
Accessories	Rows, triceps, rear delts, lats

Secondary Bench or Volume Example

Variation	Use
Paused bench	Competition skill and control
Close-grip bench	Triceps and lockout
Spoto press	Tightness near the chest
Incline press	Upper chest and shoulder-friendly volume
Dumbbell bench	Hypertrophy with freer shoulder path

Accessory Priorities

Priority	Options
Upper back	Chest-supported row, cable row, pulldown
Triceps	Pushdowns, rolling extensions, close-grip work
Rear delts	Reverse fly, face pull, band pull-apart
Shoulders	Lateral raise, controlled overhead variation if tolerated
Pec volume	Dumbbell press, machine press, push-up

Common Weak Points and Fixes

Weak Point	Likely Training Fix
Losing tightness on chest	Paused bench, Spoto press, setup practice
Miss off chest	Paused work, dumbbell pressing, upper-back tightness
Mid-range stall	Close-grip bench, triceps volume
Lockout weakness	Board/pin work if appropriate, triceps accessories
Shoulder irritation	Review setup, reduce irritating variation, coach/clinician review

Readiness-Based Adjustments

Readiness	Bench Adjustment
Green	Complete planned top work and volume
Yellow	Keep top work submaximal; trim 1-2 volume sets
Red	Technique bench, light dumbbells, upper-back work, or move session

Copy-Ready Bench Day Template

Section	Prescription	Notes
Readiness color	Green / Yellow / Red	
Warm-up	Bench warm-up and ramp sets	
Main bench	Top set(s): _ x _ @ RPE _	
Back-off bench	_ sets x _ reps	
Secondary press	Variation: _	
Accessories	Row, triceps, rear delts, lats	
Pain/technique notes		Review if pain changes movement

Coach Review Notes

- Confirm grip width, arch tolerance, and meet pause standard.
- Confirm weekly bench frequency after the first training block.
- Review pain, injury concerns, neurological symptoms, or persistent shoulder/elbow/wrist/pec issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- NSCA technique and exercise-analysis references for main lift coaching cues and variation selection.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Deadlift Development

Source: Training/06-deadlift-system.md | Status: review

Purpose

The deadlift system builds toward a 700 lb pull while managing posterior-chain and lower-back fatigue. Deadlift progress depends on clean starts, strong lat tension, enough hinge work, and restraint when shifts, heat, or poor sleep reduce readiness.

Pain, injury concerns, neurological symptoms, or persistent back, hip, hamstring, or nerve-related symptoms should be reviewed with a qualified coach and clinician.

Baseline and Target

Item	Number
Past meet deadlift	255 kg / about 562 lb
Long-term target	700 lb

Technical Priorities

Priority	Standard
Start position	Hips close enough to push, shoulders slightly in front of bar
Lat tension	Pull slack out and keep bar close
Wedge	Load hips and legs before the bar leaves the floor
Floor speed	Push floor away without yanking position loose
Bar path	Vertical and close; no drift around knees
Lockout	Hips through, ribs down, no over-leaning
Lower-back fatigue	Manage total heavy pulls and hinge accessories

Weekly Deadlift Exposure

Day	Deadlift Role
Monday heavy lift	Minimal pulling after squat, if any
Wednesday heavy lift	Secondary hinge or technique pull
Friday heavy lift	Main deadlift day

Main Deadlift Day Example

Block	Work
Warm-up	Deadlift warm-up, wedge practice, ramp sets
Top work	1-3 top sets of 1-4 reps at planned RPE
Back-off pulls	2-4 sets of 2-5 reps if bar speed holds
Secondary hinge	RDL, block pull, paused deadlift, or good morning
Accessories	Hamstrings, glutes, lats, trunk

Secondary Pull or Hinge Variation Example

Variation	Use
Paused deadlift	Position off floor and lat tightness
Block pull	Lockout and overload with less range
Deficit deadlift	Floor speed if position is excellent
Romanian deadlift	Hamstrings, glutes, hinge volume
Trap-bar deadlift	Lower stress pull option when appropriate

Accessory Priorities

Priority	Options
Hamstrings	RDL, hamstring curl, glute-ham raise
Glutes	Hip thrust, back extension, split squat
Lats/upper back	Pulldown, row, straight-arm pulldown
Trunk	Side plank, carry, anti-rotation press
Grip	Holds, rows without straps when appropriate

Common Weak Points and Fixes

Weak Point	Likely Training Fix
Bar slow off floor	Paused pulls, leg drive practice, quad/hamstring work
Hips shoot up	Wedge practice, tempo off floor, reduce load
Bar drifts forward	Lat tension drills, close-bar cues, rows
Miss near lockout	RDL, block pull, glute work
Lower back always fatigued	Reduce heavy pull volume, manage hinges, review work stress

Readiness-Based Adjustments

Readiness	Deadlift Adjustment
Green	Complete planned top work and back-offs
Yellow	Keep top set smooth; reduce back-offs or hinge accessories
Red	Technique singles, light hinge, or move deadlift day

Copy-Ready Deadlift Day Template

Section	Prescription	Notes
Readiness color	Green / Yellow / Red	
Warm-up	Deadlift warm-up and ramp sets	
Main deadlift	Top set(s): _ x _ @ RPE _	
Back-off deadlift	_ sets x _ reps	
Secondary hinge	Variation: _	
Accessories	Hamstrings, glutes, lats, trunk	
Pain/fatigue notes		Review if symptoms persist or alter movement

Coach Review Notes

- Confirm conventional versus sumo preference and current best gym pull.
- Confirm how lower-back fatigue responds to Wednesday hinge work.
- Review pain, injury concerns, neurological symptoms, or persistent back/hip/hamstring symptoms with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- NSCA technique and exercise-analysis references for main lift coaching cues and variation selection.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Accessory Training

Source: Training/07-accessory-training.md | Status: review

Purpose and Scope

Accessory training builds the muscle, positions, and work capacity that support the squat, bench, and deadlift. It should help the main lifts move better without burying recovery for a rotating-shift oilfield athlete training three heavy days per week.

Pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues should be reviewed with a qualified coach and clinician.

Accessory Training Principles

Principle	Coaching Standard
Accessories support the main lifts	Pick movements that solve a strength, muscle, or position need
Do not chase fatigue for its own sake	Soreness is not the scoreboard
Fit recovery and equipment	Use what can be done consistently after long shifts
Progress one variable at a time	Add reps, load, control, range, or consistency
Keep technique clean	Accessories should not teach sloppy positions

Accessory Priorities by Main Lift

Main Lift	Support Priorities	Example Movements
Squat	Quads, adductors, glutes, hamstrings, trunk, upper back	Leg press, split squat, hack squat, RDL, hamstring curl, rows
Bench	Upper back/lats, triceps, rear delts, shoulders, pec volume	Rows, pulldowns, pushdowns, close-grip bench, rear-delt raise
Deadlift	Hamstrings, glutes, lats, upper back, trunk, grip	RDL, back extension, hamstring curl, hip thrust, pulldown, carries

Muscle-Group Priorities

Muscle Group	Why It Matters	Good Defaults
Quads	Squat drive and leg size	Leg press, hack squat, split squat
Hamstrings	Deadlift strength and knee balance	RDL, hamstring curl, glute-ham raise
Glutes	Squat/deadlift lockout and hip drive	Hip thrust, back extension, split squat
Upper back/lats	Bench shelf, deadlift bar path, squat stability	Rows, pulldowns, chest-supported rows
Triceps	Bench lockout and pressing volume	Pushdowns, close-grip bench, rolling extensions
Rear delts/shoulders	Bench balance and shoulder tolerance	Face pulls, rear-delt raises, lateral raises
Arms	Optional hypertrophy if recovery allows	Curls, pushdowns, hammer curls

Weekly Placement

Day	Main Lift Emphasis	Accessory Bias
Monday	Squat emphasis	Quads, hamstrings, trunk, upper back
Wednesday	Bench emphasis	Rows, lats, triceps, rear delts, light lower variation
Friday	Deadlift emphasis	Hamstrings, glutes, lats, trunk, optional arms

Keep accessories after the main lift and any planned secondary lift. If time is short, keep the highest-value movement and cut the rest.

Set and Rep Guidelines

Goal	Sets	Reps/Time	Effort
Strength support	2-4	4-8 reps	Heavy but clean, 1-3 reps in reserve
Hypertrophy	2-5	8-15 reps	Controlled, near but not at failure
Technique/control	2-4	6-10 reps	Slow, precise, repeatable
Joint-friendly pump work	1-4	12-25 reps	Smooth, lower load, no grinding

Readiness-Based Accessory Adjustments

Readiness	Adjustment
Green	Complete planned accessories and progress one variable if form is clean
Yellow	Keep 1-2 priority accessories; reduce sets or avoid failure
Red	Skip accessories or use light pump work that improves recovery
Pain changes movement	Stop the aggravating exercise and get coach/clinician review

Copy-Ready Accessory Menu

Category	Movement Options
Squat support	Leg press, hack squat, split squat, step-up, adductor machine
Bench support	Chest-supported row, pulldown, close-grip bench, pushdown, rear-delt raise
Deadlift support	RDL, hamstring curl, hip thrust, back extension if tolerated, rows
Trunk support	Pallof press, plank, side plank, carries
Low-stress pump	Cable row, pushdown, curl, lateral raise, sled drag

Copy-Ready Accessory Selection Checklist

- Main lift priority for the day is clear.
- Accessories support squat, bench, or deadlift performance.
- No accessory is included only to chase fatigue.
- Movement fits available equipment and time.
- Movement does not create pain or alter technique.
- Sets and reps match the goal: strength, hypertrophy, control, or pump.
- Readiness color is used to trim or progress volume.
- Pain, injury, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues are flagged for coach/clinician review.

Coach Review Notes

- Confirm which accessories have the best carryover after the next four-week block.
- Confirm equipment access on workdays versus off days.
- Review pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Core Training

Source: Training/08-core-training.md | Status: review

Purpose and Scope

Core training for powerlifting teaches the trunk to brace, resist movement, transfer force, and stay organized under a bar. The goal is better squat, bench, and deadlift execution, not random ab soreness.

Pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues should be reviewed with a qualified coach and clinician.

Core Training for Powerlifting

Quality	Meaning	Examples
Bracing	Create trunk pressure before heavy reps	Dead bug, carry, bracing practice
Anti-extension	Resist ribs flaring and low back arching	Plank, ab wheel if tolerated
Anti-rotation	Resist twisting under load	Pallof press, cable chop/lift
Anti-lateral flexion	Resist side-bending	Side plank, suitcase carry
Carries	Brace while moving with load	Farmer carry, suitcase carry, front rack carry

How Core Work Supports the Big Three

Lift	Core Role
Squat	Keeps ribs, pelvis, and bar path organized through the descent and drive
Bench	Supports arch, leg drive, and tightness without losing position
Deadlift	Holds the wedge and keeps force moving through the bar instead of leaking through the trunk

Weekly Placement

Day	Placement	Focus
Monday	After squat/back-off work	Bracing, anti-extension
Wednesday	After bench/accessories	Anti-rotation, carries
Friday	After deadlift or accessories	Anti-lateral flexion, lighter trunk work

Avoid fatiguing the trunk before heavy squats or deadlifts. Core work should make heavy lifts more stable, not steal the brace.

Exercise Menu

Exercise	Best Use	Notes
Dead bug	Bracing and rib control	Start here when fatigue is high
Plank variations	Anti-extension	Stop before position breaks
Side plank	Anti-lateral flexion	Good on lower-equipment days
Pallof press	Anti-rotation	Use cable or band
Cable chop/lift	Anti-rotation with movement	Keep ribs down
Hanging knee raise	Trunk and hip flexor control	Use only if shoulders tolerate
Weighted carries	Bracing under load	Farmer or suitcase carry
Back extension	Posterior chain/trunk	Use only if tolerated and programmed intelligently

Set and Rep/Time Guidelines

Goal	Prescription
Bracing practice	2-4 sets of 5-8 controlled breaths/reps
Planks/side planks	2-4 sets of 20-45 seconds
Pallof/chop/lift	2-4 sets of 8-15 reps per side
Carries	2-5 trips of 20-40 yards
Back extensions	2-4 sets of 8-15 if tolerated

What to Avoid Before Heavy Squats/Deadlifts

Avoid	Why
Hard plank or ab circuits to failure	Can reduce bracing quality
Heavy back extensions before pulling	Can pre-fatigue the posterior chain
New core exercises on max-effort days	Unknown soreness or irritation
Movements that trigger symptoms	Requires coach/clinician review

Readiness-Based Core Adjustments

Readiness	Core Move
Green	Complete planned trunk work after main lifts
Yellow	Keep lower-fatigue drills: dead bug, Pallof press, side plank
Red	Use light bracing practice only or skip
Back/hip symptoms	Avoid aggravating work and seek coach/clinician review

Copy-Ready Core Training Template

Day	Core Slot	Exercise	Sets/Reps
Monday	After squat work	Dead bug or plank	2-4 sets
Wednesday	After bench work	Pallof press or carry	2-4 sets
Friday	After deadlift work	Side plank or light carry	2-3 sets

Copy-Ready Core Checklist

- Core work is placed after heavy main lifts.
- Exercise trains bracing, anti-extension, anti-rotation, anti-lateral flexion, or carries.
- Trunk work does not reduce squat/deadlift bracing quality.
- Sets stop before position breaks.
- Back extensions are used only if tolerated and programmed intelligently.
- Pain, injury, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues are flagged for coach/clinician review.

Coach Review Notes

- Confirm which trunk drills improve bracing on video.
- Confirm whether carries fit the athlete's gym layout and recovery.
- Review pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Conditioning

Source: Training/09-conditioning.md | Status: review

Purpose and Scope

Conditioning supports work capacity, recovery between sets, heart-rate control during long shifts, and general fitness without crushing lower-body recovery. The default structure is Tuesday and Thursday intervals, with optional easy walking on recovery days.

Chest pain, severe shortness of breath, fainting, unusual dizziness, neurological symptoms, pain, injury concerns, or persistent joint issues should be reviewed with a qualified coach and clinician.

Why Conditioning Matters for Strength Athletes

Benefit	Training Meaning
Better work capacity	More productive lifting sessions without dragging between sets
Better recovery habits	Easy movement can support rest days without adding heavy stress
Better shift tolerance	Long workdays feel less like a separate sport
Body composition support	Helps the lean-gain phase stay athletic

Current Weekly Conditioning Structure

Day	Conditioning
Tuesday	Cardio intervals
Thursday	Cardio intervals
Saturday	Optional easy walk or mobility
Sunday	Optional easy walk or light movement

Conditioning Rules

Rule	Standard
Improve work capacity without ruining lower-body recovery	Leave enough legs for squat and deadlift days
Keep intervals short and repeatable	Quality beats dramatic one-off tests
Do not turn every cardio day into a test	Most sessions should be repeatable next week
Adjust for heat and shift fatigue	Reduce intensity, duration, or total rounds when readiness is low

Recommended Modes

Mode	Best Use	Notes
Bike	Default interval choice	Low joint impact and easy to control
Sled push/pull	Strength-friendly conditioning	Manage quad fatigue before squat day
Incline walk	Low-readiness or recovery day	Easy to scale
Row/air bike	Hard intervals if tolerated	Watch back/hip fatigue
Sprinting	Advanced option only	Use only if joints and recovery are ready

Interval Templates

Template	Prescription	Use
Beginner/rebuild intervals	6-8 rounds: 20 sec hard / 100 sec easy	Return from layoff or poor readiness
Standard interval day	8-10 rounds: 30 sec hard / 90 sec easy	Normal Tuesday/Thursday option
Low-readiness interval day	10-20 min incline walk or easy bike	Poor sleep, heavy shift, sore legs
Heat-adjusted interval day	5-8 shorter rounds, longer rest, lower output	Hot shift or outdoor heat exposure

Placement Around Heavy Squat/Deadlift Days

Timing	Rule
Tuesday after Monday squat	Keep intervals hard enough to train, not so hard they ruin Wednesday
Thursday before Friday deadlift	Avoid high-soreness modes; bike or incline walk works well
After night shift	Start with low-readiness template unless warm-up says otherwise
During heat waves	Lower output and prioritize hydration, electrolytes, and recovery

Readiness-Based Conditioning Adjustments

Readiness	Conditioning Adjustment
Green	Complete planned interval template
Yellow	Reduce rounds by 20-30% or use lower-impact mode
Red	Easy walk/bike only or skip conditioning
Symptoms or unusual breathing/chest signs	Stop and seek medical review

Copy-Ready Conditioning Templates

Day	Template	Mode	Notes
Tuesday	Standard intervals	Bike or sled	Keep Wednesday lift in mind
Thursday	Standard or low-readiness intervals	Bike or incline walk	Protect Friday deadlift
Saturday	Optional easy walk	Walk	Recovery only
Sunday	Optional light movement	Walk/mobility	Prepare for Monday

Copy-Ready Conditioning Checklist

- Tuesday/Thursday conditioning selected before the week starts.
- Mode is joint-friendly and recoverable.
- Intervals are short and repeatable.
- Session is not treated like a weekly test.
- Heat exposure and shift fatigue are considered.
- Lower-body soreness is checked before Thursday intervals.
- Chest pain, severe shortness of breath, fainting, unusual dizziness, neurological symptoms, pain, injury concerns, or persistent joint issues are flagged for coach/clinician review.

Coach Review Notes

- Confirm whether bike, sled, incline walk, rower, or air bike is most available.
- Confirm how Tuesday intervals affect Wednesday lifting.
- Review chest pain, severe shortness of breath, neurological symptoms, pain, injury concerns, or persistent joint issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- ACSM conditioning guidance and concurrent-training research for balancing aerobic work with strength goals.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Deloads and Fatigue Management

Source: [Training/10-deloads.md](#) | Status: review

Purpose and Scope

Deloads keep training moving when fatigue climbs faster than adaptation. The goal is to reduce stress long enough for performance, joints, motivation, and technique to rebound without abandoning the program.

Pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues should be reviewed with a qualified coach and clinician.

What a Deload Is and Is Not

Deload Is	Deload Is Not
A planned reduction in training stress	A punishment
A way to restore performance	A week of doing nothing by default
A chance to sharpen technique	A max-out week
A recovery tool during hard work blocks	A reason to slash food aggressively

Planned vs Reactive Deloads

Type	When It Happens	Best Use
Planned deload	Scheduled after a hard block	Predictable fatigue management
Reactive deload	Triggered by performance/readiness decline	Shift work, heat exposure, poor sleep, joint pain trend

When to Deload

Signal	What It Means
Performance drop	Loads that should move well keep slowing down
Joint pain trend	Repeated irritation or worsening pain needs review
Poor sleep accumulation	Several short or low-quality nights stack up
High work stress	Long shifts or overtime reduce training quality
Heat exposure accumulation	Hot workdays create repeated dehydration/fatigue stress
Motivation/focus crash	Normal sessions feel unusually hard to start
Technique breakdown	Positions fail at loads that are normally clean

Deload Options

Option	How To Use
Reduce load	Keep movement pattern, lower weight 10-20%
Reduce volume	Keep some intensity, cut sets by 30-50%
Reduce intensity	Avoid top sets above RPE 6-7
Technique-only week	Use clean, fast reps and leave the gym fresher
Extra recovery days	Move or skip sessions during severe work/sleep stress

Sample 1-Week Deload Structure

Day	Session	Deload Focus
Monday	Squat/bench technique	2-4 easy work sets, no grinders, light accessories
Tuesday	Conditioning	Easy bike or walk; no hard intervals
Wednesday	Bench plus light hinge	Technique reps, upper-back pump, low fatigue
Thursday	Conditioning	Easy incline walk or bike
Friday	Deadlift technique	Light singles/triples, no heavy back-offs
Saturday	Recovery	Walk, mobility, food prep, sleep
Sunday	Recovery	Light movement optional

How Cardio Changes During Deload

Conditioning should help recovery during a deload. Replace hard intervals with easy walking, easy bike, or shorter low-readiness intervals. Do not use deload week to test conditioning.

How Nutrition Changes During Deload

Rule	Standard
Do not aggressively cut food because training is lighter	Recovery still needs fuel
Keep protein stable	Protein target does not change
Carbs	Slight reduction only if activity is lower and appetite drops
Hydration/electrolytes	Keep aligned with heat exposure and shifts

Returning From Deload

Return Step	Rule
First week back	Resume normal structure but avoid bonus volume
Main lifts	Start with planned submaximal work and assess bar speed
Accessories	Add back priority movements first
Conditioning	Return to standard intervals only if legs and readiness are good
If symptoms persist	Do not force progression; get coach/clinician review

Copy-Ready Deload Decision Checklist

- Performance has dropped for more than one session.
- Joint pain trend is worsening or repeated.
- Sleep debt has accumulated.
- Work stress or heat exposure has stacked up.
- Motivation/focus is unusually low.
- Technique is breaking down at normal loads.
- Deload option selected: reduce load / reduce volume / reduce intensity / technique-only / extra recovery days.
- Cardio changed to easy recovery work.
- Protein kept stable.
- Food not aggressively cut.
- Pain, injury, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues flagged for coach/clinician review.

Coach Review Notes

- Confirm whether deloads should be planned every block or triggered by readiness.
- Confirm the best deload option after night-shift blocks and high-heat weeks.
- Review pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- NSCA and peer-reviewed fatigue-management literature for deloads, recovery weeks, and volume reduction.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Six-Week Strength Block

Source: Training/11-four-week-training-block.md | Status: review

Purpose and Scope

This chapter turns the Volume II training system into the first usable 6-week strength block. The repository filename still says four-week-training-block as legacy project-structure wording, but the chapter content now uses a 6-week block.

The block uses RPE-based loading instead of fixed percentages. Confirm current training maxes before percentage-based work is added later. Do not test true maxes inside this block. Pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues should be reviewed with a qualified coach and clinician.

Block Goal

Build productive strength practice in the squat, bench, and deadlift while keeping recovery realistic for rotating oilfield day/night shifts. Weeks 1-4 build, Week 5 consolidates, and Week 6 reduces stress.

Training Assumptions

Assumption	Rule
Loading	Use RPE and bar speed; no fixed percentages yet
Max testing	No true max testing; no routine RPE 9-10 work
Readiness	Green/yellow/red rules apply every session
Top work	Main lifts usually stay RPE 6-8 in Weeks 1-5
Deadlift volume	Conservative; quality beats total tonnage
Current entry point	If already around Week 4, use the Week 4 section below

Six-Week Layout

Week	Goal	Main Lift Top Work	Volume Direction	Conditioning
Week 1	Base/entry week	RPE 6-7	Establish clean work	Rebuild/standard intervals
Week 2	Small increase	RPE 6.5-7.5	Add a small load or rep increase	Standard intervals if legs recover
Week 3	Productive build	RPE 7-8	Strong but repeatable volume	Keep recoverable
Week 4	Strongest productive week	Up to RPE 8	Best quality work, no max testing	Adjust around heat/work stress
Week 5	Consolidate or slight taper	RPE 7-7.5	Hold or trim volume	Lower if readiness dips
Week 6	Deload/low-stress consolidation	RPE 5-6	Reduce load, volume, and intensity	Easy work only

Weekly Schedule

Day	Session	Focus
Monday	Heavy lifting	Squat emphasis
Tuesday	Conditioning	Intervals, recoverable
Wednesday	Heavy lifting	Bench emphasis
Thursday	Conditioning	Intervals, protect Friday deadlift
Friday	Heavy lifting	Deadlift emphasis
Saturday	Recovery	Walk, mobility, sleep, food prep
Sunday	Recovery/light movement	Easy movement if it improves readiness

Monday: Squat-Emphasis Day

Warm-up: use the full squat warm-up on green/yellow days and the 10-minute minimum on low-time days. Ramp until bar speed and depth confirm the readiness color.

Block	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Main squat	3x5 @ RPE 6-7	4x4 @ RPE 7	1x4 @ RPE 7.5, then 3x4 easier	1x3 @ RPE 8, then 3x3 easier	3x3 @ RPE 7	2-3x3 @ RPE 5-6
Secondary bench	Paused bench 3x6 @ RPE 6	4x5 @ RPE 6-7	4x4 @ RPE 7	3x4 @ RPE 7	3x5 @ RPE 6-7	2x5 @ RPE 5-6
Accessories	Quad + hamstring + upper back	Same, add 1 rep/set if clean	Same	Keep only productive sets	Trim 1 set each	Light pump only
Core	Dead bug or plank 2-4 sets	Same	Same	Same	2-3 sets	Easy bracing only

Readiness modification: yellow means cap squat at RPE 7 and remove 1-2 back-off/accessory sets. Red means technique squat only, RPE 5-6, or move the session.

Tuesday: Conditioning

Readiness	Mode Options	Template
Green	Bike, sled, air bike if tolerated	8-10 rounds: 30 sec hard / 90 sec easy
Yellow	Bike, incline walk, light sled	6-8 rounds: 20 sec hard / 100 sec easy
Red	Walk or easy bike	10-20 minutes easy
Heat-adjusted	Indoor bike or walk	5-8 shorter rounds, longer rest, lower output

Do not turn Tuesday into a test. Wednesday bench should still feel trainable.

Wednesday: Bench-Emphasis Day

Warm-up: bench warm-up, upper-back activation, empty-bar pause practice, and ramp sets. Keep shoulder position and leg drive consistent before adding load.

Block	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Main bench	4x5 @ RPE 6-7	5x4 @ RPE 7	1x4 @ RPE 7.5, then 4x4 easier	1x3 @ RPE 8, then 4x3 easier	4x3 @ RPE 7	2-3x4 @ RPE 5-6
Secondary squat/hinge	Pause squat 3x5 @ RPE 6	Pause squat 3x4 @ RPE 6-7	Tempo squat 3x4 @ RPE 7	Pause squat 3x3 @ RPE 7	Light squat 2-3x4 @ RPE 6	Technique squat 2x3 @ RPE 5
Accessories	Row + triceps + rear delts	Same, small rep increase	Same	Keep quality, no failure	Trim 1-2 sets	Light pump only
Core	Pallof press or carry 2-4 sets	Same	Same	Same	2-3 sets	Easy anti-rotation

Readiness modification: yellow means main bench stays submaximal and accessories are trimmed. Red means technique bench, upper-back pump work, and no heavy secondary squat/hinge.

Thursday: Conditioning

Readiness	Mode Options	Template
Green	Bike or incline walk	8 rounds: 30 sec hard / 90 sec easy
Yellow	Bike or easy sled drag	6 rounds: 20 sec hard / 100 sec easy
Red	Walk or easy bike	10-20 minutes easy
Heat-adjusted	Indoor bike or walk	Reduce rounds and keep effort conversational-to-moderate

Thursday conditioning must protect Friday deadlift. Avoid sprinting, high-soreness sled work, or all-out air-bike work if legs or low back are not recovered.

Friday: Deadlift-Emphasis Day

Warm-up: deadlift warm-up, lat tension, wedge practice, and conservative ramp sets. Stop load jumps if the bar drifts or the start position changes.

Block	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Main deadlift	3x3 @ RPE 6-7	4x3 @ RPE 7	1x3 @ RPE 7.5, then 2x3 easier	1x2-3 @ RPE 8, then 2x2 easier	3x2 @ RPE 7	2x2-3 @ RPE 5-6
Secondary bench	Close-grip bench 3x6 @ RPE 6	4x5 @ RPE 6-7	4x4 @ RPE 7	3x4 @ RPE 7	3x5 @ RPE 6	2x5 @ RPE 5
Accessories	Hamstring + lat + glute	Same	Same, no grinding	Conservative volume	Trim volume	Optional light pump
Core	Side plank or light carry 2-3 sets	Same	Same	Same	2 sets	Easy only

Readiness modification: yellow means deadlift top work caps at RPE 7 and back-offs are reduced. Red means light technique pulls or move deadlifts. Do not grind deadlifts after poor sleep or a hot shift.

Saturday Recovery

Option	Target
Easy walk	10-30 minutes if it improves recovery
Mobility	Hips, ankles, T-spine, shoulders, lats
Prep	Food, hydration, sleep setup
Skip	Any recovery work that becomes another workout

Sunday Recovery or Light Movement

Option	Target
Light walk	Easy pace
Mobility reset	10 minutes
Training review	Check readiness trends, missed reps, soreness, and schedule
Plan Monday	Choose training window around day/night shift

If Starting This Plan Around Week 4

Do not guess loads from old maxes. Enter with RPE and bar speed.

Question	Decision
Warm-ups move fast and technique is stable	Run Week 4 as the strongest productive week, up to RPE 8
Warm-ups feel average or sleep/work stress is high	Treat Week 4 like Week 3: productive build, RPE 7-7.5
Bar speed is slow, joints are irritated, or heat stress is high	Treat Week 4 like Week 5: consolidate, RPE 6.5-7
Pain, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues appear	Stop the concerning work and seek coach/clinician review

Weeks 5-6 after a Week 4 start should consolidate and deload. Week 5 should not force a max. Week 6 should reduce load, volume, and intensity even if Week 4 went well.

Day-Shift and Night-Shift Scheduling Notes

Work Pattern	Training Move
Day shift	Train after shift only if readiness is green/yellow; otherwise move heavy work
Night shift	Prefer training after waking, before the shift drains hydration and focus
Long hot shift	Start yellow until warm-ups prove otherwise
Rotation change	Use recovery day, technique session, or repeat the week instead of forcing load

Copy-Ready Week 4-6 Continuation Table

Week	Squat	Bench	Deadlift	Conditioning
Week 4	Strong productive work, cap RPE 8	Strong productive work, cap RPE 8	Strong but conservative, cap RPE 8	Recoverable intervals
Week 5	Consolidate, RPE 7-7.5	Consolidate, RPE 7-7.5	Consolidate, RPE 7	Reduce if readiness dips
Week 6	Deload, RPE 5-6	Deload, RPE 5-6	Deload, RPE 5-6	Easy only

Coach Review Notes

- Confirm current training maxes before percentage work is added.
- Confirm whether the athlete is truly entering around Week 4 or should restart at Week 1.
- Review pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- Peer-reviewed RPE/autoregulation literature for using session readiness and perceived effort without max testing.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Progression Rules

Source: Training/12-progression-rules.md | Status: review

Purpose and Scope

These rules explain how to progress the first 6-week strength block without fixed percentages or max testing. The goal is to add stress only when RPE, bar speed, technique, readiness, and recovery support it.

Pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues should be reviewed with a qualified coach and clinician.

Main Lift Progression Across 6 Weeks

Week	Progression Target
Week 1	Find clean entry loads at RPE 6-7
Week 2	Add a small load or one rep per set if Week 1 was clean
Week 3	Build to RPE 7-8 with repeatable technique
Week 4	Strongest productive work, cap around RPE 8, no max testing
Week 5	Consolidate; hold load, trim volume, or slightly taper
Week 6	Deload/low-stress consolidation at RPE 5-6

Progress only when the previous week had clean reps, stable technique, no missed reps, and recovery was acceptable.

Back-Off Work Progression

Signal	Rule
Back-offs move fast	Add 5-10 lb next week or add 1 rep to one set
Back-offs are correct but slow	Repeat the same work next week
Back-offs change technique	Reduce load or cut a set
Main top set was harder than planned	Reduce back-off volume that day

Accessory Progression

Goal	Progression
Strength support	Add small load when all sets hit the top of rep range cleanly
Hypertrophy	Add reps before load; stop 1-3 reps before failure
Technique/control	Improve range, tempo, or consistency before load
Pump/recovery	Keep load modest and chase clean blood flow, not failure

Accessories should support the main lifts. If accessories make the next main-lift day worse, reduce them.

Conditioning Progression

Current Response	Progression
Intervals repeatable and legs recover	Add 1 round or small output increase
Wednesday or Friday lifting suffers	Reduce Thursday intensity or total rounds
Heat/work stress high	Use heat-adjusted or low-readiness template
Low readiness	Walk or easy bike instead of intervals

Conditioning progression is successful when it improves work capacity without stealing strength performance.

Using RPE and Bar Speed

Signal	Meaning
RPE lower than planned and bar speed fast	Small increase is allowed
RPE on target and technique clean	Keep the plan
RPE higher than planned	Hold load, reduce back-offs, or repeat week
Bar speed drops suddenly	Stop adding load and check readiness
Technique changes	Reduce load or end the top-work climb

Green/Yellow/Red Progression Rules

Readiness	Main Lift	Back-Offs	Accessories	Conditioning
Green	Run planned progression	Complete planned work	Full priority work	Planned intervals
Yellow	Cap top work 0.5-1 RPE lower	Trim 1-2 sets	Keep only priorities	Reduce rounds/intensity
Red	Technique work or move session	Minimal	Skip or light pump	Easy walk/bike or skip

After Missed Reps or Bad Sessions

Situation	Next Move
One missed rep from bad judgment	Do not add load next week; repeat or reduce slightly
Multiple missed reps	Reduce next exposure and review readiness
Technique breakdown	Lower load and rebuild clean reps
Bad session after long/hot shift	Treat as readiness noise before rewriting the program
Same issue repeats	Coach review before pushing progression

Poor Sleep, Long Shifts, and Hot Workdays

Stressor	Adjustment
Poor sleep	Start yellow; no grinders or max attempts
Long shift	Reduce accessory volume first
Hot workday	Extend warm-up, hydrate, and cap top work if bar speed is slow
Multiple hard workdays	Repeat the week or move the heaviest session
Night shift transition	Use technique work or consolidation instead of forcing load

When to Repeat a Week

Repeat the week instead of increasing load when:

- Top sets hit the target RPE but bar speed was slower than expected.
- Readiness was yellow/red for more than one heavy day.
- Conditioning interfered with lifting.
- Work stress or heat exposure distorted the week.
- Technique was clean but not strong enough to justify more load.

When to Deload Early

Deload early when performance drops across multiple sessions, joint pain trends worse, sleep debt accumulates, heat stress stacks up, motivation/focus crashes, or technique breaks down at normal loads. Medical red flags should not be treated as a programming problem.

Transition After the 6-Week Block

Outcome	Next Step
Block went well	Start a new block with slightly higher entry loads
Week 4 was strong but Week 5/6 felt needed	Keep the same structure and progress slowly
Fatigue was high	Add a longer deload or lower volume in the next block
Technique improved but loads did not	Keep RPE approach and build volume quality
Pain or medical red flags appeared	Coach/clinician review before continuing

If Starting at Week 4

Use Week 4 as an entry assessment, not a test. Select loads by ramping to the prescribed RPE with clean technique. If RPE 8 arrives at a lower load than expected, stop there. If readiness is yellow, run Week 4 like a Week 3 build. If readiness is red, run a consolidation week and then proceed to Week 5/6.

Copy-Ready Progression Checklist

- No true max testing planned.
- Main lift top work stays around RPE 6-8 in Weeks 1-5.
- Week 4 is productive, not a max-out.

- Week 5 consolidates instead of forcing a peak.
- Week 6 reduces load, volume, and/or intensity.
- Bar speed and technique support any load increase.
- Missed reps trigger repeat/reduce decisions.
- Poor sleep, long shifts, and heat exposure modify the day.
- Conditioning does not hurt Wednesday bench or Friday deadlift.
- Accessories support the main lifts without failure training.
- Pain, injury, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues are flagged for coach/clinician review.

Coach Review Notes

- Confirm current training maxes before any percentage progression is added.
- Confirm whether Week 4 should be treated as an entry week, build week, or strongest productive week.
- Review pain, injury concerns, neurological symptoms, chest pain, severe shortness of breath, or persistent joint issues with a coach and clinician.

References

- NSCA Essentials of Strength Training and Conditioning for progressive overload, exercise selection, and fatigue management.
- ACSM resistance-training guidance for strength training frequency, volume, intensity, and safe progression.
- Peer-reviewed RPE/autoregulation literature for using session readiness and perceived effort without max testing.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Volume III - Recovery

Recovery Overview

Source: Recovery/01-recovery-overview.md | Status: review

Purpose and Scope

Recovery is the system that lets nutrition and training work under rotating 3-to-7 day/night oilfield shifts, heat exposure, heavy lifting, and inconsistent sleep. This volume gives practical rules for sleep, shift planning, hydration, soreness, heat stress, and weekly recovery decisions.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention. Do not treat those as normal training fatigue.

Recovery Connections

Area	Recovery Connection
Nutrition	Consistent meals, protein, carbs, and calories support training adaptation
Training	Readiness decides when to push, hold, trim, or deload
Hydration	Fluids and electrolytes matter most after hot shifts and sweaty training
Tracking	Bodyweight, sleep, soreness, pain notes, heat exposure, and performance guide decisions

Main Recovery Priorities

Priority	Standard
Sleep	Protect the main sleep block and use naps when shifts demand it
Food consistency	Do not let long shifts erase meals
Hydration/electrolytes	Match fluids and electrolytes to heat, sweat, and work duration
Stress management	Control what can be controlled: schedule, prep, caffeine, wind-down
Soreness control	Use movement, warm-ups, and volume management
Heat management	Cool down, rehydrate, and adjust training readiness
Shift-work planning	Move heavy sessions when sleep, heat, or work stress make the plan unrealistic

Green/Yellow/Red Recovery Status

Status	Signals	Training Move
Green	Sleep acceptable, normal soreness, hydrated, focused	Run plan
Yellow	Short sleep, high heat, heavy soreness, stressful shift	Trim accessories, cap RPE, lower conditioning
Red	Severe fatigue, pain changes movement, heat illness signs, medical red flags	Stop/modify session; seek urgent care for red flags

Recovery Rules by Day Type

Day Type	Recovery Rule
Heavy lifting days	Eat before/after, hydrate, warm up fully, protect sleep after training
Cardio days	Keep intervals recoverable; do not damage squat/deadlift readiness
Rest days	Walk, mobility, food prep, sleep catch-up, and stress reduction

Coach/Clinician Red Flags

Urgent medical attention is needed for heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms. Coach/clinician review is appropriate for pain that changes technique, persistent joint issues, repeated sharp pain, or symptoms that do not improve with conservative training adjustments.

Copy-Ready Recovery Overview Table

Check	Green	Yellow	Red
Sleep	Main block protected	Short or broken	Multiple poor nights plus poor function
Heat	Managed	Heavy exposure	Heat illness symptoms
Soreness	Normal	High but manageable	Changes movement
Hydration	Steady	Behind	Dizziness/confusion/rapid worsening
Plan	Run	Trim	Stop or seek help

Coach Review Notes

- Confirm which recovery markers will be tracked weekly.
- Confirm how shift rotation will be logged against training performance.
- Review red flags conservatively and do not rely on routine recovery habits for urgent symptoms.

References

- American Academy of Sleep Medicine and CDC sleep-health guidance for sleep duration, consistency, and fatigue context.
- NIOSH shift-work and long-work-hour guidance for fatigue-aware scheduling.
- CDC/NIOSH/OSHA heat-stress guidance for heat exposure, cooling/rest, and urgent symptom escalation.
- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for conservative pain/symptom escalation.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Sleep System

Source: *Recovery/02-sleep-system.md* | Status: review

Purpose and Scope

Sleep is the highest-return recovery habit for heavy lifting, conditioning, heat exposure, and lean weight gain. This chapter focuses on practical sleep protection for rotating day/night shifts.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Sleep Goals

Goal	Standard
Main sleep block	Protect the longest realistic sleep window
Naps	Use 20-90 minutes when shifts compress sleep
Consistency	Anchor sleep around shifts instead of chasing perfect clock times
Training support	Avoid hard training when sleep debt is stacked and readiness is red

Day-Shift Sleep Strategy

Timing	Action
Before bed	Pack meals/cooler, lower lights, stop work scrolling
Sleep block	Keep room dark, cool, and quiet
Wake-up	Hydrate, eat, and avoid rushing into heavy training
After shift	If training, use readiness score before loading heavy

Night-Shift Sleep Strategy

Timing	Action
After shift	Eat a planned meal, rehydrate, cool down, reduce light
Sleep block	Dark room, cool temperature, phone quiet
Before shift	Wake, eat, hydrate, get light exposure
Training	Prefer after waking before the shift if possible

Sleep After Training and Hot Shifts

Situation	Recovery Move
Training late	Eat, hydrate, wind down, avoid extra stimulants
Heavy squat/deadlift day	Plan extra wind-down time and reduce late caffeine
Hot shift	Cool down, replace fluids/electrolytes with food, then sleep
Poor sleep after heat	Start next training day yellow until warm-ups prove otherwise

Caffeine, Light, Room, Wind-Down, Naps

Tool	Practical Rule
Caffeine	Use early in the wake window; avoid it close to planned sleep
Light exposure	Bright light after waking; reduce light before sleep
Room setup	Cool, dark, quiet, and predictable
Wind-down	Repeatable 20-30 minute routine
Naps	Short nap for alertness; longer nap when sleep debt is high

After a Bad Sleep Night

Signal	Action
One poor night	Train if warm-ups are normal; cap top work if needed
Poor sleep plus heat exposure	Start yellow and trim volume
Multiple poor nights	Move heavy session, repeat week, or deload early
Red-flag symptoms	Seek urgent medical attention when appropriate

Copy-Ready Sleep Checklist

- Main sleep block planned.
- Caffeine cutoff set before the shift starts.
- Room made dark, cool, and quiet.
- Meals and hydration handled before sleep.
- Light exposure used after waking.
- Nap planned if sleep block is short.
- Readiness adjusted after poor sleep.
- Urgent symptoms treated as medical, not routine fatigue.

Coach Review Notes

- Confirm normal day-shift and night-shift sleep windows.
- Confirm caffeine timing and tolerance.
- Review ongoing insomnia, severe fatigue, or urgent symptoms with an appropriate clinician.

References

- American Academy of Sleep Medicine and CDC sleep-health guidance for sleep duration, consistency, and fatigue context.
- NIOSH shift-work and long-work-hour guidance for fatigue-aware scheduling.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Shift-Work Recovery

Source: *Recovery/03-shift-work-recovery.md* | Status: review

Purpose and Scope

Rotating oilfield shifts change sleep, meals, hydration, caffeine timing, and training readiness. The goal is to keep the weekly structure intact while moving heavy work away from the worst recovery windows.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

How Rotating Shifts Affect Readiness

Stressor	Training Effect
Short sleep	Lower bar speed and focus
Night shift	Harder caffeine/sleep timing
Hot shifts	Higher hydration and recovery cost
Long workdays	Less time for warm-up, meals, and downshift
Rotation change	Increased need for yellow/red modifications

Day-Shift Strategy

Anchor	Action
Before shift	Pack meals, water, electrolytes, and training bag
During shift	Eat on schedule, sip fluids, avoid late caffeine
After shift	Train only if readiness is green/yellow
Before sleep	Wind down quickly; do not turn prep into another late-night task

Night-Shift Strategy

Anchor	Action
After waking	Eat, hydrate, get light exposure
Before shift	Train here when possible
During shift	Use meals, fluids, and caffeine intentionally
After shift	Cool down, eat, reduce light, sleep

Rotation-Change Strategy

Situation	Move
Days to nights	Use a lighter technique day if sleep is compressed
Nights to days	Protect the first real sleep block before heavy lifting
Multiple bad transitions	Repeat the training week or deload early

Training Placement Around Shifts

Training Day	Best Placement
Heavy squat	After best sleep window of the week
Heavy bench	More flexible, but still readiness-based
Heavy deadlift	Avoid after worst sleep/hottest shift when possible
Conditioning	Reduce if it threatens the next heavy session

Meal, Hydration, and Caffeine Anchors

Anchor	Rule
Meals	Keep protein/carbs packed before shift starts
Hydration	Bring enough fluid for the full shift
Electrolytes	Use when heat/sweat is high
Caffeine	Use early in wake window, not near planned sleep

Move or Repeat Rules

Signal	Decision
One poor shift but warm-ups normal	Train with yellow cap
Poor sleep plus slow warm-ups	Move heavy session
Two or more compromised heavy days	Repeat the week
Red flags	Stop and seek urgent medical attention if symptoms match

Copy-Ready Shift-Work Recovery Checklist

- Current shift pattern identified.
- Heavy sessions placed near best sleep windows.
- Meals and fluids packed before shift.
- Caffeine timing set.
- Heat exposure logged.
- Readiness color chosen before training.
- Heavy session moved if warm-ups are slow and sleep is poor.
- Urgent symptoms treated as medical red flags.

Coach Review Notes

- Confirm the most common rotation pattern.
- Confirm which heavy day is most affected by shift fatigue.
- Review persistent readiness problems or red-flag symptoms with coach/clinician support.

References

- American Academy of Sleep Medicine and CDC sleep-health guidance for sleep duration, consistency, and fatigue context.
- NIOSH shift-work and long-work-hour guidance for fatigue-aware scheduling.
- CDC/NIOSH/OSHA heat-stress guidance for heat exposure, cooling/rest, and urgent symptom escalation.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Heat-Stress Recovery

Source: [Recovery/04-heat-stress-recovery.md](#) | Status: review

Purpose and Scope

Heat exposure is a recovery cost. Hot outdoor shifts can reduce readiness, worsen sleep, increase fluid/electrolyte needs, and make normal training loads feel heavier. This chapter is conservative and does not treat heat illness.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, confusion, stopped sweating with overheating, or rapidly worsening symptoms require urgent medical attention.

Heat as Recovery Cost

Heat Stress Signal	Recovery Meaning
Heavy sweat	Higher fluid/electrolyte need
Sudden bodyweight drop	Rehydrate and review heat exposure
Headache/cramps after heat	Add food/electrolytes and lower training stress
Poor sleep after heat	Start next training day yellow
Confusion/fainting/rapid worsening	Urgent medical attention

Hot-Shift Recovery Strategy

Timing	Action
Before shift	Pack water, electrolyte packets, salty meals, cooling towel
During shift	Sip fluids, eat planned meals, use shade/breaks when available
After shift	Cool down, drink with food, replace electrolytes conservatively
Before training	Use readiness score and warm-up bar speed

Post-Shift Cooling and Rehydration

Step	Move
Cool	Get out of heat, use cool shower/towel/fan if available
Rehydrate	Fluids plus food; avoid water-only overload
Electrolytes	Use when sweat was high or food was delayed
Sleep	Lower room temperature and reduce light

Heat Changes Training, Conditioning, and Sleep

Area	Adjustment
Heavy lifting	Start yellow after major heat exposure; cap RPE if bar speed is slow
Conditioning	Use heat-adjusted intervals or easy walking
Sleep	Cool down before bed and expect higher recovery need
Accessories	Trim first when fatigue is high

Cooler and Hydration Recovery Setup

Item	Use
Water jug	Main fluid supply
Cooler bottles	Cold fluids through shift
Electrolyte packets	Use when heat/sweat justify them
Salty meal/snack	Supports fluid retention during long shifts
Cooling towel/fan access	Helps post-shift downshift

Copy-Ready Heat Recovery Checklist

- Heat exposure logged.
- Water jug and cooler bottles packed.
- Electrolyte packets packed.

- Meals include sodium-containing foods.
- Post-shift cooling done before training or sleep.
- Training starts yellow after major heat exposure.
- Conditioning adjusted for heat.
- Heat illness symptoms or rapidly worsening symptoms treated as urgent.

Coach Review Notes

- Confirm job-site heat exposure and break access.
- Confirm hydration/electrolyte plan with any blood pressure, kidney, heart, or medication concerns.
- Do not use routine recovery habits as a substitute for urgent medical care.

References

- CDC/NIOSH/OSHA heat-stress guidance for heat exposure, cooling/rest, and urgent symptom escalation.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Soreness and Pain Management

Source: *Recovery/05-soreness-pain-management.md* | Status: review

Purpose and Scope

Soreness is expected during hard training. Pain that changes technique, worsens, radiates, or comes with medical red flags is not treated like normal soreness. This chapter gives tracking and decision rules, not diagnosis.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Soreness vs Pain

Signal	Likely Category	Action
General muscle soreness	Expected soreness	Warm up, move, train if technique is normal
Stiffness that improves with warm-up	Manageable	Start yellow and reassess
Pain that changes movement	Training concern	Stop/modify and seek coach/clinician review
Sharp/worsening/radiating symptoms	Red flag	Review urgently when appropriate

Soreness Rules by Training Source

Source	Rule
Squat	If depth or bracing changes, reduce load/volume
Bench	If shoulder/pec/elbow pain changes bar path, stop that variation
Deadlift	Treat low-back fatigue conservatively; reduce pulls first
Accessories	Cut the movement if it hurts the next main lift
Conditioning	Lower intensity if soreness affects squat/deadlift readiness

High Soreness Plan

Step	Move
Warm-up	Longer general warm-up and ramp sets
Load	Cap RPE lower than planned
Volume	Trim accessories first
Conditioning	Use easy bike/walk
Next week	Repeat week if soreness disrupted multiple sessions

Pain That Changes Technique

Stop the aggravating movement, choose a non-aggravating alternative if available, and document the pattern. Persistent pain, sharp/worsening pain, neurological symptoms, or pain with medical red flags needs coach/clinician review.

What to Track

Field	Entry
Location	Joint/muscle/side
Severity	0-10 and whether it changes movement
Movement affected	Squat, bench, deadlift, accessory, conditioning
Duration	When it started and how long it lasts
Pattern	Better/worse with warm-up, load, sleep, heat, shift stress

Red Flags

Red Flag	Action
Sharp or worsening pain	Coach/clinician review
Numbness/tingling, weakness, radiating symptoms	Clinician review
Chest pain, fainting, severe shortness of breath	Urgent medical attention
Heat illness symptoms	Urgent medical attention
Neurological symptoms or rapidly worsening symptoms	Urgent medical attention

Copy-Ready Soreness/Pain Checklist

- Soreness location logged.
- Severity and movement effect logged.
- Warm-up response checked.
- Technique stayed normal or session was modified.
- Accessories trimmed before main lift quality was compromised.
- Pain that changes movement flagged.
- Red flags treated as urgent when present.

Coach Review Notes

- Confirm how pain/soreness will be logged with training.
- Confirm which movements repeatedly create joint issues.
- Review persistent joint issues, neurological symptoms, or medical red flags with a clinician.

References

- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for conservative pain/symptom escalation.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Weekly Recovery Planning

Source: *Recovery/06-weekly-recovery-planning.md* | Status: review

Purpose and Scope

Weekly recovery planning turns shift schedule, sleep, heat, soreness, and training performance into decisions before fatigue piles up. The goal is to keep the 6-week training block moving without forcing heavy work into bad recovery windows.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Weekly Rhythm

Day	Recovery Focus
Monday	Fuel squat day, hydrate, protect post-training sleep
Tuesday	Keep intervals recoverable
Wednesday	Support bench day; monitor lower-body soreness
Thursday	Protect Friday deadlift with conservative conditioning
Friday	Recover after deadlift; start weekend downshift
Saturday	Sleep, meal prep, easy walk, mobility
Sunday	Review week, plan shift/training changes

Saturday Priorities

Priority	Action
Sleep	Extend sleep or nap if shift week was short
Food prep	Prepare meals for the next training/work block
Mobility	Hips, ankles, T-spine, shoulders, lats
Stress	Reduce chores that can be done earlier

Sunday Priorities

Priority	Action
Review	Sleep, heat, soreness, pain, performance, missed meals
Plan	Place heavy sessions near best sleep windows
Adjust	Decide maintain, trim, move, repeat, or deload
Pack	Cooler, training bag, recovery supplies

Previous Week Review

Question	Decision Help
Did bar speed drop?	Repeat week or trim volume
Did heat exposure stack up?	Lower conditioning and cap RPE
Was sleep poor more than once?	Move heavy session or repeat week
Did pain change technique?	Stop aggravating movement and seek review
Were urgent symptoms present?	Urgent medical attention

Next Week Shift Planning

Shift Pattern	Plan
Day shift	Train after shift only if readiness supports it
Night shift	Prefer training after waking
Rotation change	Use technique day, recovery day, or moved heavy day
High heat forecast	Plan more fluids, electrolytes, and lower conditioning

Recovery Decision Table

Signal	Decision
Sleep and readiness stable	Maintain plan
Soreness high but technique normal	Trim accessories
Conditioning hurts lifting	Lower conditioning
Poor sleep plus slow warm-ups	Move heavy session
Multiple compromised sessions	Repeat week
Performance drop plus joint pain trend	Deload early and seek review

Copy-Ready Weekly Recovery Planner

Item	Notes
Work shifts	
Heat exposure forecast	
Best heavy training windows	
Sleep risk days	
Meal prep plan	
Hydration/electrolyte plan	
Soreness/pain notes	
Conditioning adjustment	
Decision	Maintain / trim / lower conditioning / move heavy / repeat / deload

Coach Review Notes

- Confirm weekly review day and tracking format.
- Confirm which work shifts most often require moved heavy sessions.
- Review red flags conservatively and seek urgent medical care when symptoms warrant.

References

- American Academy of Sleep Medicine and CDC sleep-health guidance for sleep duration, consistency, and fatigue context.
- NIOSH shift-work and long-work-hour guidance for fatigue-aware scheduling.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Recovery Checklists

Source: *Recovery/07-recovery-checklists.md* | Status: review

Purpose and Scope

These checklists consolidate the daily and weekly recovery actions that support the nutrition plan, the 6-week strength block, conditioning, and rotating shift work.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Daily Recovery Checklist

- Main sleep block or nap plan set.
- Protein and carb meals packed or planned.
- Water and electrolytes matched to heat/sweat.
- Caffeine timed away from planned sleep.
- Readiness color checked before training.
- Soreness/pain notes logged.

Heavy Lifting Day Recovery Checklist

- Pre-training meal and fluids handled.
- Full warm-up completed.
- RPE capped by readiness.
- Accessories trimmed if sleep/heat stress is high.
- Post-training meal and fluids completed.
- Sleep window protected after training.

Cardio Day Recovery Checklist

- Intervals selected by readiness.
- Thursday conditioning protects Friday deadlift.
- Heat-adjusted option used when needed.
- Session did not become a max test.
- Hydration and meal timing completed.

Rest Day Checklist

- Easy walk or mobility only if it improves recovery.
- Meal prep completed.
- Sleep debt addressed.
- Heat/soreness/pain notes reviewed.
- Next training day planned.

Bad Sleep Checklist

- Identify cause: shift, caffeine, heat, stress, pain, schedule.
- Start training day yellow.
- Cap top work if bar speed is slow.
- Trim accessories before main lifts.
- Plan nap or earlier wind-down.

Hot Shift Checklist

- Water jug and cooler bottles packed.
- Electrolyte packets packed.
- Salty meal/snack included.
- Post-shift cooling planned.
- Training adjusted if heat exposure was high.
- Heat illness symptoms treated as urgent.

Night Shift Checklist

- Train after waking when possible.
- Bright light after waking.
- Caffeine used early in wake window.
- Post-shift meal, hydration, and light reduction planned.
- Sleep room dark, cool, and quiet.

Pain/Red Flag Checklist

- Pain location and severity logged.
- Movement affected logged.
- Pain changing technique triggers modification.
- Sharp/worsening pain flagged for review.
- Numbness, tingling, weakness, or radiating symptoms flagged.
- Chest pain, fainting, severe shortness of breath, heat illness symptoms, neurological symptoms, or rapidly worsening symptoms treated as urgent.

Weekly Recovery Review Checklist

- Review sleep average and worst sleep days.
- Review heat exposure and hydration issues.
- Review heavy lift performance and bar speed.
- Review conditioning effect on lifting.
- Review soreness and pain patterns.
- Decide: maintain, trim accessories, lower conditioning, move heavy session, repeat week, or deload early.
- Plan meals, cooler, fluids, and sleep windows for next week.

Coach Review Notes

- Confirm which checklist format will be easiest to use weekly.
- Confirm how red flags will be logged and communicated.
- Seek urgent medical attention for urgent symptoms rather than trying to checklist through them.

References

- American Academy of Sleep Medicine and CDC sleep-health guidance for sleep duration, consistency, and fatigue context.
- NIOSH shift-work and long-work-hour guidance for fatigue-aware scheduling.
- CDC/NIOSH/OSHA heat-stress guidance for heat exposure, cooling/rest, and urgent symptom escalation.
- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for conservative pain/symptom escalation.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Volume IV - Tracking

Tracking Overview

Source: *Tracking/01-tracking-overview.md* | Status: review

Purpose and Scope

Tracking turns Nutrition, Training, and Recovery into weekly decisions. The goal is enough information to adjust bodyweight, strength work, conditioning, sleep, hydration, and shift planning without making the system too complicated to use during long oilfield shifts.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Why Tracking Matters

Decision Area	What Tracking Shows
Nutrition	Whether bodyweight and waist are moving toward a lean 275 lb
Training	Whether load, reps, RPE, and technique are progressing
Recovery	Whether sleep, heat, soreness, pain, and shift stress are limiting readiness
Weekly planning	Whether to maintain, add carbs, trim volume, repeat a week, or deload

What to Track Daily

Daily Item	Quick Entry
Bodyweight	Morning scale weight when practical
Sleep	Duration and quality
Shift context	Day, night, off, rotation change
Training/readiness	Green, yellow, red
Heat exposure	Low, moderate, high
Meals/hydration	Hit plan, partial, missed
Soreness/pain	Location and whether it changed movement

What to Track Weekly

Weekly Item	Use
7-day average bodyweight	Main nutrition trend
Waist	Body composition guardrail
Main lift trends	Training progression
Conditioning effect	Whether Tuesday/Thursday is recoverable
Sleep and heat exposure	Recovery decision context
Compliance	Whether the plan was actually followed

What Not to Overtrack

Do not track every possible variable. Skip data that does not change decisions. The dashboard should answer: bodyweight trend, waist trend, training quality, recovery quality, red flags, and next week's adjustment.

Cross-Volume Decision Links

Tracking Signal	Nutrition Decision	Training Decision	Recovery Decision
Weight gain too slow and performance flat	Add carbs	Maintain or progress if ready	Check sleep/heat first
Weight gain fast plus waist jump	Reduce carbs	Do not add extra volume	Review stress and appetite
Bar speed down after hot shifts	Hold food changes	Cap RPE/repeat week	Prioritize hydration/sleep
Pain changes technique	Maintain basics	Modify/stop movement	Coach/clinician review

Weekly Status Overview

Status	Signals	Decision
Green	Weight/waist on track, training progressing, recovery stable	Progress normally
Yellow	One area off: sleep, heat, waist, missed meals, slow bar speed	Adjust one lever
Red	Red flags, repeated missed lifts, worsening pain, heat illness symptoms	Stop chasing progression; seek urgent care when symptoms warrant

Copy-Ready Tracking Overview Table

Metric	Daily	Weekly	Decision
Bodyweight	Morning weight	7-day average	Nutrition adjustment
Waist	No	1x/week	Maintain/reduce/add carbs context
Training	Session log	Trend review	Progress/repeat/trim/deload
Recovery	Sleep/heat/soreness	Pattern review	Move sessions/reduce conditioning
Red flags	Log immediately	Review with coach/clinician	Urgent medical attention if severe

Coach Review Notes

- Confirm the simplest dashboard format the athlete will actually use.
- Confirm weekly review day.
- Treat urgent symptoms as medical issues, not tracking problems.

References

- ACSM and NSCA training-monitoring guidance for logging load, RPE, readiness, and recovery context.
- ISSN and sports nutrition guidance for trend-based bodyweight and nutrition adjustments.
- General bodyweight-trend practice from sports nutrition coaching: use averages and waist trends rather than single-day scale changes.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Bodyweight and Waist Tracking

Source: [Tracking/02-bodyweight-waist.md](#) | Status: review

Purpose and Scope

Bodyweight and waist tracking show whether the lean-gain plan is moving toward a strong 275 lb without drifting too fast at the waist. Use trends, not single-day reactions.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Consistent Weigh-In Method

Step	Standard
Timing	Morning, after bathroom, before food/fluid when practical
Scale	Same scale, same location
Clothing	Same condition each time
Frequency	Daily when possible; minimum 4 weigh-ins/week
Use	Calculate 7-day average bodyweight

Why the 7-Day Average Matters

Single weigh-ins swing from sodium, carbs, sleep, hard training, heat, and hydration. The 7-day average smooths noise and is the number used for nutrition decisions.

Day	Weight
Monday	
Tuesday	
Wednesday	
Thursday	
Friday	
Saturday	
Sunday	
7-day average	

Waist Measurement

Step	Standard
Frequency	1x/week
Timing	Same morning conditions as weigh-in
Location	Same marked spot each week
Method	Relaxed stance, tape level, no aggressive pulling
Use	Compare with bodyweight trend

Interpreting Gain Toward 275 lb

Trend	Meaning
Slow gain plus strong performance	May maintain briefly
Slow gain plus flat performance/recovery	Consider adding carbs
Moderate gain and stable waist	Keep plan
Fast gain plus waist jump	Pull carbs slightly

Starting Adjustment Rules

Weekly Trend	Nutrition Move
Weekly average gain is less than 0.25 lb and performance/recovery are not improving	Add 25-40 g carbs/day
Weekly average gain is 0.5-1.0 lb and waist is stable or only slightly up	Maintain
Weekly average gain is more than 1.0 lb with noticeable waist jump	Reduce 25-40 g carbs/day
Performance is dropping with poor sleep/heat stress	Review recovery before cutting food

Scale Distortion Factors

Factor	What It Can Do
Heat exposure	Drop bodyweight through fluid loss
Poor sleep	Increase water retention or appetite changes
Sodium	Temporarily raise scale weight
Carbs	Increase stored water with glycogen
Hard training	Raise inflammation/soreness-related water

Copy-Ready Bodyweight/Waist Log

Week	7-Day Avg BW	Change	Waist	Change	Performance	Recovery	Decision
1							
2							
3							
4							

Coach Review Notes

- Confirm waist measurement site.
- Confirm whether the weekly target rate should be tightened after four weeks of data.
- Review rapid unexplained changes or urgent symptoms conservatively.

References

- ACSM and NSCA training-monitoring guidance for logging load, RPE, readiness, and recovery context.
- ISSN and sports nutrition guidance for trend-based bodyweight and nutrition adjustments.
- General bodyweight-trend practice from sports nutrition coaching: use averages and waist trends rather than single-day scale changes.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Training Log

Source: *Tracking/03-training-log.md* | Status: review

Purpose and Scope

The training log shows whether the 6-week block is building strength or whether readiness, heat, sleep, soreness, or programming stress is limiting progress. Log enough to make next week's decision.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

What to Log Each Session

Field	Entry
Date	
Shift context	Day / night / off / rotation change / hot shift
Readiness color	Green / yellow / red
Main lift	Squat / bench / deadlift
Sets/ reps/load	Work sets and back-offs
RPE	Top set and notable back-offs
Bar speed/technique	Fast / normal / slow; key technique note
Accessories completed	Yes / partial / skipped
Conditioning completed	Template and mode
Pain/soreness notes	Location, severity, movement affected

Using the Log for Progression

Log Signal	Decision
RPE on target, bar speed good, technique clean	Progress next week
RPE high but reps completed	Repeat load or trim back-offs
Bar speed slow after heat/poor sleep	Use readiness modification before changing program
Accessories hurt main lift recovery	Trim accessories

Missed Reps

Miss Pattern	Next Move
One miss from bad load choice	Repeat or reduce next week
Multiple misses	Reduce load and review readiness
Miss after poor sleep/hot shift	Treat as context, not automatic program failure
Pain changed technique	Stop/modify and seek coach/clinician review

Poor Sleep or Hot Shifts

Log sleep/heat context beside the session. If a bad session follows poor sleep, long shift, or heat exposure, consider repeating the week, moving a heavy session, or trimming accessories before cutting nutrition.

Copy-Ready Training Log Table

Date	Shift/Heat	Readiness	Lift	Sets/Reps/Load	RPE	Bar Speed/Technique	Accessories	Conditioning	Pain/Soreness	Decision
------	------------	-----------	------	----------------	-----	---------------------	-------------	--------------	---------------	----------

Coach Review Notes

- Confirm how RPE and bar speed will be recorded.
- Review repeated misses, pain that changes technique, or urgent symptoms conservatively.
- Do not use the training log as a substitute for medical care.

References

- ACSM and NSCA training-monitoring guidance for logging load, RPE, readiness, and recovery context.

- ISSN and sports nutrition guidance for trend-based bodyweight and nutrition adjustments.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Recovery Log

Source: *Tracking/04-recovery-log.md* | Status: review

Purpose and Scope

The recovery log explains why training and nutrition numbers move. It connects sleep, shift work, heat exposure, hydration, soreness, pain, stress, appetite, and digestion to daily readiness and weekly decisions.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

What to Log Daily

Recovery Field	Entry
Sleep duration/quality	Hours plus good / okay / poor
Shift type	Day / night / off / rotation change
Heat exposure	Low / moderate / high
Hydration/electrolytes	Hit / partial / missed
Soreness	Low / moderate / high
Pain/red flags	Location, movement affected, urgent symptoms
Stress level	Low / moderate / high
Appetite/digestion	Normal / low / high / upset

Before Training

Recovery Signal	Readiness Decision
Sleep good, heat low, soreness normal	Green possible
One factor off	Yellow start
Multiple factors off	Yellow/red; cap RPE and trim volume
Red-flag symptoms	Stop and seek urgent medical attention when warranted

Weekly Review Use

Pattern	Decision
Poor sleep before missed lifts	Move heavy sessions or repeat week
High heat before weight drop	Review hydration before nutrition cut
Soreness high after accessories	Trim accessory volume
Appetite poor plus weight flat	Review meal timing and recovery

Copy-Ready Recovery Log Table

Date	Sleep	Shift	Heat	Hydration/Electrolytes	Soreness	Pain/Red Flags	Stress	Appetite/Digestion	Readiness
------	-------	-------	------	------------------------	----------	----------------	--------	--------------------	-----------

Coach Review Notes

- Confirm daily recovery fields are quick enough to use.
- Review red-flag symptoms urgently rather than waiting for weekly review.
- Use recovery trends to adjust training before assuming motivation is the issue.

References

- ACSM and NSCA training-monitoring guidance for logging load, RPE, readiness, and recovery context.
- ISSN and sports nutrition guidance for trend-based bodyweight and nutrition adjustments.
- AASM/CDC sleep guidance and CDC/NIOSH heat-stress resources for recovery-log context.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.

- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Weekly Review

Source: *Tracking/05-weekly-review.md* | Status: review

Purpose and Scope

The weekly review turns daily logs into decisions for Nutrition, Training, and Recovery. Review once per week, ideally Sunday, before the next shift/training week starts.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Weekly Review Rhythm

Step	Action
1	Calculate 7-day bodyweight average
2	Record waist
3	Review main lift trends and conditioning effect
4	Review sleep, heat, shift stress, soreness, pain
5	Check nutrition compliance
6	Choose next week's nutrition/training/recovery decisions

What to Review

Area	Questions
7-day bodyweight average	Up, down, or flat? By how much?
Waist	Stable, slightly up, noticeable jump?
Training performance	Are top sets cleaner, heavier, or slower?
Main lift trends	Squat, bench, deadlift moving as expected?
Conditioning effect	Did Tuesday/Thursday hurt lifting readiness?
Sleep	How many poor nights?
Heat exposure	Did hot shifts distort weight or readiness?
Shift stress	Did work force missed meals or moved sessions?
Soreness/pain	Normal soreness or pain changing movement?
Nutrition compliance	Meals, carbs, protein, hydration hit?

Decision Rules

Volume	Green Decision	Yellow Decision	Red Decision
Nutrition	Maintain	Add or reduce 25-40 g carbs/day based on trend	Hold major changes if red flags/recovery chaos dominate
Training	Progress	Repeat week, trim accessories, or cap RPE	Deload/modify; stop for urgent red flags
Recovery	Maintain	Adjust sleep, move sessions, reduce conditioning	Seek urgent care for red flags; reduce stress load

Copy-Ready Weekly Review Template

Review Item	Entry
Week dates	
7-day average bodyweight	
Bodyweight change	
Waist	
Squat trend	
Bench trend	
Deadlift trend	
Conditioning effect	
Sleep summary	
Heat exposure	
Shift stress	
Soreness/pain	
Nutrition compliance	
Nutrition decision	Maintain / add carbs / reduce carbs
Training decision	Progress / repeat / trim / deload
Recovery decision	Maintain / adjust sleep / move sessions / reduce conditioning

Coach Review Notes

- Confirm whether review happens Sunday or after the final work shift.
- Review red flags urgently instead of waiting for trend data.
- Use one main decision per category to keep the system usable.

References

- ACSM and NSCA training-monitoring guidance for logging load, RPE, readiness, and recovery context.
- ISSN and sports nutrition guidance for trend-based bodyweight and nutrition adjustments.
- General bodyweight-trend practice from sports nutrition coaching: use averages and waist trends rather than single-day scale changes.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Dashboard Template

Source: [Tracking/06-dashboard-template.md](#) | Status: review

Purpose and Scope

The dashboard is the one-page weekly control panel for Nutrition, Training, and Recovery. It should be copied, filled out, reviewed, and used to plan the next week.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Weekly Dashboard Layout

Section	Purpose
Daily check-in	Capture the week without overthinking
Weekly metrics	Bodyweight, waist, performance, recovery
Nutrition decision	Maintain/add/reduce carbs
Training decision	Progress/repeat/trim/deload
Recovery decision	Sleep/heat/shift adjustments
Next-week plan	One clear plan before Monday

Daily Check-In Table

Day	BW	Sleep	Shift/Heat	Training/Cardio	Meals/Hydration	Soreness/Pain	Readiness
Mon				Squat			
Tue				Intervals			
Wed				Bench			
Thu				Intervals			
Fri				Deadlift			
Sat				Recovery			
Sun				Recovery/light			

Weekly Metrics Table

Metric	Entry	Decision Signal
7-day average bodyweight		Nutrition trend
Change from last week		Add/maintain/reduce carbs
Waist		Body composition guardrail
Best training performance		Progression signal
Worst readiness day		Recovery planning
Heat exposure count		Hydration/recovery adjustment
Missed meals		Compliance before macro changes
Red flags		Urgent medical attention if present

Nutrition Decision

Signal	Decision
Gain <0.25 lb/week and performance/recovery not improving	Add 25-40 g carbs/day
Gain 0.5-1.0 lb/week and waist stable/slightly up	Maintain
Gain >1.0 lb/week with noticeable waist jump	Reduce 25-40 g carbs/day
Heat/sleep distorted scale	Review recovery before changing food

Training Decision

Signal	Decision
Bar speed and RPE on target	Progress
One rough session from shift/heat	Repeat or hold
Accessories hurting main lifts	Trim
Multiple poor sessions	Repeat week or deload

Recovery Decision

Signal	Decision
Sleep stable, heat managed	Maintain
Poor sleep	Adjust caffeine, light, nap, training placement
High heat	Increase recovery focus and lower conditioning
Pain/red flags	Modify training; urgent care for urgent symptoms

Next-Week Plan

Planning Item	Entry
Work shifts	
Best heavy training windows	
Nutrition change	
Training change	
Recovery focus	
Conditioning adjustment	
Coach questions	

Copy-Ready Full Dashboard Template

Category	This Week	Next Week Decision
7-day average bodyweight		
Waist		
Nutrition compliance		
Squat/bench/deadlift trend		
Conditioning effect		
Sleep		
Heat/shift stress		
Soreness/pain/red flags		
Nutrition decision		Maintain / add carbs / reduce carbs
Training decision		Progress / repeat / trim / deload
Recovery decision		Maintain / adjust sleep / move sessions / reduce conditioning

Coach Review Notes

- Confirm dashboard fits on one page when exported.
- Confirm whether coach review happens weekly or every block.
- Do not delay urgent medical attention for red-flag symptoms.

References

- ACSM and NSCA training-monitoring guidance for logging load, RPE, readiness, and recovery context.
- ISSN and sports nutrition guidance for trend-based bodyweight and nutrition adjustments.
- General bodyweight-trend practice from sports nutrition coaching: use averages and waist trends rather than single-day scale changes.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.

- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Exercise Library

Library Overview

Source: *Exercise-Library/01-library-overview.md* | Status: review

Purpose and Scope

The exercise library gives practical options for running the 6-week strength block when equipment, readiness, soreness, pain flags, time, or shift fatigue changes the plan. Substitutions should keep the training intent intact instead of turning a planned session into random work.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

How to Use the Library

- Start with the programmed movement and its intent.
- Check equipment, readiness, time, and pain/symptom notes.
- Choose the closest option that trains the same pattern with a similar fatigue cost.
- Keep RPE and volume inside the plan for the day.
- Log the substitution and the reason in the training log.

Movement Categories

Category	Primary Intent	Examples
Squat pattern	Knee-dominant strength and position	Squat, paused squat, leg press, split squat
Bench/press pattern	Horizontal press strength	Bench, close-grip bench, dumbbell press, push-up
Deadlift/hinge pattern	Hip hinge strength and posterior chain	Deadlift, Romanian deadlift, block pull, hip thrust
Row/pull pattern	Upper-back and lat support	Row, pulldown, chin-up, cable row
Single-leg pattern	Leg strength with lower axial load	Split squat, lunge, step-up
Core/bracing	Trunk control under load	Dead bug, plank, Pallof press, carries
Conditioning	Work capacity without wrecking lifting	Bike, incline walk, sled, air bike

Preserve Training Intent

Programmed Intent	Good Substitution	Poor Substitution
Heavy squat pattern	Safety-bar squat or high-bar squat at planned RPE	Max-effort leg press
Light technique bench	Paused bench with lighter load	Heavy close-grip grinder
Low-back friendly hinge	Hip thrust or light Romanian deadlift	Heavy rack pull
Low-impact conditioning	Bike or incline walk	Sprints on low readiness

Equipment-Based Logic

Missing Item	First Choice	Second Choice
Squat rack	Leg press or hack squat	Split squat
Bench station	Dumbbell press	Machine press or push-up
Deadlift platform	Trap-bar deadlift	Romanian deadlift or hip thrust
Cable stack	Dumbbell row or band row	Machine row
Sled	Bike	Incline walk

Readiness-Based Logic

Readiness	Substitution Rule
Green	Use the programmed lift or closest full-intent variation
Yellow	Keep pattern, reduce load/sets, choose lower skill or lower impact
Red	Do not chase progression; use recovery work or stop when symptoms warrant

Pain and Symptom Red Flags

Pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms should be reviewed with a qualified coach and clinician. Do not force a listed variation if it creates or worsens pain.

Copy-Ready Exercise Selection Checklist

Question	Answer
What is the programmed movement?	
What is today's intent? Strength / technique / volume / conditioning / recovery	
What equipment is unavailable?	
What is readiness? Green / yellow / red	
Any pain or red-flag symptoms?	
Best substitution that preserves pattern and intent	
Load/volume adjustment	
Log note	

Coach Review Notes

- Confirm substitutions match the athlete's technique, equipment, and recovery profile.
- Review pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms with a qualified coach and clinician.
- Do not treat this library as a medical diagnosis or treatment plan.

References

- NSCA strength-training references for movement patterns, assistance work, and fatigue-cost management.
- ACSM resistance-training guidance for exercise selection, progression, and conservative modification.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Squat Variations

Source: *Exercise-Library/02-squat-variations.md* | Status: review

Purpose and Scope

Squat variations support the Monday squat emphasis by preserving the knee-dominant strength intent while adjusting for equipment, readiness, fatigue, and tolerance. The primary squat remains the competition-style or main programmed squat unless the day requires a closer-fit variation.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Main Squat Pattern

Use the primary squat when equipment is available, readiness is green or solid yellow, technique is repeatable, and the planned RPE can be respected. Do not turn a planned technique or volume day into a max test.

Variations and Use Cases

Variation	What It Trains	When to Use	When to Avoid
Paused squat	Bottom position, control, staying tight	Technique work, confidence in depth, lighter strength work	If pauses cause pain or turn every rep into a grinder
Tempo squat	Position, bracing, bar path, patience	Lighter loading, technique correction, reduced load with high intent	If fatigue makes positions degrade
High-bar squat	Quads, upright torso, depth	More quad emphasis, lower load than low-bar, variation block	If it aggravates knees/hips or cannot be braced well
Safety-bar squat	Upper-back position, quads, bracing	If available; useful when shoulder position limits straight bar	If it overloads back fatigue or changes technique painfully
Box squat	Consistent depth, controlled eccentric, confidence	Limited range target, technique reset, equipment-supported variation	If bouncing, relaxing on box, or pain appears
Front squat	Quads, upright torso, bracing	Lighter squat pattern with strong trunk demand	If wrist/shoulder position or breathing is limiting
Leg press	Quad volume with lower skill demand	Rack unavailable, time-limited accessory replacement	If it replaces heavy squat intent too often
Hack squat	Quad volume, fixed path	Machine available, controlled quad work	If knee/hip position feels poor
Split squat	Single-leg strength, stability, quad/glute work	Low equipment, lower axial load, accessory slot	If balance limits loading or pain changes mechanics

Substitution Priorities

Need	Best Fit
Preserve heavy squat strength	Primary squat, safety-bar squat, high-bar squat
Preserve technique with less load	Paused squat, tempo squat
Reduce setup demand	Leg press, hack squat
Reduce axial load	Split squat, leg press
Limited time	Primary squat top work plus one simple quad movement

Copy-Ready Squat Substitution Table

Programmed Squat Intent	Preferred Substitution	Load/Volume Guardrail	Notes
Heavy squat	Safety-bar squat or high-bar squat	Same RPE cap, fewer sets if needed	Keep bracing and depth consistent
Technique squat	Paused squat or tempo squat	Lighter load, crisp reps	No grinders
Quad volume	Leg press or hack squat	Moderate reps, controlled tempo	Do not chase machine PRs
Low equipment	Split squat	Moderate reps each side	Keep torso and knee path consistent
Yellow readiness	Paused squat, tempo squat, or leg press	Reduce load or sets	Preserve movement quality

Coach Review Notes

- Review pain that changes squat depth, stance, bracing, or bar path with a qualified coach and clinician.
- Persistent joint issues, neurological symptoms, or rapidly worsening symptoms should not be managed by swapping variations alone.
- Choose the variation that preserves the training intent with the least unnecessary fatigue.

References

- NSCA strength-training references for movement patterns, assistance work, and fatigue-cost management.
- ACSM resistance-training guidance for exercise selection, progression, and conservative modification.
- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for pain that changes movement.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Bench Variations

Source: *Exercise-Library/03-bench-variations.md* | Status: review

Purpose and Scope

Bench variations support the Wednesday bench emphasis by preserving horizontal pressing strength while adjusting for equipment, shoulder/elbow tolerance, readiness, and time. The main bench pattern is the competition-style or primary programmed bench.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Main Bench Pattern

Use the primary bench when the bench station is available, setup is stable, technique is repeatable, and the planned RPE can be respected. Keep substitutions close to the planned intent: strength, technique, volume, or accessory pressing.

Variations and Use Cases

Variation	What It Trains	When to Use	When to Avoid
Paused bench	Chest position, control, competition skill	Technique work, clean strength practice	If pauses create shoulder pain or grind every rep
Close-grip bench	Triceps, lockout, upper-back tightness	Supplemental pressing, shoulder-friendly option for some lifters	If elbows/wrists become irritated
Spoto press	Control just off chest, tightness	Technique work without full chest contact	If position is inconsistent or painful
Incline press	Upper chest, shoulders, triceps	Pressing volume away from flat bench	If shoulder position feels poor
Dumbbell press	Range control, left/right balance	Bench station unavailable, volume work	If stability limits safe loading
Floor press	Lockout, reduced range	Shoulder-friendly range option for some lifters, no bench available	If it changes intent from volume to ego loading
Machine press	Simple pressing volume	Time-limited or high-fatigue day	If machine path causes joint irritation
Push-up variation	Low equipment, pressing volume	Travel, warm-up, light volume, recovery day	If wrists/shoulders object or load is too low for intent

Substitution Priorities

Need	Best Fit
Preserve primary bench skill	Paused bench or close-grip bench
Reduce shoulder range demand	Floor press or machine press if tolerated
No bench station	Dumbbell press, floor press, push-up
More triceps emphasis	Close-grip bench, floor press
Yellow readiness	Paused bench light, machine press, push-up

Copy-Ready Bench Substitution Table

Programmed Bench Intent	Preferred Substitution	Load/Volume Guardrail	Notes
Heavy bench	Paused bench or close-grip bench	Same RPE cap; reduce load as needed	Keep setup tight
Technique bench	Paused bench or Spoto press	Crisp reps only	No forced reps
Pressing volume	Dumbbell press or machine press	Moderate reps	Stop before form breaks
Bench unavailable	Dumbbell press, floor press, push-up	Match effort, not load	Log equipment reason
Yellow readiness	Machine press or lighter paused bench	Trim sets first	Preserve bar speed

Coach Review Notes

- Review pain that changes bench setup, grip, bar path, or lockout with a qualified coach and clinician.
- Neurological symptoms, persistent joint issues, or rapidly worsening symptoms require more than a substitution decision.
- Preserve the press intent without turning technique work into max-effort pressing.

References

- NSCA strength-training references for movement patterns, assistance work, and fatigue-cost management.
- ACSM resistance-training guidance for exercise selection, progression, and conservative modification.

- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for pain that changes movement.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Deadlift Variations

Source: [Exercise-Library/04-deadlift-variations.md](#) | Status: review

Purpose and Scope

Deadlift variations support the Friday deadlift emphasis by preserving hinge strength while managing fatigue, equipment, range of motion, and readiness. The main pattern is the athlete's primary deadlift stance unless the day calls for a closer-fit option.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Main Deadlift Pattern

Use the primary deadlift stance when setup, bracing, bar path, and speed match the plan. Do not use substitutions as a reason to test max strength on a poor-readiness day.

Variations and Use Cases

Variation	What It Trains	When to Use	When to Avoid
Conventional deadlift	Full hinge strength, floor position	Primary stance if programmed/tolerated	If back fatigue changes position
Sumo deadlift	Hip position, shorter pull for some lifters	If used and tolerated in training	If stance causes hip or knee irritation
Paused deadlift	Position off floor, patience, lats	Technique work and lighter strength practice	If pauses break position
Tempo deadlift	Control, bracing, bar path	Lighter day, technical reset	If fatigue causes rounding or hitching
Romanian deadlift	Hamstrings, hinge pattern, eccentric control	Supplemental hinge, less floor setup demand	If low-back fatigue dominates
Block pull	Position from set height, overload with control	Range-specific work, floor setup limited	If it becomes a heavy ego lift
Rack pull	Lockout exposure with conservative loading	Rare use; controlled range and clear purpose	If loading jumps beyond planned fatigue
Trap-bar deadlift	Hinge/leg drive, simpler setup	Equipment available, reduced skill demand	If it no longer matches deadlift intent
Hip thrust	Glutes, hip extension with less spinal loading	Posterior-chain work when hinge fatigue is high	If setup irritates hips/back
Back extension	Hinge endurance, glutes/hamstrings	Light accessory or warm-up if tolerated	If symptoms appear or range is forced

Lower-Back Fatigue Management

If lower-back fatigue changes bracing, start position, or bar path, reduce load, reduce sets, move to a lower-fatigue hinge, or end the hinge slot. Do not stack heavy pulls, heavy rows, and high-impact conditioning when readiness is already low.

Copy-Ready Deadlift Substitution Table

Programmed Deadlift Intent	Preferred Substitution	Load/Volume Guardrail	Notes
Heavy deadlift	Primary stance, trap-bar, or block pull	Same RPE cap, no max testing	Keep setup repeatable
Technique pull	Paused deadlift or tempo deadlift	Lighter load	Stop if position breaks
Posterior-chain volume	Romanian deadlift or hip thrust	Moderate reps	Keep fatigue recoverable
Low-back fatigue high	Hip thrust, light back extension, or skip hinge volume	Reduce axial fatigue	Preserve next-week training
Platform unavailable	Trap-bar deadlift, Romanian deadlift, hip thrust	Match intent, not load	Log equipment reason

Coach Review Notes

- Review pain that changes bracing, start position, hip position, or lockout with a qualified coach and clinician.
- Neurological symptoms, persistent joint issues, or rapidly worsening symptoms require conservative review.
- Choose hinge substitutions that preserve intent without adding reckless fatigue.

References

- NSCA strength-training references for movement patterns, assistance work, and fatigue-cost management.
- ACSM resistance-training guidance for exercise selection, progression, and conservative modification.
- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for pain that changes movement.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Accessories

Source: *Exercise-Library/05-accessories.md* | Status: review

Purpose and Scope

Accessories build the muscle, positions, and weak links that support squat, bench, and deadlift performance. They should support the main lifts without stealing recovery from the next heavy session.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Accessory Categories

Category	Purpose	Exercise Menu
Quads	Squat support and leg drive	Leg press, hack squat, split squat, lunge, step-up, leg extension
Hamstrings	Hinge support and knee flexion	Romanian deadlift, hamstring curl, glute-ham raise, back extension
Glutes	Hip extension and squat/deadlift support	Hip thrust, cable pull-through, glute bridge, split squat
Upper back/lats	Bench stability and deadlift position	Chest-supported row, cable row, pulldown, chin-up, dumbbell row
Triceps	Bench lockout	Close-grip bench, dip if tolerated, cable pressdown, skull crusher, band pressdown
Rear delts/shoulders	Shoulder balance and upper-back support	Face pull, rear-delt fly, lateral raise, machine shoulder press
Arms	Elbow-friendly volume and balance	Curl, hammer curl, cable curl, rope pressdown
Carries/grip	Bracing, grip, trunk control	Farmer carry, suitcase carry, trap-bar carry, timed holds

Equipment Substitutions

Equipment Available	Best Accessory Choices
Barbell	Romanian deadlift, close-grip bench, row, hip thrust
Dumbbell	Split squat, row, press, lateral raise, curl, carry
Cable	Row, pulldown, pressdown, curl, Pallof press, pull-through
Machine	Leg press, hack squat, chest-supported row, machine press, hamstring curl
Bands	Pressdown, row, face pull, pull-apart, Pallof press
Bodyweight	Push-up, split squat, lunge, plank, side plank, chin-up if available

Set and Rep Intent

Accessory Intent	Typical Target	Guardrail
Strength support	3-5 sets of 5-8	Stop before form breaks
Muscle volume	2-4 sets of 8-15	Leave 1-3 reps in reserve
Joint-friendly pump	2-4 sets of 12-25	Smooth reps, low joint stress
Grip/carry	Short timed or distance sets	No sloppy posture

Readiness-Based Trimming

Readiness	Accessory Move
Green	Complete planned accessories
Yellow	Keep the highest-value accessory; trim 1-2 sets
Red	Skip accessories or choose light recovery work
Hot shift or poor sleep	Avoid adding new exercises or chasing accessory PRs

Copy-Ready Accessory Menu

Need	First Choice	Substitution	Notes
Quad volume	Leg press	Split squat or hack squat	Keep knees tracking well
Hamstring volume	Hamstring curl	Romanian deadlift or back extension	Manage low-back fatigue
Upper-back support	Chest-supported row	Cable row or dumbbell row	Do not turn into a lower-back lift
Triceps	Cable pressdown	Close-grip bench or band pressdown	Keep elbows calm
Rear delts	Face pull	Rear-delt fly or band pull-apart	Controlled reps
Grip/bracing	Farmer carry	Suitcase carry or timed hold	Stop if posture collapses

Coach Review Notes

- Review pain that changes exercise mechanics with a qualified coach and clinician.
- Persistent joint issues, neurological symptoms, or rapidly worsening symptoms should not be ignored for accessory volume.
- Pick accessories for their job, not novelty.

References

- NSCA strength-training references for movement patterns, assistance work, and fatigue-cost management.
- ACSM resistance-training guidance for exercise selection, progression, and conservative modification.
- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for pain that changes movement.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Core and Conditioning

Source: *Exercise-Library/06-core-conditioning.md* | Status: review

Purpose and Scope

Core and conditioning work should improve bracing, work capacity, and recovery without interfering with the Monday/Wednesday/Friday heavy lifting structure. Choose options that match readiness, joint tolerance, heat exposure, and the next lifting day.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Core and Bracing Options

Exercise	What It Trains	Use Case	Avoid/Modify When
Dead bug	Bracing with limb movement	Warm-up, low-fatigue trunk work	Low back arches or symptoms appear
Plank	Basic trunk endurance	Short bracing sets	Shoulder/back position breaks
Side plank	Lateral trunk control	Warm-up or accessory core	Shoulder or hip discomfort changes form
Pallof press	Anti-rotation	Cable/band core work	Setup causes twisting or pain
Cable chop/lift	Controlled rotation	Athletic trunk control	Load pulls position apart
Carries	Bracing under load	Grip, trunk, posture	Posture collapses or grip fails early
Hanging knee raise	Anterior core and hip flexion	If tolerated and controlled	Shoulder, grip, or hip symptoms appear

Conditioning Options

Option	Best Use	Fatigue Cost
Bike	Intervals or easy conditioning	Low impact, controllable
Incline walk	Recovery conditioning, hot-shift weeks	Low impact, easy to dose
Sled push/pull	Work capacity with low eccentric stress	Moderate, leg-heavy
Row/air bike	Intervals if tolerated	Moderate to high
Sprinting	Rare high-readiness option	High impact and high recovery cost

Choose Low-Impact Conditioning When

Situation	Better Choice
Legs are sore from heavy squats or deadlifts	Bike or incline walk
Readiness is yellow	Bike, incline walk, or easy sled
Heat exposure was high	Shorter low-impact session or recovery walk
Joint irritation is present	Low-impact option that does not worsen symptoms

Avoid High-Impact Conditioning When

Avoid sprinting or aggressive intervals when readiness is low, joints are irritated, technique is altered, sleep is poor, heat exposure is high, or the next heavy lift would likely suffer. High-impact work should not replace low-impact conditioning on a poor-readiness day.

Heat and Readiness Adjustment

After hot shifts, reduce conditioning duration, lower intensity, move conditioning away from heavy lifting, or choose recovery movement. Do not use conditioning to "make up" for missed training when red-flag symptoms are present.

Copy-Ready Core/Conditioning Substitution Table

Programmed Work	Preferred Option	Substitution	Guardrail
Core warm-up	Dead bug	Pallof press	Smooth, controlled reps
Bracing finisher	Carry	Plank or side plank	Stop before posture breaks
Intervals	Bike	Air bike or row if tolerated	Keep recovery for heavy lifts
Easy conditioning	Incline walk	Bike	Conversational effort
Sled unavailable	Bike	Incline walk	Match effort, not exact minutes
Low readiness	Incline walk	Easy bike	No sprints

Coach Review Notes

- Review pain that changes bracing, gait, impact tolerance, or breathing strategy with a qualified coach and clinician.
- Chest pain, fainting, severe shortness of breath, neurological symptoms, heat illness symptoms, or rapidly worsening symptoms require urgent medical attention.
- Conditioning choices should support the training week, not compete with it.

References

- NSCA strength-training references for movement patterns, assistance work, and fatigue-cost management.
- ACSM resistance-training guidance for exercise selection, progression, and conservative modification.
- ACSM conditioning guidance and concurrent-training literature for low-impact conditioning choices.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Substitution Rules

Source: *Exercise-Library/07-substitution-rules.md* | Status: review

Purpose and Scope

Substitution rules keep the program usable when the exact plan is not available. The goal is not variety for its own sake. The goal is to preserve the training intent with the best available option and a recoverable fatigue cost.

Heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Substitution Hierarchy

Priority	Rule	Example
1	Same movement pattern	Squat to safety-bar squat
2	Same training intent	Technique bench to paused dumbbell press, not heavy grinder
3	Similar fatigue cost	Low-impact bike to incline walk, not sprints
4	Similar equipment setup if possible	Cable row to machine row before a complex barbell row

What Not to Substitute

Do Not	Why
Replace a light technique movement with a max-effort movement	It changes intent and fatigue
Replace low-impact conditioning with sprints when readiness is low	It adds impact and recovery cost
Replace pain-free movement with a painful variation just because it is listed	The list is optional, not mandatory
Add extra accessories because the main lift was trimmed	It can hide the same fatigue problem

Equipment-Limited Rules

Limitation	Rule
Rack unavailable	Use closest squat pattern available; keep RPE plan
Bench unavailable	Use dumbbell, machine, floor press, or push-up variation
Platform unavailable	Use trap-bar, Romanian deadlift, hip thrust, or back extension if tolerated
Cable unavailable	Use dumbbell, machine, band, or bodyweight pull/press option

Time-Limited Rules

Time Available	Decision
45-60 minutes	Main lift, one back-off, one accessory
30-45 minutes	Main lift priority plus one simple accessory
Under 30 minutes	Warm up, perform the highest-priority work, log what was skipped
Post-shift exhausted	Reduce setup complexity and preserve technique

Readiness-Limited Rules

Readiness	Substitution Decision
Green	Keep planned lift or close variation
Yellow	Keep pattern, reduce load/sets, choose lower-impact options
Red	Stop, recover, or choose very light movement if appropriate
Heat/sleep compromised	Reduce conditioning intensity and avoid adding heavy variations

Pain and Symptom Rules

Pain that changes technique, persistent joint issues, neurological symptoms, or rapidly worsening symptoms should be reviewed with a qualified coach and clinician. Do not self-select a harder variation to train through symptoms. Chest pain, fainting, severe shortness of breath, heat illness symptoms, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Examples

Scenario	Preserve Intent With	Avoid
Squat variation missing	Primary squat, safety-bar squat, high-bar squat, or leg press depending on intent	Random max-effort machine work
Bench station unavailable	Dumbbell press, floor press, machine press, push-up	Heavy dips if not planned/tolerated
Deadlift fatigue high	Hip thrust, light Romanian deadlift, or skip hinge accessory	Heavy rack pulls
Cable machine unavailable	Dumbbell row, band row, machine row, band pressdown	Adding unrelated volume
Hot shift before conditioning	Incline walk, easy bike, shorter sled	Sprints or extra intervals

Copy-Ready Substitution Decision Checklist

Step	Check	Entry
1	Programmed movement and intent	
2	Reason for substitution	Equipment / time / readiness / pain / shift fatigue
3	Same movement pattern chosen?	
4	Same intent preserved?	Strength / technique / volume / conditioning / recovery
5	Fatigue cost similar or lower?	
6	Any pain or red-flag symptoms?	
7	Load/sets/ reps adjusted?	
8	Logged for weekly review?	

Coach Review Notes

- Review repeated substitutions that appear for the same movement pattern.
- Review pain that changes technique, neurological symptoms, persistent joint issues, or rapidly worsening symptoms with a qualified coach and clinician.
- Substitutions should make the plan easier to execute, not easier to ignore.

References

- NSCA strength-training references for movement patterns, assistance work, and fatigue-cost management.
- ACSM resistance-training guidance for exercise selection, progression, and conservative modification.
- NIH MedlinePlus, Mayo Clinic, and Cleveland Clinic general red-flag resources for pain that changes movement.

Final-Review Notes

- Confirm exact editions, dates, and links during final layout.
- Keep this chapter in review until coaching and safety review are complete.
- Keep safety language conservative; these sources support guidance context, not individualized medical care.

Back Matter

The Promise

Source: Generated review placeholder | Status: review

Review placeholder. Approved promise copy is not present in the repository.

Review placeholder - final copy needed.

Approved closing promise copy is not present yet.

Leave this section as a placeholder until the author supplies final wording.

This placeholder is intentionally preserved for review and does not mark the manuscript final.

About the Author

Source: Generated review placeholder | Status: review

Review placeholder. Approved author bio is not present in the repository.

Review placeholder - final copy needed.

Approved author biography copy is not present yet.

Leave this section as a placeholder until the author supplies final wording.

This placeholder is intentionally preserved for review and does not mark the manuscript final.

References

Source: REFERENCES.md | Status: review

This consolidated list supports the Version 1.0 review manuscript. It is intentionally concise and should remain in review until exact edition, URL, and citation formatting are approved for final publication.

Training, Programming, And Exercise Selection

Used in Implementation, Training, and Tracking. Standard exercise-prescription reference; confirm edition for final layout.

- American College of Sports Medicine. ACSM's Guidelines for Exercise Testing and Prescription. Wolters Kluwer, 2021. <https://www.acsm.org/education-resources/books/guidelines-exercise-testing-prescription>

Used in Training, Exercise Library, and Tracking. Supports conservative progression, volume, intensity, and exercise selection context.

- American College of Sports Medicine. Progression Models in Resistance Training for Healthy Adults. Medicine & Science in Sports & Exercise, 2009. <https://doi.org/10.1249/MSS.0b013e3181915670>

Used in Training and Exercise Library. Standard strength and conditioning text; confirm edition for final layout.

- Haff GG, Triplett NT, editors. Essentials of Strength Training and Conditioning, 4th ed. National Strength and Conditioning Association / Human Kinetics, 2016. <https://us.humankinetics.com/products/essentials-of-strength-training-and-conditioning-4th-edition-with-hkpropel-access>

Nutrition, Hydration, And Supplements

Used in Nutrition, Implementation, and Tracking. Joint Academy of Nutrition and Dietetics, Dietitians of Canada, and ACSM position paper.

- Thomas DT, Erdman KA, Burke LM. Nutrition and Athletic Performance. Medicine & Science in Sports & Exercise, 2016. <https://doi.org/10.1249/MSS.0000000000000852>

Used in Nutrition and Tracking. Supports protein-distribution and athlete fueling context.

- Jager R, Kerksick CM, Campbell BI, et al. International Society of Sports Nutrition Position Stand: Protein and Exercise. Journal of the International Society of Sports Nutrition, 2017. <https://doi.org/10.1186/s12970-017-0177-8>

Used in Nutrition and Implementation. Supports practical protein/carbohydrate timing context without requiring rigid meal timing.

- Kerksick CM, Arent S, Schoenfeld BJ, et al. International Society of Sports Nutrition Position Stand: Nutrient Timing. Journal of the International Society of Sports Nutrition, 2017. <https://doi.org/10.1186/s12970-017-0189-4>

Used in Nutrition and Supplements. Supports conservative creatine discussion; does not replace clinician review.

- Kreider RB, Kalman DS, Antonio J, et al. International Society of Sports Nutrition Position Stand: Safety and Efficacy of Creatine Supplementation in Exercise, Sport, and Medicine. Journal of the International Society of Sports Nutrition, 2017. <https://doi.org/10.1186/s12970-017-0173-z>

Used in Nutrition, Supplements, and Shift Work. Supports caffeine context and caution around sleep and individual tolerance.

- Guest NS, VanDusseldorp TA, Nelson MT, et al. International Society of Sports Nutrition Position Stand: Caffeine and Exercise Performance. Journal of the International Society of Sports Nutrition, 2021. <https://doi.org/10.1186/s12970-020-00383-4>

Used in Nutrition, Hydration, and Recovery. Supports exercise hydration context; final copy should avoid individualized medical prescriptions.

- Sawka MN, Burke LM, Eichner ER, et al. American College of Sports Medicine Position Stand: Exercise and Fluid Replacement. Medicine & Science in Sports & Exercise, 2007. <https://doi.org/10.1249/mss.0b013e31802ca597>

Heat, Shift Work, Sleep, And Recovery

Used in Nutrition, Recovery, and Implementation. Current workplace heat-stress guidance for hydration, rest, acclimatization, training, and escalation context.

- National Institute for Occupational Safety and Health. Workplace Recommendations: Heat Stress. Centers for Disease Control and Prevention, 2026. <https://www.cdc.gov/niosh/heat-stress/recommendations/index.html>

Used in Recovery and Implementation. OSHA heat-exposure overview for planning, prevention, and worker-safety context.

- Occupational Safety and Health Administration. Heat: Working in Outdoor and Indoor Heat Environments. U.S. Department of Labor, 2026. <https://www.osha.gov/heat-exposure>

Used in Recovery, Shift Work, and Tracking. Supports adult sleep-duration context; not used as medical diagnosis or treatment guidance.

- Watson NF, Badr MS, Belenky G, et al. Recommended Amount of Sleep for a Healthy Adult: A Joint Consensus Statement of the AASM and Sleep Research Society. Journal of Clinical Sleep Medicine, 2015. <https://doi.org/10.5664/jcsm.4758>

Used in Recovery and Tracking. Public-health sleep overview for sleep consistency and insufficient-sleep context.

- Centers for Disease Control and Prevention. About Sleep. Centers for Disease Control and Prevention, 2024. <https://www.cdc.gov/sleep/about/index.html>

Used in Recovery, Shift Work, and Implementation. Fatigue-aware scheduling and shift-work education context; apply conservatively outside healthcare.

- National Institute for Occupational Safety and Health. NIOSH Training for Nurses on Shift Work and Long Work Hours. Centers for Disease Control and Prevention, 2020. <https://www.cdc.gov/niosh/work-hour-training-for-nurses/longhours/default.html>

Medical-Information And Red-Flag Context

Used in Nutrition, Hydration, and Recovery. General medical-information source for dehydration symptoms and urgent escalation context.

- National Library of Medicine. Dehydration. MedlinePlus, 2023. <https://medlineplus.gov/dehydration.html>

Used in Recovery, Heat Stress, and Safety Language. General medical-information source for heat illness red flags and emergency escalation context.

- National Library of Medicine. Heat Emergencies. MedlinePlus Medical Encyclopedia, 2025. <https://medlineplus.gov/ency/article/000056.htm>

Final-Review Notes

- Confirm exact editions, access dates, and citation style before final publication.
- Keep these sources tied to the claims they actually support.
- Do not use this reference list to imply individualized medical, dietetic, or clinical care.
- Preserve urgent medical attention language for heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms.
- Keep the manuscript in review status until a later issue explicitly authorizes final release.

Version History

Source: Publishing/04-back-matter.md | Status: review

Version History Review Placeholder

Version	Date	Status	Notes
Review draft	2026-07-09	review	Publication assembly, table of contents, and PDF export prep added

PDF Export Checklist

Source: *Publishing/05-pdf-export-checklist.md* | Status: review

Purpose

Use this checklist before a later PDF build issue. This file prepares the manuscript for export, but it does not create the final PDF and does not mark any chapter as final.

Export Readiness Checklist

- Confirm every chapter intended for export is listed in MANUSCRIPT.md (../MANUSCRIPT.md).
- Confirm chapter order matches Publishing/02-table-of-contents.md (02-table-of-contents.md).
- Confirm front matter is present and includes a conservative safety disclaimer near the front.
- Confirm back matter is present and includes references, review notes, glossary placeholder, version history placeholder, and appendix list placeholder.
- Confirm every exported chapter has the intended status and no chapter is marked final unless a future approval issue explicitly authorizes that change.
- Confirm references are formatted consistently and are relevant to the chapter where they appear.
- Confirm tables render properly in narrow and full-width layouts.
- Confirm page breaks do not split important tables, checklists, or safety notes in confusing places.
- Confirm links and internal references work, or decide which links should become plain text in the PDF.
- Confirm urgent-symptom safety language appears near the front of the manuscript.
- Confirm no placeholder text remains except approved publication placeholders.
- Confirm copied checklists remain readable when exported.
- Confirm headings follow a consistent hierarchy.
- Confirm file names and links use the real repository paths listed in the table of contents.

Safety Placement Check

The front matter should include clear language that heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms require urgent medical attention.

Export Notes For Later Build

- Use the linked manuscript assembly as the source order.
- Keep a separate review PDF and final PDF process.
- Proof the review PDF before changing publication status.
- Do not create the final PDF from this issue.

Final Review Checklist

Source: Publishing/06-final-review-checklist.md | Status: review

Purpose

Use this checklist before any final publication decision. Completing this checklist prepares the manuscript for a later finalization step; it does not itself make the manuscript final.

Content Review

- Confirm each chapter has a clear purpose and fits the manuscript order.
- Confirm duplicated or outdated chapter references have been removed from the publication assembly.
- Confirm no unsupported claims were added during publication prep.
- Confirm examples, tables, checklists, and decision rules are consistent across chapters.

Coaching And Programming Review

- Confirm training progression is RPE-based and does not require max testing.
- Confirm readiness rules, substitutions, deloads, and conditioning guidance preserve training intent.
- Confirm pain, injury, persistent joint symptoms, and neurological symptoms are routed to coach/clinician review or urgent care language where appropriate.

Nutrition Review

- Confirm macro, meal timing, hydration, electrolyte, supplement, grocery, and adjustment chapters remain practical and conservative.
- Confirm supplement language does not overstate outcomes or replace medical advice.
- Confirm nutrition adjustments connect to the Tracking volume.

Recovery And Heat-Stress Safety Review

- Confirm sleep, shift-work, heat-stress, soreness, and weekly recovery guidance are consistent.
- Confirm heat illness symptoms are treated as urgent symptoms, not normal training fatigue.
- Confirm recovery decisions connect to Training, Nutrition, and Tracking.

Medical Red-Flag Language Review

- Confirm the front matter includes urgent medical attention language for heat illness symptoms, chest pain, fainting, severe shortness of breath, neurological symptoms, or rapidly worsening symptoms.
- Confirm pain, injury concerns, persistent symptoms, or persistent joint issues are routed to appropriate coach/clinician review.
- Confirm no chapter implies diagnosis, treatment, or guaranteed prevention of medical conditions.

Copy Edit

- Confirm spelling, punctuation, capitalization, and heading style.
- Confirm tone is practical, direct, and professional.
- Confirm table labels and checklist wording are consistent.
- Confirm links point to real repository files where possible.

Layout Review

- Confirm headings produce a useful PDF outline.
- Confirm tables and checklists are readable after export.
- Confirm page breaks do not separate safety notes from the context needed to understand them.
- Confirm front matter, volume starts, back matter, and appendices have clean spacing.

PDF Proof Review

- Generate a review PDF only after export readiness is complete.

- Proof the review PDF for broken links, page breaks, table rendering, heading hierarchy, and visible placeholders.
- Confirm the proof is not labeled final until the publication approval step is complete.

Version Tag And Release Checklist

- Choose a version tag only after final review is approved.
- Record version history in the back matter.
- Confirm release notes describe what changed without overstating finality.
- Keep the current manuscript in review status until a later issue explicitly authorizes final release.